Koopman Spectral Theory for Indivisible Stochastic Processes

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Abstract

The operator-theoretic analysis of stochastic dynamics standardly assumes a transition kernel satisfying the Chapman–Kolmogorov (CK) factorization, which equips the two-time operator family with the cocycle structure on which generator-based spectral theory rests. We develop spectral machinery for two-time transition families that violate this condition irreducibly—indivisible families—for which no time-ordered-exponential generator in the Dyson (Bochner-integrable) class exists. The framework operates on the forward family $\{\mathcal{P}(t,s)\}_{s\le t}$ on $L^2(\Omega,\mu)$ through the singular structure of its composition deficit $\Delta(t,s,r) = \mathcal{P}(t,s) - \mathcal{P}(t,r)\mathcal{P}(r,s)$, not through eigenexpansions. Two results organize the paper. Locally, an indivisibility tensor $\beta(s) = C(s) - L(s)^2 - L'(s)$, computed from second-order Taylor coefficients at the diagonal, governs the exact leading-order scaling $\Delta(t,s,r) = (t-r)(r-s)\,\beta(s) + o((t-s)^2)$; its vanishing is necessary for divisibility of a jointly C^2 family, and $\beta(s_0) \neq 0$ forces CK failure near s_0. Globally, closed deficit subspaces $V_{\mathrm{in}}, V_{\mathrm{out}}$ concentrate the failure of factorization—every deficit operator vanishes on $V_{\mathrm{in}}^{\perp}$—and, under a standing contraction assumption, a universal three-pair bound places every divisible comparison family at operator-norm distance at least $D(t,s,r)/3$ from the family at the worst single time pair, a constant that cannot be improved. Covariant eigenfamilies with non-multiplicative multipliers witness indivisibility; an integrated deficit on the trajectory space $L^2([0,T]\times\Omega)$ vanishes exactly under divisibility; each two-time map admits a Stinespring dilation on a fixed dilation space, jointly continuous in (t,s), with a measurable formal Hamiltonian generator under explicit regularity, and divisible families admit an exact cocycle dilation on every finite time grid. In any selected unistochastic gauge the deficit splits exactly into an interference cross-term plus a non-composition term; only the sum is gauge-invariant. Worked two- and three-state families, including one with $V_{\mathrm{in}} \neq V_{\mathrm{out}}$, exhibit the constructions in closed form.

Keywords. Koopman operator; dynamic mode decomposition; non-Markovian dynamics; Chapman–Kolmogorov; indivisible stochastic processes; Stinespring dilation; spectral theory; non-self-adjoint operators.

AMS Subject Classifications. 37M10, 37A50, 37N40, 47A10, 47B33, 47L30, 60J25, 65P20.

1 Introduction

The Koopman operator formalism, introduced by Koopman in 1931 [1] and revived as a computational tool through dynamic mode decomposition [2, 3, 4] and the broader operator-theoretic program in dynamical systems [5, 6, 7, 27], provides a linear lift of nonlinear dynamics: nonlinear evolution on the state space is replaced by a linear evolution of observables, whose spectral

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analysis exposes the modes, frequencies, and growth rates of the underlying flow. For deterministic dynamics the formalism is by now standard [9]. The non-autonomous extension to two-time families $\{\mathcal{K}(t,s)\}_{s\le t}$ has been developed by Maćešić, Črnjarić-Žic and Mezić [30] for linear non-autonomous systems and by Črnjarić-Žic, Maćešić and Mezić [31] for random dynamical systems; the parallel non-autonomous transfer-operator program of Froyland and collaborators [33, 32, 34] has supplied the dual side. For stochastic dynamics the natural extension proceeds through transition kernels and the Chapman–Kolmogorov factorization condition [10, 8]: if a process $\{X_t\}_{t\ge 0}$ on a state space Ω admits a transition kernel $P(t,x \mid s,y)$ satisfying

$$P(t,x \mid s,y) = \int_\Omega P(t,x \mid r,z)\, P(r,z \mid s,y)\, dz \qquad \text{for all } s<r<t, \tag{1}$$

then the forward transition operator $\mathcal{P}(t,s)$ acting on densities $g\colon \Omega \to \mathbb{C}$ as

$$\big(\mathcal{P}(t,s)g\big)(x) \;=\; \int_\Omega P(t,x \mid s,y)\, g(y)\, dy$$

inherits the cocycle (semigroup) structure $\mathcal{P}(t,s) = \mathcal{P}(t,r)\,\mathcal{P}(r,s)$ and, under suitable regularity, possesses an infinitesimal generator $\mathcal{L}$ via $\mathcal{P}(t,s) = \exp\big((t-s)\,\mathcal{L}\big)$. Its Hilbert adjoint, the Koopman operator on observables, $(\mathcal{K}(t,s)f)(y) = \int_\Omega f(x)\, P(t,x \mid s,y)\, dx = \mathbb{E}[f(X_t) \mid X_s = y]$, composes contravariantly, $\mathcal{K}(t,s) = \mathcal{K}(r,s)\,\mathcal{K}(t,r)$. The spectral theory of $\mathcal{L}$ then governs the long-time behavior of the process. Throughout this paper we work with the forward family $\{\mathcal{P}(t,s)\}$ and record the Koopman family as its adjoint (Remark 3.2); the composition deficits of the two conventions are mutual adjoints with equal norms, so no spectral content depends on the choice. A word on the title: "Koopman spectral theory" is used here as the umbrella term for the operator-theoretic program—spectral analysis of dynamics through linear transition and observable operators—rather than as a reference to the Koopman (adjoint) side alone; the paper extends that program's spectral machinery beyond the divisible regime, and every statement transfers between the forward and Koopman conventions by adjunction.

This paper concerns transition families for which (1) does not hold. We refer to such families, and to processes inducing them, as *indivisible* following Milz and Modi [36] and recent work of Barandes [11, 12]. Barandes's stochastic–quantum correspondence associates to a stochastic dynamics a dilated unitary description, with indivisibility of the stochastic family reflected in the failure of the dilated evolution to compose along intermediate times; we invoke the correspondence purely in this dilational form, as motivation for the Stinespring construction of Section 5. Two structural cautions, developed in Section 2, delimit what a same-dimension unitary description can do. First, a column-stochastic matrix admits a same-dimension unitary representation $\Gamma_{ij} = |U_{ij}|^2$ only if it is unistochastic, a strict subset even of the doubly stochastic matrices for $n \ge 3$ (Proposition 2.11). Second, even where such representations exist pairwise, the unitary cocycle property of a selected family of representatives neither implies nor is implied by CK factorization of the squared moduli (Remark 2.12); the gap is precisely the interference cross-term analysed in Section 4. We do not adopt or endorse Barandes's broader interpretational program; the operator-theoretic content of this paper is independent of any reading of indivisibility as the source of quantum-mechanical features, and the spectral and dilation results of Sections 4–7 stand as theorems on column-stochastic two-time families regardless of how the underlying physical content is interpreted.

An independent foundational argument [16] motivates considering indivisibility as a structural condition that arises naturally from physical considerations; we refer the reader to that work for the foundational discussion and treat the operator-theoretic content of the position—singular structure of the two-time transition family without semigroup property, Hilbert-space lift via Stinespring dilation, causal organization on $L^2([0,T]\times\Omega)$—as the subject of the present paper.

The contributions are as follows. First, we develop spectral theory for the two-time family $\{\mathcal{P}(t,s)\}_{s\le t}$ of an indivisible transition family directly, without semigroup structure: the composition-deficit operator $\Delta(t,s,r) = \mathcal{P}(t,s) - \mathcal{P}(t,r)\mathcal{P}(r,s)$ replaces the generator as the carrier of dynamical information, and no time-ordered-exponential generator in the strongly measurable, Bochner-integrable class exists when the deficit is nonzero (Proposition 3.6; the proposition is an elementary consequence of the multiplicativity of time-ordered exponentials and is recorded to fix the scope of the paper). Second—and, with the deficit geometry below, the principal contribution of the paper—we identify a local infinitesimal indivisibility tensor $\beta(s) = C(s) - L(s)^2 - L'(s)$ and prove the exact leading-order scaling law $\Delta(t,s,r) = (t-r)(r-s)\,\beta(s) + o((t-s)^2)$ near the diagonal (Proposition 4.12): β is computable from the second-order Taylor coefficients of the family alone, vanishes identically for divisible jointly-C^2 families, and furnishes a sufficient pointwise condition for indivisibility; its largest singular value calibrates the local deficit, and its leading right singular vector identifies the locally most deficit-active direction. Third, we construct closed input and output deficit subspaces $V_{\mathrm{in}}, V_{\mathrm{out}} \subset L^2(\Omega,\mu)$ from the singular subspaces of the deficit family and prove a concentration-and-obstruction theorem (Theorem 4.5): every deficit operator vanishes on $V_{\mathrm{in}}^{\perp}$ and attains its norm on a leading right singular vector in V_{in}; under a standing contraction assumption, no divisible family approximates the action of $\mathcal{P}(t,s)$ on V_{in} uniformly across intermediate time pairs, and, under the same assumption, a universal three-pair bound (Corollary 4.6) places every divisible comparison family at operator-norm distance at least $D(t,s,r)/3$ at the worst single time pair, for every triple with positive deficit; the constant $1/3$ cannot be improved (Proposition 4.7), and it is approached, though not attained, by two-state families with deterministic intermediate kernels. Fourth, spectral modes are recast in singular-spectral terms: a mode of the two-time family is a covariant eigenfamily (Definition 4.2), asserted only where it exists, and for divisible families every covariant eigenfamily has a multiplicative multiplier (Lemma 4.3), so a non-multiplicative multiplier witnesses indivisibility. Fifth, we make the causal structure of the framework explicit through one-sided operators on $L^2([0,T]\times\Omega)$ (Section 7, Theorem 7.4); the integrated kernel deficit on the retarded transition operator is a single scalar measure of indivisibility that vanishes exactly under divisibility. Sixth, starting from the known existence of completely positive trace-preserving representations of stochastic matrices [57, 58], we construct (Theorem 5.4) a Stinespring dilation [17, 18] of the empirical transition family whose channel is the identity on the diagonal $t = s$: a unitary family $\{W(t,s)\}$ on a fixed dilation space, strongly continuous in (t,s), reproducing the transition statistics by partial trace and admitting a measurable formal Hamiltonian generator under an absolute-continuity hypothesis; for divisible families a cocycle dilation exists on every finite time grid, with a fresh environment factor per step and environment dimension growing with the partition, and the fixed-dimension continuum cocycle question is posed as open. The construction makes no detailed-balance hypothesis on the kernel and applies to any column-stochastic family with the stated regularity, including the irreversible families generic to indivisible dynamics. Seventh, in any selected unistochastic gauge the deficit matrix decomposes exactly into an interference cross-term and a squared-modulus non-composition term (Theorem 4.8); the identity is gauge-relative, CK holds at a triple if and only if the two terms cancel, and the vanishing of either term separately is a property of the selected lifts rather than of the family—a point we fix with explicit counterexamples. Finally, we exhibit (Section 6) two- and three-state non-divisible families for which the deficit, the subspace pair, the tensor $\beta(s)$, and the dilation can be computed in closed form, including a non-doubly-stochastic family with $V_{\mathrm{in}} \neq V_{\mathrm{out}}$, together with a product rule (Proposition 6.1) under which the deficit of a product family is a Leibniz-type sum and the indivisibility tensor is a Kronecker sum, which supplies closed-form families in dimension 2^k; and we record the precise hierarchy of the two-time CK condition relative to Markovianity (Proposition 2.6, Corollary 2.7): Markov processes are divisible, but the converse fails, so indivisibility is a strictly stronger restriction on the two-time kernel than non-Markovianity, and the framework is not

redundant with standard memory-kernel approaches to non-Markov dynamics. The hierarchy itself is classical—part (a) of the proposition is the tower property, part (b) is Feller's 1959 example—and is presented as scope-fixing background rather than claimed as a contribution.

Causal structure. The construction throughout this paper operates on $L^2([0,T]\times\Omega)$ with one-sided causal operators: the transition family on densities is time-forward by construction, and no symmetry under time reversal is assumed or required. The retarded transition operator introduced in Section 7 is causal—adapted to the natural filtration of past subspaces, in the sense of Proposition 7.3—and the kernel-level cocycle structure is a strict consequence of forward-in-time propagation. We make this explicit at the outset to forestall a reading of Hilbert-space methods as implying time-reversal symmetry where none is intended.

Relationship to prior work. The framework of this paper sits at the intersection of three traditions, with which we now briefly position the contributions.

Non-autonomous Koopman/transfer-operator theory. Two-time operator families have been developed in the non-autonomous setting by Maćešić–Črnjarić-Žic–Mezić [30] and Črnjarić-Žic–Maćešić–Mezić [31] on the Koopman side, and on the dual transfer-operator side by Froyland and collaborators [33, 32, 34]. Both programs presuppose the cocycle property $\mathcal{P}(t,s) = \mathcal{P}(t,r)\mathcal{P}(r,s)$ (equivalently, the Chapman–Kolmogorov factorization of the underlying transition kernel). The present paper extends both lines by removing this presupposition and characterizing the structure that survives: the deficit operator $\Delta(t,s,r)$ becomes the non-trivial object whose spectral content the framework recovers.

Process-tensor formalism. A parallel line of work in operational quantum stochastic processes [35, 36, 37, 38] develops a multi-time Stinespring/dilation framework—the process tensor—for non-Markovian quantum dynamics. Two structural distinctions to the present paper are worth noting. First, that program is fundamentally discrete-time: the process tensor describes a finite sequence of control operations at fixed time steps, and the dilation theorem (Pollock et al. [35, Thm. 2], Milz–Modi [36]) is a statement about discrete unitary circuits with environment dimension growing with the number of steps. Second, the process-tensor construction handles arbitrary CPTP control operations, of which our setting (Definition 5.1) is a classical-readout specialization. Theorem 5.4 below works at the complementary continuous two-time level: each pair (t,s) separately receives a dilation on a fixed environment ($\dim\mathcal{K} \le n^2$ uniformly in (t,s)), jointly continuous in (t,s) under explicit regularity. For divisible families, the grid cocycle dilation of Theorem 5.4(iii) has exactly the discrete fresh-environment-per-step structure of the process tensor—equivalently, the standard repeated-interaction (collision-model) structure [54]—with environment dimension growing with the number of steps; we make no fixed-dimension continuum cocycle claim and pose that question as open. The two programs are compatible: in the discrete-time limit, our construction specializes to a process-tensor representation on K time steps with classical instruments; in the joint-continuity direction, the process-tensor literature makes no statement.

Non-Markovianity measures. The functional $J(\mathcal{R})$ of Theorem 7.4 is the classical analogue of the divisibility-based non-Markovianity measures developed in the open-quantum-systems literature [41, 42, 43, 44, 45, 46], particularly the Rivas–Huelga–Plenio measure [43, 44], which similarly integrates a divisibility deficit over time. Two structural differences distinguish $J(\mathcal{R})$ from these quantum measures: it operates on the column-stochastic family directly rather than on its CP-divisible quantum analogue, and it integrates over triples (s,r,t) on a single trajectory space $\mathcal{H}_T = L^2([0,T]\times\Omega)$, providing a single scalar functional that depends on the full two-time family rather than on a one-parameter dynamical map.

Stochastic-to-quantum lifts. The existence of a completely positive trace-preserving repre-

sentation of a stochastic matrix, and the role of Chapman–Kolmogorov divisibility for the lifted family, precede this paper. Canturk [57] represents every stochastic matrix on an N-level system by N Kraus operators, proves uniqueness within that class, and identifies CP-divisibility as the extension of the Chapman–Kolmogorov equation; Doukas [58] lifts an arbitrary finite-dimensional stochastic kernel to a map on $\mathcal{B}(\mathcal{H})$ with an explicit dictionary between the kernel and CPTP maps, and isolates Chapman–Kolmogorov divisibility of the lifted family as the condition under which the lift admits a Lindblad form. Section 5 does not claim the basic lift. Its content is the diagonal-retention channel of Definition 5.1, chosen so that the channel is the identity on the diagonal $t = s$; the jointly continuous unitary completion with $W(s, s) = I$ on a fixed environment of dimension $n^2 - n + 1$; the formal generator under (R3′); the grid cocycle for divisible families; and the place of the lift in the deficit framework.

Approximate flows and sewing. The rough-path literature [59, 60, 63] and the non-linear sewing lemma of Brault and Lejay [61, 62] treat two-parameter families that are only approximately multiplicative, with explicit control of the composition defect $\mu_{ts} - \mu_{tu} \circ \mu_{us}$, and construct an exact flow from an almost-flow whose defect is small at fine scales. Measuring the failure of a composition law by a defect is common to that literature and to this paper. The purposes differ: there the defect is an error to be controlled so that an exact flow can be reconstructed; here the defect is the object of study, and its singular geometry (Section 4) and local tensor (Section 4.5) have no counterpart there.

Theorem-level comparison. Table 1 sets the results of this paper against the three programs: what each program presupposes, which results here have a counterpart in it, and which have none.

Program	Presupposes	Counterpart in this paper	No counterpart
Non-autonomous Koopman and transfer-operator theory [30, 31, 33, 32, 34]	the cocycle $\mathcal{P}(t, s) = \mathcal{P}(t, r)\mathcal{P}(r, s)$	Definition 4.2 reduces to the non-autonomous eigenfunction of [30] under the cocycle; Lemma 4.3 is that case	Theorem 4.5, Corollary 4.6, Proposition 4.12, Theorem 7.4: each object vanishes identically under the cocycle. Theorem 4.8: the invariant sum $\mathfrak{C} + \mathfrak{D} = \Delta$ vanishes under the cocycle, while the gauge-dependent summands need not (Theorem 4.8(c), Remark 4.9)
Divisibility-based non-Markovianity measures [41, 42, 43, 44, 45, 46]	a one-parameter family of dynamical maps	$J(\mathcal{R})$ of Theorem 7.4 is the classical analogue of the RHP integral; $D(t, s, r)$ is the pairwise deficit; $\beta(s)$ of Definition 4.11 is a second-order partial analogue of the RHP local rate	the subspace concentration of Theorem 4.5(a); the approximation obstruction of Theorem 4.5(b) and Corollary 4.6 (the measures quantify non-divisibility but do not bound divisible approximants away from the family); the trajectory-space formulation (Lemma 7.2, Proposition 7.3) and the kernel-recovery Lemma D.1
Process tensors [35, 36, 37, 38]	discrete time steps, with an environment factor per step	Theorem 5.4(iii) is the multi-time dilation with a fresh factor per step; Theorem 5.4(iv) is Stinespring uniqueness	Theorem 5.4(i)–(ii): a fixed environment of dimension $n^2 - n + 1$, joint continuity in (t, s), and the formal generator; the two-time kernel-level results (Proposition 2.11, Remark 2.12, Theorems 4.5 and 4.8, Corollary 4.6, Proposition 4.12)

Table 1: Theorem-level comparison with the three neighbouring programs.

Outline. Section 2 establishes the framework of indivisible two-time transition families, records the unistochasticity facts used later, and states the dilational form of the stochastic–quantum correspondence. Section 3 fixes the operator conventions and standing assumptions, reviews the divisible semigroup formulation, and identifies the precise point at which it fails for indivisible families. Section 4 develops the singular structure of the deficit family: concentration and obstruction (Theorem 4.5, Corollary 4.6), the gauge-relative cross-term decomposition, and the local tensor $\beta(s)$. Section 5 constructs the Stinespring dilation under explicit regularity hypotheses. Section 6 works out explicit two- and three-state non-divisible examples and a product rule that supplies closed-form families in higher dimension. Section 7 treats the causal structure via one-sided operators on $L^2([0,T]\times\Omega)$. Section 8 discusses implications and open questions. Full proofs of the four main theorems are given in Appendices A–D.

2 Indivisible two-time transition families: framework

2.1 Transition kernels and the Chapman–Kolmogorov condition

Let (Ω,Σ,μ) be a measure space, with $\Omega\subseteq\mathbb{R}^d$ a Polish state space and μ a σ-finite reference measure. Let $\{X_t\}_{t\in[0,T]}$ be an Ω-valued stochastic process on a probability space $(\Theta,\mathcal{F},\mathbb{P})$, adapted to a filtration $\{\mathcal{F}_t\}$. We denote by ρ_s the law of X_s on (Ω,Σ) and assume $\rho_s\ll\mu$ with density $\pi_s=d\rho_s/d\mu\in L^1(\Omega,\mu)$ throughout.

Definition 2.1 (Transition kernel)**.** A measurable function $P\colon[0,T]^2_{\leq}\times\Omega\times\Omega\to\mathbb{R}_{\geq 0}$, where $[0,T]^2_{\leq}=\{(t,s):0\leq s\leq t\leq T\}$, is a *transition kernel* for the process $\{X_t\}$ if for all $s\leq t$ and ρ_s-a.e. $y\in\Omega$,

$$\mathbb{P}(X_t\in A\mid X_s=y)\;=\;\int_A P(t,x\mid s,y)\,d\mu(x),\qquad A\in\Sigma,$$

and $\int_\Omega P(t,x\mid s,y)\,d\mu(x)=1$.

Definition 2.2 (Chapman–Kolmogorov condition)**.** A transition kernel P satisfies the *Chapman–Kolmogorov* (CK) factorization condition if for all $0\leq s<r<t\leq T$ and ρ_s-a.e. y,

$$P(t,x\mid s,y)\;=\;\int_\Omega P(t,x\mid r,z)\,P(r,z\mid s,y)\,d\mu(z)\qquad\mu\text{-a.e. } x. \tag{2}$$

Definition 2.3 (Two-time transition family)**.** A *two-time transition family* on (Ω,Σ,μ) is a measurable family $\{P(t,\cdot\mid s,\cdot)\}_{(t,s)\in[0,T]^2_{\leq}}$ of non-negative kernels satisfying $\int_\Omega P(t,x\mid s,y)\,d\mu(x)=1$ for a.e. y and the boundary condition that $P(s,\cdot\mid s,\cdot)$ acts as the identity. The family is *process-induced* if there exists a stochastic process $\{X_t\}$ for which each $P(t,\cdot\mid s,\cdot)$ is a transition kernel in the sense of Definition 2.1.

Definition 2.4 (Divisible / indivisible)**.** A two-time transition family is *divisible* if it satisfies (2) at every triple $0\leq s<r<t\leq T$, and *indivisible* otherwise. (For a family that is not process-induced, the qualifier "ρ_s-a.e." in (2) is read as μ-a.e.) A process $\{X_t\}$ is *divisible* if it admits a transition kernel whose induced two-time family is divisible, and *indivisible* if no such transition kernel exists. The *indivisibility deficit* of a family at the triple (t,s,r) is

$$D(t,s,r)\;=\;\Big\|P(t,\cdot\mid s,\cdot)\;-\;\int_\Omega P(t,\cdot\mid r,z)\,P(r,z\mid s,\cdot)\,d\mu(z)\Big\|_{\mathrm{op}},$$

where $\|\cdot\|_{\mathrm{op}}$ is the operator norm on $L^2(\Omega,\mu)\to L^2(\Omega,\mu)$ (well defined under the boundedness conventions fixed in Section 3).

The indivisibility deficit measures the failure of factorization at intermediate time r. For divisible families $D(t,s,r) \equiv 0$; an indivisible family has $D(t,s,r) > 0$ at some triple, and for operator-norm-continuous families the deficit is then positive on a set of positive measure in $\{(t,s,r) : 0 \le s < r < t \le T\}$.

Remark 2.5 (Realizability; results are stated for transition families). Except where a process is explicitly hypothesized, the results of this paper are statements about two-time transition families and require no assumption that the family is process-induced: nothing below reconstructs, or needs, consistent finite-dimensional joint distributions. Process-induced families are the special case that motivates the framework. The distinction matters in both directions. The worked families of Section 6 are specified directly at the two-time level, which is exactly what the spectral framework analyses; whether a given two-time family is realizable by a genuine process is a separate structural question, not treated here. Conversely, the one place in this paper where the distinction does work is Proposition 2.6, which concerns processes, since Markovianity is a property of joint laws rather than of two-time kernels.

The indivisibility condition of Definition 2.4 is a strictly stronger restriction on the two-time kernel structure than non-Markovianity. The following proposition records the precise hierarchy and is used in the discussion of related notions immediately below; both parts are classical—(a) is the tower property of conditional expectation, (b) is Feller's example [51]—and the proofs are included only to keep the paper self-contained.

Proposition 2.6 (Markov property implies Chapman–Kolmogorov; the converse fails)**.** *Let $X = \{X_t\}_{t\in[0,T]}$ be a stochastic process on (Ω, Σ) admitting a two-time transition kernel $P(t, x \mid s, y)$ in the sense of Definition 2.1. The following hold.*

(a) If X is Markov—i.e., for all $0 \le s \le t \le T$ and bounded measurable $f : \Omega \to \mathbb{R}$, $\mathbb{E}[f(X_t) \mid \mathcal{F}_s] = \mathbb{E}[f(X_t) \mid X_s]$ almost surely—then P satisfies the Chapman–Kolmogorov factorization (2).

(b) The converse fails: there exist non-Markov processes X on Ω whose two-time transition kernel P satisfies (2).

Proof. For (a), the tower property and the Markov hypothesis give, for $0 \le s < r < t$ and ρ_s-a.e. $y \in \Omega$,

$$\int_\Omega f(x)\, P(t, x \mid s, y)\, d\mu(x) = \mathbb{E}[f(X_t) \mid X_s = y] = \mathbb{E}\big[\,\mathbb{E}[f(X_t) \mid \mathcal{F}_r] \;\big|\; X_s = y\,\big] = \mathbb{E}\big[\,\mathbb{E}[f(X_t) \mid X_r] \;\big|\; X_s = y\,\big],$$

where the last equality uses Markovianity. Iterated conditioning then gives this equal to $\int_\Omega \int_\Omega f(x)\, P(t, x \mid r, z)\, P(r, z \mid s, y)\, d\mu(x)\, d\mu(z)$. Equating for all bounded measurable f yields (2) ρ_s-a.e.

For (b), the canonical witness is Feller's construction [51] of a non-Markovian process with the semigroup property: a process on a finite state space whose two-time transition probabilities satisfy the Chapman–Kolmogorov equation (2)—indeed compose as a stationary transition semigroup—while the joint distributions at three or more times are inconsistent with any Markov chain sharing those two-time transitions. The Chapman–Kolmogorov condition constrains only the pairwise conditional kernels; the Markov property constrains all finite-dimensional distributions, and Feller's example populates the gap. (Projections of Markov chains under lumpable partitions do not witness the converse: a lumpable partition yields an observed process that is itself Markov.) □

Corollary 2.7 (Indivisibility is strictly stronger than non-Markovianity)**.** *If X is indivisible (Definition 2.4), then X is non-Markov. The converse fails: non-Markov processes whose two-time kernel satisfies (2) are divisible.*

Proof. Contrapositive of Proposition 2.6(a) gives the first claim. Proposition 2.6(b) gives the second. □

The framework of this paper operates at the two-time level: it requires only the two-time kernel family, not the full hierarchy of finite-dimensional distributions, and it captures exactly the indivisible class—including non-Markov processes whose memory structure is concentrated at the two-time level. Standard memory-kernel approaches to non-Markov dynamics—Mori–Zwanzig projections, Nakajima–Zwanzig master equations, time-convolutionless expansions—typically operate on processes whose non-Markovianity is compatible with two-time CK (after augmentation or projection); the indivisible class lies outside this regime and is the structural focus here.

Remark 2.8 (Distinctions among related notions). The notions of *non-Markov*, *Chapman–Kolmogorov-violating*, *indivisible* (in the sense of this paper), *memory-bearing*, and *absence of an infinitesimal generator on* Ω are tightly related but mathematically distinct, and we wish to be precise about the distinction.

Augmentation does not trivialize the framework. A standard response to non-Markov dynamics is to enlarge the state space to record relevant memory variables, producing a Markov process on a larger configuration space $\widetilde{\Omega} \supsetneq \Omega$ whose marginal on Ω is the original non-Markov process. The classical projection-operator formalism of Mori [23] and Zwanzig [24] systematizes this construction in the setting of statistical mechanics; analogous augmentations include hidden-Markov embeddings of partially observed systems [25] and, in the open-quantum-systems literature, dilation of CP-divisible Lindblad dynamics from a system to a system-plus-environment model [26]. The result of augmentation is a divisible kernel on $\widetilde{\Omega}$, but the construction does not eliminate the spectral phenomena studied in the present paper. Two reasons.

First, the spectral framework is configuration-space relative. Theorems 4.5 and 7.4 are statements about the two-time operator family $\{\mathcal{P}(t,s)\}$ acting on $L^2(\Omega,\mu)$ for a chosen Ω; the deficit operator $\Delta(t,s,r)$ lives in $\mathcal{B}(L^2(\Omega,\mu))$, and the deficit subspaces $V_{\text{in}}, V_{\text{out}}$ are subspaces of $L^2(\Omega,\mu)$. Passage to $\widetilde{\Omega}$ produces a different operator family $\{\widetilde{\mathcal{P}}(t,s)\}$ on the larger space $L^2(\widetilde{\Omega},\tilde{\mu})$, whose deficit operator vanishes by construction; this does not refute the non-vanishing of the deficit on the original Ω. The spectral content recovered by the present framework is the content of the two-time dynamics restricted to the chosen configuration space, not a feature of the underlying probability law in the abstract.

Second, the question that the framework is designed to answer is empirical: given two-time transition data on a chosen Ω, do these data come from a CK-factorizing kernel? Theorem 7.4 provides a quantitative test ($J(\mathcal{R}) > 0$) for this question that operates entirely on Ω, regardless of whether some larger $\widetilde{\Omega}$ might exist on which a CK-factorizing extension is available. For the analyst who has committed to Ω as the relevant configuration space—because the data are about Ω, or because Ω is the natural domain of physical observables—the spectral framework characterizes the structure that is in fact present in those data.

Constrained augmentation may be unavailable. Unrestricted measurable augmentation, by contrast, is always available: enlarging the state to the realized history (the path up to the present time) Markovizes any process, so non-augmentability cannot be meant unconditionally. The substantive question is whether an augmentation exists that respects the operational constraints of the modeling context. For some classes of dynamics—most notably the indivisible dynamics arising in the stochastic-quantum correspondence [11, 12]—two no-go theorems on the unitary side of the correspondence exclude augmentations that respect the corresponding constraint. Bell's theorem [13], in Fine's form [14], states that in the two-setting, two-outcome scenario a joint probability distribution over the outcomes of all measurement settings—equivalently, a hidden-variable augmentation that is local in Bell's sense, its augmented state factorizing across the spacelike-separated subsystems—exists if and only if the

CHSH inequalities hold; quantum correlations violate them, so no Bell-local augmentation exists for such composite systems. The Kochen–Specker theorem [15] excludes noncontextual augmentations, those assigning each observable a value independent of the measurement context, in Hilbert-space dimension at least three. Both theorems leave unconstrained history augmentations untouched. We do not establish any such constrained non-augmentability as part of the present paper, and the spectral framework does not depend on it: the framework is a tool for analysing whatever non-CK two-time kernel family is presented, on whatever configuration space the analyst has chosen.

Generator absence. CK violation on Ω implies the absence of a one-parameter generator family for $\{\mathcal{P}(t,s)\}$ on $L^2(\Omega,\mu)$ (Proposition 3.6). The converse implication—that absence of a generator implies CK violation—requires the standard regularity hypotheses of Hille–Yosida and is not used in this paper.

Summary. The present paper treats indivisibility in the strict configuration-space sense (Definition 2.4): no transition kernel on Ω itself satisfies the CK condition. The spectral framework characterizes this condition through structures defined on $L^2(\Omega,\mu)$, is independent of the question of whether a divisible augmentation on a larger state space exists, and applies whenever the analyst is committed to a particular configuration space. We emphasize that, on this view, indivisibility is a property of the pair (process, configuration space), not of the process in the abstract: the same physical evolution can be indivisible relative to one observable algebra and divisible relative to another. The present paper studies the structural invariants of the two-time operator family conditional on a fixed configuration-space choice; the choice itself is taken as given by the modeling context.

Remark 2.9. For an indivisible family, the kernels $\{P(t,\cdot \mid s,\cdot)\}_{s\le t}$ are intrinsically two-time in their *composition structure*: they cannot be generated by a one-parameter semigroup $\{P_\tau\}$ composing as $P_{\tau+\sigma}=P_\tau P_\sigma$, because (2) fails. Note that dependence on absolute times is not the issue: a family may depend on the elapsed time $\tau=t-s$ only and still be indivisible, because its elapsed-time kernels need not compose—the two-state example of Section 6 is exactly of this kind. We use the notation $\mathcal{P}(t,s)$ throughout for the corresponding two-time transition operator (Definition 3.1).

2.2 Unistochasticity, unitary representations, and the dilational correspondence

For a finite configuration space $\Omega=\{1,2,\dots,n\}$ and times $s\le t$, the transition kernel reduces to a column-stochastic matrix $\Gamma(t\leftarrow s)\in\mathbb{R}^{n\times n}_{\ge 0}$ with $\sum_i \Gamma_{ij}(t\leftarrow s)=1$ for each j, encoding $\mathbb{P}(X_t=i\mid X_s=j)=\Gamma_{ij}(t\leftarrow s)$ in the process-induced case. The CK condition (2) becomes the matrix factorization

$$\Gamma(t_2\leftarrow t_0) \;=\; \Gamma(t_2\leftarrow t_1)\,\Gamma(t_1\leftarrow t_0), \qquad t_0\le t_1\le t_2.$$

Indivisibility is the failure of this factorization for some intermediate time t_1.

It is natural to ask when a stochastic matrix admits a same-dimension unitary representation, since such representations underlie the cross-term decomposition of Section 4 and motivate the dilation of Section 5. The precise facts are as follows; they are strictly weaker than a representation theorem for arbitrary stochastic matrices, and they carry no composition structure in time.

Definition 2.10 (Unistochastic matrix)**.** A column-stochastic matrix $\Gamma\in\mathbb{R}^{n\times n}_{\ge 0}$ is *unistochastic* if there exists a unitary $U\in U(n)$ with $\Gamma_{ij}=|U_{ij}|^2$ for all i,j.

Proposition 2.11 (Pairwise unistochastic representations; strict inclusions)**.** *(a) For any unitary $U \in U(n)$, the matrix $(|U_{ij}|^2)_{ij}$ has unit row and column sums. Hence every unistochastic matrix is doubly stochastic.*

(b) For $n = 2$ the converse holds: every 2×2 doubly stochastic matrix is unistochastic, realized e.g. by a rotation $U = R(\theta)$ with $\cos^2\theta$ equal to the diagonal entry.

(c) For $n \geq 3$ the inclusions

$$\{\textit{unistochastic}\} \subsetneq \{\textit{doubly stochastic}\} \subsetneq \{\textit{column-stochastic}\}$$

are strict. For $n = 3$ the first inclusion is witnessed by $\frac{1}{2}(J_3 - I_3)$, with J_3 the 3×3 all-ones matrix, which is doubly stochastic but not unistochastic; for $n > 3$ it is witnessed by the padded matrix $\frac{1}{2}(J_3 - I_3) \oplus I_{n-3}$. (The unpadded $\frac{1}{2}(J_n - I_n)$ has row sums $(n-1)/2$ and is doubly stochastic only for $n = 3$.)

(d) If every member of a two-time family $\{\Gamma(t \leftarrow s)\}$ is unistochastic, then for each pair (t, s) separately *there exists a unitary $U(t, s)$ with $\Gamma_{ij}(t \leftarrow s) = |U_{ij}(t, s)|^2$. Any such assignment $(t, s) \mapsto U(t, s)$ is called a* gauge*; it is non-unique (at minimum, $U \mapsto D_1 U D_2$ for diagonal unitaries D_1, D_2 preserves the squared moduli), and no composition law across triples is implied.*

Proof. (a) The rows and columns of a unitary are unit vectors. (b) For $\Gamma = \begin{pmatrix} a & 1-a \\ 1-a & a \end{pmatrix}$ with $a \in [0, 1]$, take $U = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$ with $\cos^2\theta = a$. (c) For $n = 3$: suppose $\frac{1}{2}(J_3 - I_3) = (|U_{ij}|^2)$ for some $U \in U(3)$. Then $|U_{ii}| = 0$ and $|U_{ij}| = 1/\sqrt{2}$ for $i \neq j$. Orthogonality of the first two rows requires $\sum_j U_{1j}\overline{U_{2j}} = 0$, a vanishing sum of three complex numbers whose moduli are $|U_{11}||U_{21}| = 0$, $|U_{12}||U_{22}| = 0$, and $|U_{13}||U_{23}| = \frac{1}{2}$; no such sum vanishes. For $n > 3$: suppose $\frac{1}{2}(J_3 - I_3) \oplus I_{n-3} = (|U_{ij}|^2)$ for some $U \in U(n)$. The zero pattern forces $U_{ij} = 0$ whenever exactly one of i, j lies in $\{1, 2, 3\}$, so $U = U_1 \oplus U_2$ with U_1 a 3×3 block and U_2 an $(n-3)\times(n-3)$ block; unitarity of U makes each block unitary (its columns are orthonormal and, the off-block entries vanishing, span the corresponding coordinate subspace), and $|(U_1)_{ij}|^2 = \frac{1}{2}(J_3 - I_3)_{ij}$ contradicts the $n = 3$ case. The second inclusion is witnessed by any column-stochastic matrix with unequal row sums. (d) is (a)–(c) applied pairwise; the rephasing freedom is immediate, and Remark 4.9 below exhibits distinct gauges of the same divisible family with materially different cross-term decompositions. For the structure of the unistochastic set see [20, 21]. □

Remark 2.12 (Unitary cocycles neither imply nor follow from CK). The squared-modulus map $U \mapsto |U|^2$ does not intertwine unitary composition with matrix composition, in either direction. Let $R(\theta)$ be the rotation by θ. Then $R(90°) = R(45°)\,R(45°)$ is an exact unitary cocycle, while $|R(45°)|^2 = F$ (the fair-coin matrix with all entries $\frac{1}{2}$) satisfies $F^2 = F$, and $|R(90°)|^2 = X$ (the swap matrix): a perfect cocycle whose squared moduli violate CK maximally, the gap being the interference cross-term made explicit in Theorem 4.8. Conversely, CK constrains only the squared moduli, and a selected gauge of a divisible family need not compose (Remark 4.9). Divisibility of $\{\Gamma(t \leftarrow s)\}$ and the cocycle property of a selected unitary gauge are logically independent properties. What survives, and what this paper uses, is the *dilational* correspondence: every column-stochastic family—unistochastic or not—admits a unitary representation on an enlarged Hilbert space $\mathcal{H}' = \mathcal{H} \otimes \mathcal{K}$ via Stinespring dilation, developed in Section 5. This is the form in which the stochastic–quantum correspondence [11, 12] enters the present framework; the interpretive content—identifying the unitary side with quantum mechanics—is not used as a premise of any theorem here.

3 Operator framework: conventions, standing assumptions, and breakdown of semigroup structure

3.1 The two-time transition operator, its Koopman adjoint, and the standing assumption

Let $L^2(\Omega, \mu)$ denote the Hilbert space of square-integrable functions on the state space.

Definition 3.1 (Two-time transition operator; Koopman adjoint)**.** For a two-time transition family with kernel P, the *(forward) transition operator* $\mathcal{P}(t,s)\colon L^2(\Omega,\mu) \to L^2(\Omega,\mu)$ is defined, whenever it is bounded, by

$$\big(\mathcal{P}(t,s)g\big)(x) \;=\; \int_\Omega P(t, x \mid s, y)\, g(y)\, d\mu(y).$$

Its Hilbert adjoint $\mathcal{K}(t,s) := \mathcal{P}(t,s)^*$, the *Koopman operator*, acts on observables by $(\mathcal{K}(t,s)f)(y) = \int_\Omega f(x)\, P(t, x \mid s, y)\, d\mu(x)$, which for process-induced families is $\mathbb{E}[f(X_t) \mid X_s = y]$.

Under the CK condition, the forward family composes covariantly, $\mathcal{P}(t,s) = \mathcal{P}(t,r)\,\mathcal{P}(r,s)$ for $s < r < t$, while the Koopman family composes contravariantly, $\mathcal{K}(t,s) = \mathcal{K}(r,s)\,\mathcal{K}(t,r)$.

Remark 3.2 (Convention; adjoint transfer of all results). All results in this paper are stated for the forward family $\{\mathcal{P}(t,s)\}$. Since

$$\mathcal{K}(t,s) - \mathcal{K}(r,s)\,\mathcal{K}(t,r) \;=\; \big(\mathcal{P}(t,s) - \mathcal{P}(t,r)\,\mathcal{P}(r,s)\big)^* \;=\; \Delta(t,s,r)^*,$$

the composition deficits of the two conventions are mutual adjoints, with equal operator norms and equal singular values; every statement below transfers verbatim to the Koopman family upon exchanging the roles of the input and output deficit subspaces ($V_{\text{in}} \leftrightarrow V_{\text{out}}$), because the right (respectively left) singular vectors of Δ are the left (respectively right) singular vectors of Δ^*.

A stochastic kernel does not automatically define a bounded operator, still less a contraction, on an arbitrary fixed $L^2(\Omega,\mu)$; the quantitative results below therefore carry the following explicit hypothesis.

Assumption 3.3 (Standing contraction assumption)**.** Every two-time operator family appearing in a quantitative bound—the family $\{\mathcal{P}(t,s)\}$ under study and every divisible comparison family $\{\tilde{\mathcal{P}}(t,s)\}$—consists of bounded linear operators on $L^2(\Omega,\mu)$ with $\|\mathcal{P}(t,s)\|_{\text{op}} \le 1$ and $\big\|\tilde{\mathcal{P}}(t,s)\big\|_{\text{op}} \le 1$ for all $(t,s) \in [0,T]^2_{\le}$.

Remark 3.4 (Scope of the assumption, and when it holds). Boundedness alone—automatic when Ω is finite—suffices for the subspace-concentration statement of Theorem 4.5(a), for Proposition 3.6, and for the local tensor of Section 4.5. Assumption 3.3 is invoked for the quantitative approximation bounds (Theorem 4.5(b), Corollary 4.6) and for the boundedness of the retarded operator in Section 7; a general-norm version of the bounds, with the contraction constants replaced by explicit operator-norm factors, is recorded in the unnumbered remark following Corollary 4.6. Sufficient conditions for the assumption: by the Schur test, $\|\mathcal{P}(t,s)\|_{\text{op}} \le \big(\sup_x \int_\Omega P(t, x \mid s, y)\, d\mu(y)\big)^{1/2}$, since the column mass $\int_\Omega P(t, x \mid s, y)\, d\mu(x) = 1$ is normalized; hence a uniform row-mass bound $\int_\Omega P(t, x \mid s, y)\, d\mu(y) \le 1$ for μ-a.e. x suffices. This holds in particular for doubly stochastic and doubly substochastic finite matrices under counting measure, and whenever μ is invariant (or subinvariant) under the forward evolution. It can genuinely fail otherwise: the three-state family of Section 6.7 has $\|\Gamma\|_{\text{op}} > 1$ under counting measure and illustrates only the boundedness-based results.

Remark 3.5 (Hilbert-bundle formulation). The fully general setting replaces the fixed space $L^2(\Omega,\mu)$ by the bundle of time-indexed spaces $\{L^2(\Omega,\rho_t)\}_{t\in[0,T]}$: Jensen's inequality makes the Koopman operator a contraction $L^2(\rho_t)\to L^2(\rho_s)$ with no hypothesis beyond kernel normalization, and the forward family acts contractively on the dual bundle. The deficit geometry developed below carries over with the subspaces indexed by time. We work on a fixed $L^2(\Omega,\mu)$ under Assumption 3.3 for simplicity of statement, and record the bundle formulation as the natural general setting.

3.2 The standard divisible case

In the divisible case the operator-theoretic formulation is standard [10, 8, 9], and it splits into two regimes. When the family is time-homogeneous—$\mathcal{P}(t,s)$ a function of the elapsed time $t-s$ alone—divisibility makes the elapsed-time family a one-parameter semigroup, and under the usual regularity (strong continuity, contractivity) it is a C_0-contraction semigroup whose infinitesimal generator $\mathcal{L}$ is supplied by the Hille–Yosida theorem: $\mathcal{P}(t,s)=e^{(t-s)\mathcal{L}}$. When the family depends genuinely on the absolute times, divisibility instead makes $\{\mathcal{P}(t,s)\}$ an evolution family (a two-parameter cocycle), represented for bounded, strongly measurable time-dependent generators by the Dyson time-ordered exponential. In either regime the spectral theory of the generator family governs long-time behavior; standard references are [10, 8, 9].

3.3 Where the standard formulation breaks for indivisible families

For an indivisible family, the operators $\{\mathcal{P}(t,s)\}_{s\le t}$ of Definition 3.1 no longer form a semigroup: the composition $\mathcal{P}(t,r)\mathcal{P}(r,s)$ is in general a different operator from $\mathcal{P}(t,s)$. The discrepancy is precisely the indivisibility deficit:

$$\big\|\mathcal{P}(t,s)\;-\;\mathcal{P}(t,r)\,\mathcal{P}(r,s)\big\|_{\mathrm{op}}\;=\;D(t,s,r), \tag{3}$$

which is positive for some intermediate $r\in(s,t)$.

The following proposition follows in two lines from the multiplicativity of time-ordered exponentials. It is recorded to fix the scope of the paper, in the sense in which Proposition 2.6 is recorded, and is not claimed as a contribution.

Proposition 3.6. *Let $\{\mathcal{P}(t,s)\}_{s\le t}$ be the two-time operator family of an indivisible transition family, with each $\mathcal{P}(t,s)$ bounded. Then there does not exist a strongly measurable, Bochner-integrable family of bounded operators $\{\mathcal{L}(r)\}_{r\in[0,T]}$ — that is, $\mathcal{L}\in L^1([0,T];\mathcal{B}(L^2(\Omega,\mu)))$ — such that*

$$\mathcal{P}(t,s)\;=\;\mathcal{T}\exp\!\left(\int_s^t\mathcal{L}(r)\,dr\right)$$

for all $s\le t$ in $[0,T]$, where $\mathcal{T}\exp$ denotes the time-ordered exponential.

Proof. For $\mathcal{L}\in L^1([0,T];\mathcal{B}(L^2(\Omega,\mu)))$ the time-ordered exponential exists — the Dyson series converges absolutely, its k-th term dominated by $\big(\int_s^t\|\mathcal{L}(r)\|\,dr\big)^k/k!$ — and automatically satisfies the cocycle identity

$$\mathcal{T}\exp\!\left(\int_s^t\mathcal{L}(r)\,dr\right)\;=\;\mathcal{T}\exp\!\left(\int_r^t\mathcal{L}(\sigma)\,d\sigma\right)\cdot\mathcal{T}\exp\!\left(\int_s^r\mathcal{L}(\sigma)\,d\sigma\right)$$

for any $s\le r\le t$, by the multiplicative property of time-ordered products. If $\{\mathcal{P}(t,s)\}$ admitted such a representation, the cocycle identity would yield $\mathcal{P}(t,s)=\mathcal{P}(t,r)\,\mathcal{P}(r,s)$, contradicting (3) on the set where $D(t,s,r)>0$. □

Proposition 3.6 formalizes the central obstruction: an indivisible two-time family admits no Dyson-type cocycle generator. The standard tools of generator-based spectral analysis through time-ordered exponentials are unavailable; alternative formalisms (memory-kernel Volterra integrals, distributional generators, projection-operator constructions on enlarged spaces) lie outside the time-ordered-exponential framework and are not the object of this proposition. The remainder of this paper develops alternative spectral machinery that operates directly on the two-time family.

3.4 Comparison with non-autonomous Koopman analysis

A substantial literature on non-autonomous and time-varying Koopman analysis already exists, addressing dynamical systems whose two-time operator family $\{\mathcal{P}(t,s)\}$ depends genuinely on the absolute times (t,s) rather than only on the elapsed time $\tau = t - s$ [5, 27, 28, 29]. In that literature the underlying flow is typically governed by a time-dependent vector field on Ω, the corresponding operator family admits a measurable time-dependent generator $\mathcal{L}(r)\colon r \in [0,T]$, and the family is reconstructed via a Dyson-style time-ordered exponential

$$\mathcal{P}(t,s) \;=\; \mathcal{T}\exp\!\left(\int_s^t \mathcal{L}(r)\,dr\right).$$

Because the time-ordered exponential is a multiplicative cocycle in (t,s,r), the family generated this way *automatically satisfies the semigroup composition* $\mathcal{P}(t,s) = \mathcal{P}(t,r)\mathcal{P}(r,s)$, equivalently CK at the kernel level. For bounded, strongly measurable time-dependent generators the evolution family is constructed directly by the Dyson series; unbounded non-autonomous generators require additional evolution-family hypotheses (of Kato or Acquistapace–Terreni type), which are not needed here. The spectral analysis of the time-dependent generator $\mathcal{L}(r)$ captures the dynamical content; tools such as time-varying DMD [27, 29] are calibrated for this setting.

The setting of the present paper is structurally different: indivisible two-time families, by Proposition 3.6, admit no Dyson-type cocycle generator—no bounded, Bochner-integrable generator family $\{\mathcal{L}(r)\}$, time-dependent or otherwise, that reproduces them via a time-ordered exponential. The CK violation that defines indivisibility is a strictly stronger failure than time-dependence of the generator. Equivalently: the non-autonomous Koopman literature treats the case in which the generator depends on time but the cocycle structure survives; the present paper treats the case in which the cocycle structure itself fails. The deficit operator $\Delta(t,s,r)$ developed in Section 4 is the quantitative measure of this stronger failure.

4 Singular structure of the deficit: concentration, obstruction, and local scaling

4.1 Singular-spectral framework and covariant eigenfamilies

For each fixed pair $(t,s) \in [0,T]^2_{\leq}$, the operator $\mathcal{P}(t,s) \in \mathcal{B}(L^2(\Omega,\mu))$ has a spectrum $\sigma(\mathcal{P}(t,s)) \subset \mathbb{C}$. Throughout this section we work under the standing assumption that $\mathcal{P}(t,s)$ is compact for $t > s$—automatic for finite Ω, and ensured on continuous state spaces, for instance, by smoothing properties of the transition kernel [9]; Section 4.2 delimits the scope of this hypothesis. Compactness guarantees that every operator in play—each $\mathcal{P}(t,s)$ and each deficit operator $\Delta(t,s,r)$—admits a singular-value (Schmidt) decomposition, and it is this singular structure, not an eigen-expansion, on which the framework of this paper is built.

Remark 4.1 (Why singular-spectral rather than eigen-spectral). Compactness alone does not provide an eigen-expansion: a compact non-normal operator may fail to be diagonalizable, may

have an incomplete system of eigenvectors, and may have no eigenvalues at all—the Volterra operator on $L^2[0,1]$ is compact with spectrum $\{0\}$ and empty point spectrum. Two-time transition operators are generically non-normal, so no biorthogonal eigenbasis expansion of $\mathcal{P}(t,s)$ is asserted anywhere in this paper. Eigenstructure enters only through the covariant eigenfamilies of Definition 4.2, asserted only where they exist; full orthonormal eigen-expansions are available in the normal case (Remark 4.4).

Eigenvalue-type content, where it exists, takes the following two-time form, aligned with the non-autonomous Koopman eigenfunction notion of Maćešić, Črnjarić-Žic and Mezić [30].

Definition 4.2 (Covariant eigenfamily; exponential and indivisible modes)**.** A *covariant eigenfamily* (a *mode*) of the two-time family $\{\mathcal{P}(t,s)\}$ is a family of nonzero functions $\{\varphi^{(u)}\}_{u\in[0,T]} \subset L^2(\Omega,\mu)$ together with a multiplier $\lambda\colon [0,T]^2_{\leq} \to \mathbb{C}$ such that

$$\mathcal{P}(t,s)\,\varphi^{(s)} \;=\; \lambda(t,s)\,\varphi^{(t)} \qquad \text{for all } 0 \leq s \leq t \leq T.$$

The mode is *exponential* if $\varphi^{(u)} \equiv \varphi$ is time-independent and $\lambda(t,s) = e^{(t-s)\mu}$ for some $\mu \in \mathbb{C}$. The mode is *indivisible* if its multiplier violates the multiplicative law $\lambda(t,s) = \lambda(t,r)\,\lambda(r,s)$ at some triple $s < r < t$.

Lemma 4.3 (Multiplicative multipliers under divisibility)**.** *If $\{\mathcal{P}(t,s)\}$ is divisible, then every covariant eigenfamily has a multiplicative multiplier: $\lambda(t,s) = \lambda(t,r)\,\lambda(r,s)$ for all $s \leq r \leq t$. Consequently, the existence of an indivisible mode witnesses indivisibility of the family.*

Proof. Divisibility gives $\lambda(t,s)\,\varphi^{(t)} = \mathcal{P}(t,s)\varphi^{(s)} = \mathcal{P}(t,r)\mathcal{P}(r,s)\varphi^{(s)} = \lambda(r,s)\,\mathcal{P}(t,r)\varphi^{(r)} = \lambda(r,s)\lambda(t,r)\,\varphi^{(t)}$, and $\varphi^{(t)} \neq 0$ forces $\lambda(t,s) = \lambda(t,r)\lambda(r,s)$. □

Remark 4.4 (The normal case). When each $\mathcal{P}(t,s)$ is normal—as in the symmetric two-state example of Section 6, where every $\mathcal{P}(t,s)$ is self-adjoint—the spectral theorem supplies a full orthonormal eigen-expansion of each two-time operator, and singular values coincide with eigenvalue moduli. No such expansion is asserted in general, and none of the results below requires one.

4.2 Scope of the compactness hypothesis

The standing assumption that $\mathcal{P}(t,s)$ is compact for $t > s$ is load-bearing for the Schmidt decomposition arguments of Section 4 and Appendix A. We collect concrete classes of transition families for which compactness holds, and classes for which it fails, so that the scope of subsequent results is unambiguous.

Compactness holds. (i) *Finite configuration spaces.* For $|\Omega| = n < \infty$, $L^2(\Omega,\mu) \cong \mathbb{C}^n$ is finite-dimensional, so every bounded operator is compact. The two-state and three-state worked examples of Section 6 fall in this class. (ii) *Diffusions on bounded domains.* For an SDE $dX_t = b_t(X_t)\,dt + \sigma_t(X_t)\,dB_t$ on a bounded domain $\Omega \subset \mathbb{R}^d$ with smooth coefficients b_t, σ_t and uniformly elliptic diffusion $\sigma_t\sigma_t^* \succeq cI > 0$, the transition kernel $P(t,x \mid s,y)$ admits smooth Lebesgue densities for $t > s$ by Hörmander or Aronson estimates. The transition operator $\mathcal{P}(t,s)$ is then a Hilbert–Schmidt integral operator on $L^2(\Omega,dx)$, hence compact, with $\|\mathcal{P}(t,s)\|^2_{\mathrm{HS}} = \int\!\!\int |P(t,x \mid s,y)|^2\,dx\,dy < \infty$. (iii) *Regular Lebesgue-density kernels.* More generally, any two-time family whose kernel admits a density $P(t,x \mid s,y) \in L^2(\Omega\times\Omega, \mu\otimes\mu)$ for $t > s$ yields a Hilbert–Schmidt transition operator. (iv) *Trace-class regularity.* Stronger conditions yield trace class: if the kernel admits a nuclear decomposition $P(t,x \mid s,y) = \sum_n a_n(x)\,b_n(y)$ with $\sum_n \|a_n\|_{L^2}\,\|b_n\|_{L^2} < \infty$ — equivalently, if $\mathcal{P}(t,s)$ factors as a product of two Hilbert–Schmidt operators, as happens for divisible families with square-integrable kernels by splitting at an

intermediate time — then $\mathcal{P}(t,s)$ is trace class. Absolute integrability of the kernel alone is not a sufficient criterion.

Compactness fails. (i) *Deterministic dynamics.* For a deterministic measure-preserving flow $X_t = \Phi_{t-s}(X_s)$ on a continuous state space, the transition operator is composition with the (inverse) flow, unitary on $L^2(\Omega,\mu)$, hence not compact whenever $L^2(\Omega,\mu)$ is infinite-dimensional. The deficit operator $\Delta(t,s,r)$ is still well-defined, but its spectrum need not be discrete. (ii) *Point-mass kernels.* Kernels of the form $P(t,x \mid s,y) = \delta(x - F_{t,s}(y))$ for some measurable family $\{F_{t,s}\}$ produce operators of the same composition-with-flow form, again not compact on continuous Ω. (iii) *Smoothing-free kernels in infinite dimensions.* Translation kernels on $\mathbb{R}^d$, multiplication kernels of unit norm, and similar non-compact Markov operators on L^2 also fall outside the standing assumption.

The results of Section 4 are stated under the compactness assumption. For non-compact two-time families, the discrete singular spectrum of the Schmidt decomposition is replaced by continuous spectrum and the singular-value structure becomes a singular-value measure rather than a sequence; extending the framework to that setting is an open question we do not address here.

4.3 Deficit concentration and obstruction to divisible approximation

The principal result of this section consists of two parts: a concentration property of the deficit operator, identifying a closed subspace on which the indivisibility content is realized, and a quantitative bound showing that the action of $\mathcal{P}(t,s)$ on this subspace cannot be uniformly approximated by any divisible family.

Before stating the theorem, we flag a conceptual distinction. Triangle inequalities through any candidate divisible $\tilde{\mathcal{P}}$ yield quantitative *bounds* on operator-norm distances (and produce Corollary 4.6 below as one such bound). The concentration result of Theorem 4.5 is structurally different: it identifies a *subspace* of $L^2(\Omega,\mu)$ on which the indivisibility content of $\mathcal{P}(t,s)$ lives, in the form of input and output deficit subspaces $V_{\text{in}}, V_{\text{out}}$ defined intrinsically from the singular-value structure of the family $\{\Delta(t,s,r)\}$. The subspace identification is qualitative content—a statement about *where* the deficit operates, not how large it is—and is independent of, but compatible with, the quantitative bounds derivable by triangle inequality. Part (a) is a structural consequence of the definitions of V_{in} and V_{out} together with the Schmidt decomposition of a compact operator; the dynamical content lies in part (b), Corollary 4.6, Proposition 4.12, and Theorem 7.4. We prove both: subspace concentration (Theorem 4.5(a), requiring only boundedness and compactness) and a universal quantitative obstruction (Theorem 4.5(b) and Corollary 4.6, which carry the contraction hypothesis of Assumption 3.3 explicitly).

Theorem 4.5 (Deficit concentration and obstruction to divisible approximation)**.** *Let $\{\mathcal{P}(t,s)\}$ be the two-time operator family of an indivisible transition family on Ω, with each $\mathcal{P}(t,s)$ bounded and compact for $t > s$. Define the deficit operator*

$$\Delta(t,s,r) \;=\; \mathcal{P}(t,s) - \mathcal{P}(t,r)\,\mathcal{P}(r,s) \;\in\; \mathcal{B}(L^2(\Omega,\mu)), \qquad 0 \le s < r < t \le T,$$

which is compact as a difference of compositions of compact and bounded operators. Define the input *(right-singular) and* output *(left-singular) deficit subspaces*

$$V_{\text{in}} \;=\; \overline{\operatorname{span} \bigcup_{0\le s<r<t\le T} \operatorname{Range}\big(\Delta(t,s,r)^*\big)} \;\subset\; L^2(\Omega,\mu), \qquad V_{\text{out}} \;=\; \overline{\operatorname{span} \bigcup_{0\le s<r<t\le T} \operatorname{Range}\big(\Delta(t,s,r)\big)} \;\subset\; L^2(\Omega,$$

Then:

(a) *(Deficit concentration; boundedness and compactness only) Both* V_{in} *and* V_{out} *have positive dimension if and only if the family is indivisible. For each triple* (s, r, t) *with* $D(t, s, r) > 0$*, the Schmidt decomposition of* $\Delta(t, s, r)$ *has discrete singular spectrum with leading singular value* $D(t, s, r)$*, attained on a unit right-singular vector* $f^*(t, s, r) \in V_{\mathrm{in}}$ *with image* $\Delta(t, s, r)f^*(t, s, r) \in V_{\mathrm{out}}$*, and* $\Delta(t, s, r)$ *vanishes on* $V_{\mathrm{in}}^{\perp}$*.*

(b) *(Obstruction to divisible approximation; requires Assumption 3.3) Assume in addition that* $\{\mathcal{P}(t, s)\}$ *and the divisible comparison family below satisfy Assumption 3.3. For any divisible two-time transition family on* Ω *with operator family* $\{\tilde{\mathcal{P}}(t, s)\}$ *and any* $(t, s) \in [0, T]^2_{\leq}$*,*

$$\sup_{\substack{f \in V_{\mathrm{in}} \\ \|f\|=1}} \left\| \mathcal{P}(t,s)f \;-\; \tilde{\mathcal{P}}(t,s)f \right\|_{L^2} \;\geq\; \sup_{r \in (s,t)} \Big[D(t,s,r) \;-\; \delta(t,r;\tilde{\mathcal{P}}) \;-\; \delta(r,s;\tilde{\mathcal{P}}) \Big], \tag{4}$$

where $\delta(\tau_1, \tau_2; \tilde{\mathcal{P}}) = \left\| \mathcal{P}(\tau_1, \tau_2) - \tilde{\mathcal{P}}(\tau_1, \tau_2) \right\|_{\mathrm{op}}$*. In particular, the bound (4) cannot vanish at any triple* (t, s, r) *for which* $\delta(t, r; \tilde{\mathcal{P}}) + \delta(r, s; \tilde{\mathcal{P}}) < D(t, s, r)$*. Quantitatively: at any triple with* $D(t, s, r) > 0$*, no divisible* $\tilde{\mathcal{P}}$ *achieves uniform single-pair approximation* $\sup_{(\tau_1,\tau_2)} \delta(\tau_1, \tau_2; \tilde{\mathcal{P}}) < D(t, s, r)/3$*.*

The proof, given in Appendix A, uses the Schmidt decomposition of the compact deficit operator to extract the leading singular vector cleanly, then bounds the operator-norm distance to any divisible $\tilde{\mathcal{P}}$ via a triangle decomposition through the deficit operator.

The complementary quantitative obstruction is the following corollary. The inequality is a triangle inequality; its content is the constant, which Proposition 4.7 shows cannot be improved. Where Theorem 4.5(b) gives an V_{in}-restricted action-distance bound, the corollary gives a global operator-norm bound, free of the V_{in} restriction and symmetric in the three time pairs of the triple: under the standing contraction assumption, every divisible family is operator-norm-distant from $\mathcal{P}$ by at least $D(t, s, r)/3$ at the worst single time pair, for any triple with positive deficit.

Corollary 4.6 (Universal three-pair lower bound on divisible approximation)**.** *Let* $\{\mathcal{P}(t, s)\}$ *be as in Theorem 4.5, and assume both it and the divisible comparison family satisfy Assumption 3.3. Then for any divisible two-time transition family on* Ω *with operator family* $\{\tilde{\mathcal{P}}(t, s)\}$ *and any triple* (t, s, r) *with* $0 \leq s < r < t \leq T$*,*

$$D(t, s, r) \;\leq\; \delta(t, s; \tilde{\mathcal{P}}) + \delta(t, r; \tilde{\mathcal{P}}) + \delta(r, s; \tilde{\mathcal{P}}), \tag{5}$$

where $\delta(\tau_1, \tau_2; \tilde{\mathcal{P}}) = \left\| \mathcal{P}(\tau_1, \tau_2) - \tilde{\mathcal{P}}(\tau_1, \tau_2) \right\|_{\mathrm{op}}$*. Equivalently, at every triple with* $D(t, s, r) > 0$*, every divisible* $\tilde{\mathcal{P}}$ *satisfies*

$$\sup_{(\tau_1,\tau_2) \in [0,T]^2_{\leq}} \delta(\tau_1, \tau_2; \tilde{\mathcal{P}}) \;\geq\; \frac{D(t, s, r)}{3}.$$

The bound (5) complements the V_{in}-restricted statement of Theorem 4.5(b) along three axes simultaneously: it operates on operator-norm distance rather than V_{in}-restricted action distance, it is symmetric in the three time pairs (t, s), (t, r), (r, s) rather than singling out the time pair (t, s), and it holds for every choice of intermediate time r—so the strongest form of the bound takes the supremum over the intermediate time, $\sup_{(\tau_1,\tau_2)} \delta(\tau_1, \tau_2; \tilde{\mathcal{P}}) \geq \sup_{r \in (s,t)} D(t, s, r)/3$. Together with Theorem 4.5, the pair (subspace concentration; quantitative obstruction) is the load-bearing content of Section 4: where the indivisibility content lives, and how far from any divisible approximation $\mathcal{P}$ must sit at the worst time pair. The obstruction concerns uniform

operator-norm approximation of the two-time family itself; it does not preclude a divisible model matching selected observables, marginals, or low-order statistics of the same data. The proof is the triangle-inequality argument given as Step 5 of Appendix A.

Remark (General-norm version). Without Assumption 3.3, the triangle decomposition of Appendix A yields, with explicit operator-norm factors,

$$D(t,s,r) \;\le\; \delta(t,s;\tilde{\mathcal{P}}) \;+\; \delta(t,r;\tilde{\mathcal{P}})\,\left\|\tilde{\mathcal{P}}(r,s)\right\|_{\mathrm{op}} \;+\; \|\mathcal{P}(t,r)\|_{\mathrm{op}}\,\delta(r,s;\tilde{\mathcal{P}}),$$

and correspondingly in (4); the clean constants above are exactly the contraction case.

The constant $1/3$ in Corollary 4.6 cannot be improved. On $\Omega = \{1,2\}$ write $v := (|1\rangle - |2\rangle)/\sqrt{2}$, so that $vv^* = \frac{1}{2}(I - X)$ with X the transposition, and set

$$m(D) \;:=\; \frac{3-\sqrt{9-4D}}{2}, \qquad 0 < D < 2, \tag{6}$$

the smaller root of $\varepsilon^2 - 3\varepsilon + D = 0$. Then $m(D) = D/3 + m(D)^2/3 > D/3$ and $m(D)/(D/3) \to 1$ as $D \to 0$.

Proposition 4.7 (The constant $1/3$ is sharp)**.** *Let $\Omega = \{1,2\}$ with counting measure, $0 < D < 2$, $T > 0$, and $\psi(\tau) := \max\{0,\, 2\tau/T - 1\}$ for $\tau \in [0,T]$. The family*

$$\Gamma(t \leftarrow s) \;=\; I \;-\; D\,\psi(t-s)\,vv^*, \qquad 0 \le s \le t \le T,$$

is jointly continuous and doubly stochastic, satisfies (R1) and Assumption 3.3, and has $D(T,0,T/2) = D$. Moreover:

(a) every divisible column-stochastic comparison family $\{\tilde{\mathcal{P}}(t,s)\}$ on Ω satisfies

$$\sup_{(\tau_1,\tau_2)} \delta(\tau_1,\tau_2;\tilde{\mathcal{P}}) \;\ge\; m(D);$$

(b) the divisible family $\tilde{\mathcal{P}}(t,s) = I - (1 - e^{-\kappa(t-s)})\,vv^*$ with $e^{-\kappa T/2} = 1 - m(D)$ satisfies Assumption 3.3 and

$$\sup_{(\tau_1,\tau_2)} \delta(\tau_1,\tau_2;\tilde{\mathcal{P}}) \;=\; m(D).$$

So the infimum over divisible comparison families of $\sup\delta$ at a triple with deficit D equals $m(D)$ for this family, and since $m(D)/D \to 1/3$, no constant $c > 1/3$ can replace $1/3$ in Corollary 4.6. For every $D > 0$ the bound is strict, $m(D) > D/3$, and the extremal family has deterministic intermediate kernels, $\Gamma(T \leftarrow T/2) = \Gamma(T/2 \leftarrow 0) = I$.

Proof. The matrix $\Gamma(t \leftarrow s)$ is symmetric with eigenvalue 1 on $(|1\rangle + |2\rangle)/\sqrt{2}$ and eigenvalue $1 - D\psi(t-s) \in [-1,1]$ on v; its entries are $1 - D\psi/2$ and $D\psi/2$, which lie in $[0,1]$ for $D \le 2$. So it is doubly stochastic with operator norm 1, jointly continuous because ψ is, and $\Gamma(s \leftarrow s) = I$ because $\psi(0) = 0$. At the triple $(T,0,T/2)$, $\Gamma(T \leftarrow T/2) = \Gamma(T/2 \leftarrow 0) = I$ and $\Gamma(T \leftarrow 0) = I - Dvv^*$, so $\Delta(T,0,T/2) = -D\,vv^*$ and $D(T,0,T/2) = D$.

(a) Let $A := \tilde{\mathcal{P}}(T,T/2)$ and $B := \tilde{\mathcal{P}}(T/2,0)$; by divisibility $\tilde{\mathcal{P}}(T,0) = AB$. Put $\varepsilon := \max\{\|I - A\|_{\mathrm{op}}\,,\, \|I - B\|_{\mathrm{op}}\,,\, \|\Gamma(T \leftarrow 0) - AB\|_{\mathrm{op}}\} \le \sup\delta$. If $\varepsilon \ge 1$ there is nothing to prove, since $m(D) < 1$. A column-stochastic 2×2 matrix $A = \left(\begin{smallmatrix} 1-p_A & q_A \\ p_A & 1-q_A \end{smallmatrix}\right)$ with $p_A, q_A \in [0,1]$ has the form $A = I + vw_A^*$ with $w_A = \sqrt{2}\,(-p_A,\, q_A)^\top$; hence $\|I - A\|_{\mathrm{op}} = \|w_A\|$ and $\langle w_A, v\rangle = -(p_A +$

$q_A) =: -\sigma_A$ with $0 \le \sigma_A \le \|w_A\| \le \varepsilon$. Similarly $B = I + v w_B^*$ with $\langle w_B, v\rangle = -\sigma_B$, $0 \le \sigma_B \le \varepsilon$. Then $AB = I + v\big(w_A + (1-\sigma_A)\, w_B\big)^*$ and $\Gamma(T \leftarrow 0) - AB = -v\big(Dv + w_A + (1-\sigma_A)\, w_B\big)^*$, so

$$\begin{aligned} \|\Gamma(T \leftarrow 0) - AB\|_{\mathrm{op}} &= \|Dv + w_A + (1-\sigma_A)\, w_B\| \\ &\ge \big\langle Dv + w_A + (1-\sigma_A)\, w_B,\ v\big\rangle \;=\; D - \sigma_A - (1-\sigma_A)\,\sigma_B. \end{aligned}$$

On $[0,1]^2$ the right-hand side is nonincreasing in σ_A and in σ_B, so for $\sigma_A, \sigma_B \le \varepsilon < 1$ it is at least $D - 2\varepsilon + \varepsilon^2$. Thus $\varepsilon \ge D - 2\varepsilon + \varepsilon^2$, that is, $\varepsilon^2 - 3\varepsilon + D \le 0$, which forces $\varepsilon \ge m(D)$.

(b) Write $a(\tau) := 1 - D\psi(\tau)$ and $\tilde{a}(\tau) := e^{-\kappa\tau}$, so that $\Gamma(t \leftarrow s) - \tilde{\mathcal{P}}(t,s) = -\big(a(t-s) - \tilde{a}(t-s)\big)\, vv^*$ and $\delta(t,s;\tilde{\mathcal{P}}) = |a(t-s) - \tilde{a}(t-s)|$; the family $\tilde{\mathcal{P}}$ is a doubly stochastic semigroup in elapsed time with operator norm 1. Set $m := m(D)$. For $\tau \le T/2$, $a(\tau) = 1$ and $0 \le 1 - \tilde{a}(\tau) \le 1 - e^{-\kappa T/2} = m$. On $[T/2, T]$ the function $f := a - \tilde{a}$ is concave, since a is affine and $\tilde{a}$ is convex, with $f(T/2) = m$ and $f(T) = 1 - D - (1-m)^2 = -(D - 2m + m^2) = -m$, and

$$f'(T/2) \;=\; -\frac{2D}{T} + \kappa\,(1-m) \;\le\; \frac{2}{T}\,(m - D) \;<\; 0,$$

using $\kappa = -(2/T)\log(1-m) \le (2/T)\, m/(1-m)$ and $D - m = m(2-m) > 0$. A concave function with negative initial slope is decreasing, so f decreases from m to $-m$ on $[T/2, T]$ and $|f| \le m$ there. Hence $\sup \delta = m$, attained at $t - s = T/2$ and at $t - s = T$. □

4.4 Gauge-relative cross-term decomposition of the deficit

When every matrix of a finite-dimensional family is unistochastic, the matrix elements of $\Delta(t,s,r)$ admit an exact algebraic decomposition through any selected gauge (Proposition 2.11(d)). The decomposition is an identity for every selection; what is intrinsic to the family is the *sum* of the two terms, which vanishes exactly under CK. The individual terms are properties of the selected lifts, not of the family—a point Remark 4.9 makes concrete.

Theorem 4.8 (Gauge-relative cross-term decomposition)**.** *Let $\{\Gamma(t \leftarrow s)\}$ be a two-time family of unistochastic matrices on the finite configuration space $\Omega = \{1, \dots, n\}$, and select any gauge: unitaries $U(t,s)$, $U(t,r)$, $U(r,s)$ with $\Gamma_{ij}(\cdot) = |U_{ij}(\cdot)|^2$ at each of the three time pairs of a triple $s < r < t$. Define the composite unitary $V(t,s;r) := U(t,r)\, U(r,s)$. The matrix elements of the deficit operator $\Delta(t,s,r) = \mathcal{P}(t,s) - \mathcal{P}(t,r)\mathcal{P}(r,s)$ in the configuration-space basis decompose as*

$$\Delta(t,s,r)_{ij} \;=\; \mathfrak{C}(t,s,r)_{ij} \;+\; \mathfrak{D}(t,s,r)_{ij}, \tag{7}$$

where the cross-term (interference) contribution

$$\mathfrak{C}(t,s,r)_{ij} \;=\; \sum_{z \ne w} U_{iz}(t,r)\, U_{zj}(r,s)\, \overline{U_{iw}(t,r)}\, \overline{U_{wj}(r,s)} \tag{8}$$

is the sum over off-diagonal pairs (z,w) in the squared-modulus expansion of the composite unitary $V(t,s;r)$, and the non-composition contribution

$$\mathfrak{D}(t,s,r)_{ij} \;=\; \big|U_{ij}(t,s)\big|^2 \;-\; \big|V_{ij}(t,s;r)\big|^2 \tag{9}$$

is the gap between the squared modulus of the selected unitary at the time pair (t,s) and the squared modulus of the composite unitary. Moreover:

(a) The identity (7) holds for every gauge selection.

(b) CK holds at the triple (t,s,r) if and only if $\mathfrak{C}(t,s,r) + \mathfrak{D}(t,s,r) = 0$.

(c) The individual vanishing of $\mathfrak{C}$ or of $\mathfrak{D}$ is gauge-dependent: for a divisible triple there exist gauges in which both terms vanish and gauges in which both are nonzero with exact cancellation (Remark 4.9), and an exact unitary cocycle ($\mathfrak{D} = 0$) can coexist with maximal CK violation carried entirely by $\mathfrak{C}$ (Remark 2.12).

The proof, given in Appendix B, is a direct algebraic identity from the squared-modulus expansion of $V(t, s; r)$; part (b) is immediate from (a) since CK at the triple is the statement $\Delta(t, s, r) = 0$.

Remark 4.9 (Gauge dependence of the individual terms). Let F denote the fair-coin matrix (all entries $\frac{1}{2}$), X the swap, H the Hadamard unitary, $R(\theta)$ the rotation, and $K := \frac{1}{\sqrt{2}}\left(\begin{smallmatrix} 1 & i \\ i & 1 \end{smallmatrix}\right)$. Take the divisible triple $\Gamma(t, s) = \Gamma(t, r) = \Gamma(r, s) = F$, for which CK holds exactly ($F^2 = F$). The gauge $(U(t, s), U(t, r), U(r, s)) = (H, R(45°), R(45°))$ gives $V = R(90°)$, hence $\mathfrak{D} = F - X \neq 0$ and $\mathfrak{C} = X - F \neq 0$, with exact cancellation $\mathfrak{C} + \mathfrak{D} = 0$; the gauge (KH, K, H) gives $|V|^2 = |KH|^2 = F$, hence $\mathfrak{D} = 0$ and $\mathfrak{C} = 0$ separately. The same family, the same triple, and the same deficit ($\Delta = 0$) thus support selections in which the two terms individually vanish and selections in which they are individually large. Any interpretation attached to $\mathfrak{C}$ or $\mathfrak{D}$ alone is therefore an interpretation of the selected lifts.

Remark 4.10 (Optional interpretive content). On the unitary side of the dilational correspondence, the cross-term contribution $\mathfrak{C}(t, s, r)$ collects the off-diagonal cross-products (interference terms) of the composite unitary $V(t, s; r) = U(t, r)U(r, s)$, and the non-composition contribution $\mathfrak{D}(t, s, r)$ measures the squared-modulus gap between $U(t, s)$ and $V(t, s; r)$. Readers who view off-diagonal terms at the intermediate time as coherences may read $\mathfrak{C}$ accordingly; by Theorem 4.8(c) any such reading is gauge-relative, and the operator-theoretic content of the decomposition does not depend on it.

4.5 Local-diagonal scaling and the indivisibility tensor

Theorem 4.5 and Corollary 4.6 are global statements about the deficit operator $\Delta(t, s, r)$ on triples (s, r, t) in the simplex $\{0 \le s < r < t \le T\}$. We now establish the asymptotic behavior of Δ near the diagonal $s = r = t$, which is governed by a local infinitesimal quantity that we identify as the *indivisibility tensor*. The result connects the global two-time framework with standard generator-based structure: even though no Hille–Yosida generator exists for an indivisible two-time family (Proposition 3.6), an explicit local obstruction is computable from the second-order Taylor coefficients of $\Gamma(t, s)$ at the diagonal.

Throughout this subsection we work on a finite configuration space $\Omega = \{1, \dots, n\}$ and assume that the column-stochastic family $\{\Gamma(t, s)\}_{s \le t}$ is *jointly* C^2 in the two time variables on the compact simplex $[0, T]^2_{\le}$, in the matrix-entry sense: each entry $(t, s) \mapsto \Gamma_{ij}(t, s)$ has continuous partial derivatives up to second order, taken one-sidedly at the boundary. Joint C^2 regularity on the compact simplex is what licenses the uniformity of the Taylor remainders below: the second partial derivatives are uniformly continuous, so the $o((t - s)^2)$ remainder in Proposition 4.12(a) is uniform in the position $\theta = (r - s)/(t - s) \in [0, 1]$ of the intermediate time and in s. Define the time-zero generator and second-order coefficient

$$L(s) \;:=\; \partial_t \Gamma(t, s)\big|_{t=s}, \qquad C(s) \;:=\; \partial_t^2 \Gamma(t, s)\big|_{t=s},$$

both viewed as elements of $\mathcal{B}(L^2(\Omega, \mu))$. The boundary condition $\Gamma(s, s) = I$ gives the second-order Taylor expansion $\Gamma(t, s) = I + (t - s)L(s) + \frac{1}{2}(t - s)^2 C(s) + o((t - s)^2)$.

Definition 4.11 (Indivisibility tensor)**.** The *indivisibility tensor* of a jointly C^2 column-stochastic family $\{\Gamma(t, s)\}$ at time $s \in [0, T]$ is the bounded operator

$$\beta(s) \;:=\; C(s) \;-\; L(s)^2 \;-\; L'(s) \;\in\; \mathcal{B}(L^2(\Omega, \mu)),$$

where $L'(s) := \partial_s L(s)$.

The expression $L(s)^2 + L'(s)$ is the second-order coefficient in the Taylor expansion at $t = s$ of any time-ordered exponential $T\exp\big(\int_s^t L(u)\,du\big)$: for a differentiable evolution family with $\partial_t U(t,s) = A(t)\,U(t,s)$ and $U(s,s) = I$, one has $\partial_t^2 U(t,s)|_{t=s} = A'(s) + A(s)^2$, the second-order term of the Dyson series and of the Magnus expansion [55, 56]. So $\beta(s)$ measures the local obstruction to the second-order Taylor coefficient of Γ matching that of a cocycle generated by L, and $\beta \equiv 0$ is the second-order compatibility condition satisfied by every jointly C^2 cocycle family. The identity is standard. This section adds the residual $C - L^2 - L'$ as a named invariant of an arbitrary jointly C^2 column-stochastic family, the factor $(t-r)(r-s)$ in the scaling law of Proposition 4.12, the reading of the singular structure of β as the locally most deficit-active direction, and the place of β in the global deficit theory of Theorem 4.5.

Proposition 4.12 (Local-diagonal scaling of the deficit)**.** *Let $\{\Gamma(t,s)\}_{s\le t}$ be a jointly C^2 column-stochastic family on a finite configuration space Ω with indivisibility tensor $\beta(s)$ (Definition 4.11). The following hold.*

(a) (Universal scaling) For any $0 \le s \le r \le t \le T$ with $\tau := t - s$,

$$\Delta(t,s,r) \;=\; (t-r)(r-s)\,\beta(s) \;+\; o(\tau^2) \quad \textit{as} \quad \tau \to 0,$$

with remainder uniform in the position $\theta := (r-s)/(t-s) \in [0,1]$ of the intermediate time and in s. Consequently

$$D(t,s,r) \;=\; (t-r)(r-s)\;\|\beta(s)\|_{\mathrm{op}} \;+\; o(\tau^2).$$

(b) (Necessary condition for divisibility) If $\{\Gamma(t,s)\}$ is divisible, then $\beta(s) \equiv 0$ on $[0,T]$.

(c) (Sufficient condition for indivisibility) If $\beta(s_0) \neq 0$ at some $s_0 \in [0,T]$, then $\{\Gamma(t,s)\}$ is indivisible: there exist $\delta > 0$ and triples (t,s,r) with $s \in (s_0 - \delta, s_0 + \delta)$ and $t - s < \delta$ on which $D(t,s,r) > 0$.

Proof. For (a), set $r = s + \theta\tau$ with $\theta \in [0,1]$ so that $t - r = (1-\theta)\tau$ and $r - s = \theta\tau$. The Taylor expansions

$$\begin{aligned}
\Gamma(t,s) &= I + \tau L(s) + \tfrac{\tau^2}{2} C(s) + o(\tau^2),\\
\Gamma(t,r) &= I + (1-\theta)\tau\, L(s+\theta\tau) + \tfrac{(1-\theta)^2\tau^2}{2} C(s+\theta\tau) + o(\tau^2),\\
\Gamma(r,s) &= I + \theta\tau\, L(s) + \tfrac{\theta^2\tau^2}{2} C(s) + o(\tau^2),
\end{aligned}$$

combined with $L(s+\theta\tau) = L(s) + \theta\tau L'(s) + o(\tau)$ and $C(s+\theta\tau) = C(s) + o(1)$, give

$$\Gamma(t,r) = I + (1-\theta)\tau L(s) + (1-\theta)\theta\tau^2 L'(s) + \tfrac{(1-\theta)^2\tau^2}{2} C(s) + o(\tau^2).$$

Multiplying $\Gamma(t,r)\Gamma(r,s)$ and collecting terms through $O(\tau^2)$:

$$\begin{aligned}
\Gamma(t,r)\Gamma(r,s) &= I + \tau L(s) + \theta(1-\theta)\tau^2 L'(s)\\
&\quad + \tfrac{(1-\theta)^2+\theta^2}{2}\tau^2 C(s) + \theta(1-\theta)\tau^2 L(s)^2 + o(\tau^2)\\
&= I + \tau L(s) + \tfrac{\tau^2}{2} C(s) + \theta(1-\theta)\tau^2\big[L'(s) + L(s)^2 - C(s)\big] + o(\tau^2),
\end{aligned}$$

where the second equality uses $(1-\theta)^2 + \theta^2 = 1 - 2\theta(1-\theta)$. Subtracting from $\Gamma(t,s)$:

$$\Delta(t,s,r) = \theta(1-\theta)\tau^2\big[C(s) - L(s)^2 - L'(s)\big] + o(\tau^2) = (t-r)(r-s)\,\beta(s) + o(\tau^2),$$

since $\theta(1-\theta)\tau^2 = (r-s)(t-r)$. The remainders above are uniform in θ and s: by joint C^2 regularity on the compact simplex, the second partial derivatives of Γ are uniformly continuous, so the mean-value forms of the Taylor remainders are $o(\tau^2)$ uniformly over $\theta \in [0,1]$ and $s \in [0,T]$. Taking operator norms gives the second statement of (a).

For (b), divisibility gives $\Delta \equiv 0$, so the leading term $\theta(1-\theta)\tau^2\beta(s)$ in (a) vanishes for all $\theta \in (0,1)$ and small τ, forcing $\beta(s) = 0$.

For (c), if $\beta(s_0) \neq 0$, continuity of β (which follows from the C^2 hypothesis on Γ) gives a neighborhood $(s_0 - \delta, s_0 + \delta)$ on which $\|\beta(s)\|_{\mathrm{op}} > c$ for some $c > 0$. By (a), $D(t,s,r) > c\,(t-r)(r-s) - o(\tau^2) > 0$ for τ small and θ bounded away from 0 and 1. □

Remark 4.13. Proposition 4.12(b) gives a necessary condition for divisibility, not a sufficient one. Higher-order indivisibility tensors $\beta_3, \beta_4, \ldots$ can be defined recursively from higher Taylor coefficients of Γ and the corresponding coefficients of any candidate Hille–Yosida cocycle; divisibility for a real-analytic family forces $\beta_k \equiv 0$ for all $k \geq 2$. The leading tensor $\beta = \beta_2$ is sufficient for the qualitative dichotomy (indivisible vs. divisible to second order at the diagonal) and is the load-bearing object for the local scaling formula in (a). We do not pursue the higher-order tensors here.

Remark 4.14. The indivisibility tensor $\beta(s)$ provides a quantitative bridge between the global two-time deficit framework of Theorem 4.5 and the standard generator-based machinery. Where Theorem 4.5 produces a closed input/output subspace pair $(V_{\mathrm{in}}, V_{\mathrm{out}})$ on which indivisibility content is concentrated globally, $\beta(s)$ is a pointwise operator on $L^2(\Omega,\mu)$ measuring the local rate at which factorization fails near the diagonal. Like the two-time operators themselves, $\beta(s)$ is in general neither self-adjoint nor normal, so its local content is carried by its singular structure rather than by eigenstructure: the largest singular value $\|\beta(s)\|_{\mathrm{op}}$ is the leading-order coefficient of the deficit as $(t-r), (r-s) \to 0$, and the corresponding right singular vector identifies the input direction in $L^2(\Omega,\mu)$ along which local indivisibility content is most strongly realized at time s, with its image under $\beta(s)$ the corresponding output direction. For divisible families with Hille–Yosida generator $L(s)$, $\beta(s)$ vanishes identically; for indivisible families, $\beta(s)$ furnishes a configuration-space-relative pointwise indicator of indivisibility computable from the second-order behavior of Γ at the diagonal alone. The pair $\big(\|\beta(s)\|_{\mathrm{op}},\ \text{leading singular vectors of } \beta(s)\big)$ is thus the local counterpart of the global pair $\big(D(t,s,r),\ (V_{\mathrm{in}}, V_{\mathrm{out}})\big)$.

5 Hilbert-space lift of the empirical kernel via Stinespring dilation

We now construct, for an indivisible transition family whose two-time transition statistics are known empirically over $[0,T]$, a unitary lift on a dilation Hilbert space that reproduces the empirical kernel via partial trace, and which admits a formal Hamiltonian generator under explicit regularity hypotheses. The construction operates directly on the empirical kernel $\Gamma(t \leftarrow s)$ without invoking a generator on the configuration-space Hilbert space, and produces a unitary family $\{W(t,s)\}$ on a dilated Hilbert space via a Stinespring–Naimark construction [17, 18], the dilation parametrised in (t,s) via an explicit Kraus representation. The principal advantage of routing through dilation rather than through a direct unitary lift on the configuration-space Hilbert space is that no detailed-balance hypothesis on the kernel is required: the construction succeeds for any column-stochastic family with the regularity stated below, including the irreversible families typical of indivisible dynamics.

The level of rigour in this section is calibrated to make explicit which statements are fully proved and which are sketched modulo standard but technical functional-analytic completions. The existence of a strongly continuous Stinespring lift is established under continuity of Γ; the

existence of a measurable formal Hamiltonian generator requires an additional strict-positivity and absolute-continuity hypothesis; for divisible families, a cocycle dilation is established on finite time grids, with a fresh environment factor per step and environment dimension growing with the partition, and no fixed-dimension continuum cocycle is claimed—that existence question is posed as open; joint (t, s)-continuity of the gauge equivalence between distinct dilations, and the extension of the construction to separable infinite-dimensional configuration spaces, require additional regularity not pursued here. We mark these distinctions explicitly throughout, and we use "Hilbert-space lift" or "Stinespring lift" rather than "Schrödinger evolution" in the theorem statements to keep operator-theoretic content and interpretive content separated.

5.1 Stinespring dilation of a column-stochastic family

Let $\Omega = \{1, \dots, n\}$ be a finite configuration space, $\mathcal{H} = \mathbb{C}^n$ the configuration-space Hilbert space with basis $\{|j\rangle\}_{j=1}^n$, and $\{\Gamma(t \leftarrow s)\}_{s \le t}$ a two-time family of column-stochastic matrices encoding the empirical transitions of a process. The finite-dimensional setting is taken for clarity; the construction extends to separable Hilbert spaces under standard nuclear-operator hypotheses [18, 19].

Definition 5.1 (Diagonal-retention lift of a column-stochastic family)**.** For each $(t, s) \in [0, T]^2_{\le}$, define the Kraus operators

$$K_0(t,s) \;=\; \sum_{j=1}^{n} \sqrt{\Gamma_{jj}(t \leftarrow s)}\; |j\rangle\langle j|, \qquad K_{(i,j)}(t,s) \;=\; \sqrt{\Gamma_{ij}(t \leftarrow s)}\; |i\rangle\langle j| \quad (i \neq j),$$

indexing the diagonal transitions collectively through K_0 and the off-diagonal transitions individually. The *diagonal-retention lift* of $\Gamma(t \leftarrow s)$ is the linear map $\Phi(t, s)\colon \mathcal{B}(\mathcal{H}) \to \mathcal{B}(\mathcal{H})$ defined by

$$\Phi(t,s)(\rho) \;=\; K_0(t,s)\,\rho\,K_0(t,s)^* \;+\; \sum_{i \neq j} K_{(i,j)}(t,s)\,\rho\,K_{(i,j)}(t,s)^*.$$

The map $\Phi(t, s)$ is completely positive by its Kraus form and trace-preserving because $K_0^* K_0 + \sum_{i \neq j} K_{(i,j)}^* K_{(i,j)} = \sum_j \big(\sum_i \Gamma_{ij}\big)|j\rangle\langle j| = I$ by column-stochasticity [19]. On populations it acts exactly as Γ: $\langle i|\Phi(t, s)(\rho)|i\rangle = \sum_j \Gamma_{ij}(t \leftarrow s)\,\langle j|\rho|j\rangle$, so it reproduces the transition statistics; on coherences it acts by damping, $\Phi(t, s)(|j\rangle\langle j'|) = \sqrt{\Gamma_{jj}\Gamma_{j'j'}}\,|j\rangle\langle j'|$ for $j \neq j'$. The decisive property is the boundary behavior: $\Gamma(s \leftarrow s) = I$ gives $K_0(s, s) = I$ and $K_{(i,j)}(s, s) = 0$, so $\Phi(s, s) = \mathrm{id}$ is the identity channel. (The fully dephasing alternative $\rho \mapsto \sum_{i,j} \Gamma_{ij}\langle j|\rho|j\rangle\,|i\rangle\langle i|$ is also CPTP and also reproduces the transition statistics, but at $t = s$ it is the completely dephasing channel rather than the identity, so no unitary dilation family of it can satisfy $W(s, s) = I$; the diagonal-retention indexing is what makes a continuous unitary lift with the correct boundary possible.) All results below depend on Γ only through the transition-statistic reproduction (11).

Definition 5.2 (Hilbert-space lift via Stinespring dilation)**.** A *Hilbert-space lift* (or *Stinespring lift*) of $\{\Gamma(t \leftarrow s)\}$ is a triple $(\mathcal{H}', V, \{W(t, s)\}_{s \le t})$ consisting of an auxiliary Hilbert space $\mathcal{H}' = \mathcal{H} \otimes \mathcal{K}$ with $\dim \mathcal{K} \le n^2$, an isometry $V\colon \mathcal{H} \to \mathcal{H}'$ of the form $V|j\rangle = |j\rangle \otimes |0\rangle$ for a fixed reference vector $|0\rangle \in \mathcal{K}$, and a measurable family $\{W(t, s)\}_{s \le t}$ of unitary operators on $\mathcal{H}'$ with $W(s, s) = I_{\mathcal{H}'}$, such that the full channel is reproduced: for every density operator ρ on $\mathcal{H}$ and all $s \le t$,

$$\mathrm{Tr}_{\mathcal{K}}\big[\, W(t,s)\,(\rho \otimes |0\rangle\langle 0|)\,W(t,s)^* \,\big] \;=\; \Phi(t,s)(\rho). \tag{10}$$

In particular, taking $\rho = |j\rangle\langle j|$ and reading off diagonal entries yields the transition-statistic

reproduction: for all i, j and $s \le t$,

$$\Gamma_{ij}(t \leftarrow s) \;=\; \sum_{k=1}^{\dim \mathcal{K}} \left| \langle i| \otimes \langle k|\, W(t,s)\, |j\rangle \otimes |0\rangle \right|^2, \tag{11}$$

where the sum is over an orthonormal basis $\{|k\rangle\}_{k=1}^{\dim \mathcal{K}}$ of $\mathcal{K}$. We say the lift is *strongly continuous* if $(t,s) \mapsto W(t,s)$ is strongly continuous on $[0,T]^2_{\le}$.

Remark 5.3 (Why the lift is defined at the channel level). Requiring only the squared-modulus relation (11) on basis inputs under-determines the lift: two unitaries reproducing the same transition statistics can dilate *different* channels. For $\Gamma = I$, take $W_1 = I_{\mathcal{H}'}$ and W_2 defined by $W_2(|j\rangle \otimes |0\rangle) = |j\rangle \otimes |e_j\rangle$ for an orthonormal set $\{|e_j\rangle\} \subset \mathcal{K}$ (extended unitarily). Both satisfy (11) with $\Gamma = I$, but W_1 dilates the identity channel while W_2 dilates the completely dephasing channel; the two induced channels differ on coherent inputs, so no Stinespring-type equivalence can relate the two lifts, and a uniqueness assertion tied to (11) alone would be false. Definition 5.2 therefore requires full-channel reproduction (10)—a strengthening that the lifts constructed in Theorem 5.4 satisfy—and the uniqueness assertion of Theorem 5.4(iv) is stated at the level of minimal dilations.

The theorem assembles standard machinery around one new indexing device, and the proof marks which is which. Imported: Stinespring's dilation theorem and the minimality classification [17, 18]; the continuity of Stinespring isometries under continuity of the channel [39], with the caveat [40] that strong convergence of isometries does not lift to unitary completions; Kato's parallel transport [50] for the absolutely continuous completion of part (ii); and the triviality of fibre bundles over a contractible base [22] for the jointly continuous completion of part (i). New: the diagonal-retention indexing of Definition 5.1, which routes every diagonal transition through a single environment vector, so that the channel is the identity on the diagonal, $V(s,s)|j\rangle = |j\rangle \otimes |0\rangle$ exactly, and a continuous completion with $W(s,s) = I$ exists; parts (i) and (ii) rest on this boundary condition. Remark 5.6 grades the parts by the level of rigour established.

Theorem 5.4 (Stinespring lift of an empirical kernel; formal generator under regularity)**.** *Let $\{\Gamma(t \leftarrow s)\}_{s \le t}$ be a two-time family of column-stochastic matrices on a finite configuration space $\Omega = \{1, \dots, n\}$, satisfying:*

(R1) (Boundary) $\Gamma(s \leftarrow s) = I$ for all $s \in [0,T]$.

(R2) (Continuity) The map $(t,s) \mapsto \Gamma(t \leftarrow s)$ is continuous in operator norm on $[0,T]^2_{\le}$.

The following hold:

(i) (Existence of a strongly continuous Stinespring lift) There exists a Stinespring lift $(\mathcal{H}', V, \{W(t,s)\})$ in the sense of Definition 5.2, with $\dim \mathcal{K} = n^2 - n + 1 \le n^2$, such that the full-channel relation (10) holds and $(t,s) \mapsto W(t,s)$ is strongly continuous on $[0,T]^2_{\le}$ with $W(s,s) = I_{\mathcal{H}'}$.

(ii) (Formal Hamiltonian generator under additional regularity) Suppose, in addition to (R1)–(R2), that

(R3′) (Strict positivity and absolute continuity) For each i, j either $\Gamma_{ij}(t \leftarrow s) \equiv 0$ on $[0,T]^2_{\le}$ or $\Gamma_{ij}(t \leftarrow s) > 0$ on the open set $\{s < t\}$, and for each fixed s the map $t \mapsto \sqrt{\Gamma_{ij}(t \leftarrow s)}$ is absolutely continuous on $[s,T]$. (The hypothesis is placed on the square roots $\sqrt{\Gamma_{ij}}$ deliberately, not on the entries Γ_{ij} themselves: absolute continuity of Γ_{ij} alone would not suffice, since the derivative of $\sqrt{\Gamma_{ij}}$ can fail to be integrable where $\Gamma_{ij} \to 0$ at the diagonal.)

Then for each fixed s there is a Stinespring lift $t \mapsto W(t,s)$, absolutely continuous in t on $[s,T]$ with $W(s,s) = I_{\mathcal{H}'}$ (in particular continuous in t; this absolutely continuous completion and the jointly continuous completion of (i) are separate constructions and are not asserted to coincide), and there exists a strongly measurable, almost-everywhere defined Hermitian-operator-valued function $\widehat{H}_s$ on $[s,T]$, integrable in operator norm — $\widehat{H}_s \in L^1([s,T]; \mathcal{B}(\mathcal{H}')_{\mathrm{Herm}})$, in general not *essentially bounded (Remark 5.8) — such that, for almost every $t \geq s$,*

$$i\,\partial_t W(t,s) \;=\; \widehat{H}_s(t)\, W(t,s),$$

with $W(s,s) = I_{\mathcal{H}'}$, in the sense of strong differentiability of $W(t,s)$ as an absolutely continuous unitary path. We refer to $\widehat{H}_s(t)$ as the formal Hamiltonian generator *of the lift.*

(iii) (Grid cocycle dilation under divisibility) Suppose $\{\Gamma(t \leftarrow s)\}$ is divisible. Then for every finite grid $\pi\colon s = t_0 < t_1 < \cdots < t_K = t$ in $[0,T]$ there exist environment factors $\mathcal{K}_1, \ldots, \mathcal{K}_K$ with $\dim \mathcal{K}_k \leq n^2$, reference vectors $|0_k\rangle \in \mathcal{K}_k$, and unitaries W_k on $\mathcal{H} \otimes \mathcal{K}_k$ — each obtained by applying the pairwise construction of (i) to the increment (t_k, t_{k-1}) — such that the composite unitaries on $\mathcal{H} \otimes \mathcal{K}_1 \otimes \cdots \otimes \mathcal{K}_K$,

$$W_\pi(t_m, t_l) \;:=\; \widetilde{W}_m \cdots \widetilde{W}_{l+1}, \qquad 0 \leq l \leq m \leq K$$

(where $\widetilde{W}_k$ acts as W_k on $\mathcal{H} \otimes \mathcal{K}_k$ and as the identity on the remaining factors), satisfy the exact cocycle law

$$W_\pi(t_m, t_l) \;=\; W_\pi(t_m, t_k)\, W_\pi(t_k, t_l), \qquad l \leq k \leq m,$$

and reproduce the transition statistics of the divisible family in the sense of (11) with initial environment vector $|0_1 \cdots 0_K\rangle$. The composite $W_\pi(t_m, t_l)$ dilates the composed channel $\Phi(t_m, t_{m-1}) \circ \cdots \circ \Phi(t_{l+1}, t_l)$, whose population action equals $\Gamma(t_m \leftarrow t_l)$ by divisibility; the environment dimension grows with the partition (at most n^{2K} after K steps), and a fresh environment factor per increment is what makes the cocycle law exact—this is precisely the repeated-interaction (collision-model) structure of the open-systems literature [54]. We emphasize that $W_\pi(t_m, t_l)$ is not *asserted to dilate the single-pair channel $\Phi(t_m, t_l)$ itself: the composed channel and the single-pair channel share the population action $\Gamma(t_m \leftarrow t_l)$, but their coherence damping differs in general. No fixed-dimension continuum statement is asserted: whether a divisible family admits a strongly continuous lift on a* fixed *finite-dimensional $\mathcal{H}'$, in the sense of Definition 5.2, satisfying $W(t,s_1) = W(t,s_2)\, W(s_2,s_1)$ for all $s_1 \leq s_2 \leq t$, is an open question. The converse question—whether existence of such a continuum cocycle lift forces divisibility—remains open as well: partial trace does not in general preserve CPTP composition, because the intermediate $\mathcal{K}$-state produced by $W(s_2,s_1)$ acting on $\mathcal{H} \otimes |0\rangle$ generally entangles with $\mathcal{H}$ rather than returning to $|0\rangle$.*

(iv) (Uniqueness at the level of minimal dilations) For each $(t,s) \in [0,T]^2_{\leq}$, any two minimal *Stinespring dilations of the channel $\Phi(t,s)$ are unitarily equivalent: if $V_a\colon \mathcal{H} \to \mathcal{H} \otimes \mathcal{K}_a$ and $V_b\colon \mathcal{H} \to \mathcal{H} \otimes \mathcal{K}_b$ are Stinespring isometries for $\Phi(t,s)$, each minimal, then there exists a unitary $u\colon \mathcal{K}_a \to \mathcal{K}_b$ with $(I_{\mathcal{H}} \otimes u)\, V_a = V_b$ [18, Theorem 4.2]. Unitary completions $W(t,s)$ of a given Stinespring isometry retain the residual freedom of an arbitrary unitary rotation of the orthogonal complement of the input subspace $\mathcal{H} \otimes |0\rangle$ in $\mathcal{H}'$; no equivalence between arbitrary completions is asserted (Remark 5.3), and whether the minimal-dilation equivalences $u(t,s)$ may be chosen jointly continuous in (t,s) requires further regularity beyond (R1)–(R3′) that we do not establish here.*

The proof, given in Appendix C, proceeds in four steps. (a) The Kraus family of Definition 5.1 yields an explicit Stinespring isometry $V(t,s)\colon \mathcal{H} \to \mathcal{H} \otimes \mathcal{K}$ with $\dim \mathcal{K} = n^2 - n + 1$,

satisfying the boundary condition $V(s,s)|j\rangle = |j\rangle \otimes |0\rangle$ exactly and strongly continuous under (R2). Continuity of the orthogonal projection $V(t,s)V(t,s)^*$ on the contractible base $[0,T]^2_{\leq}$ trivializes the associated Stiefel bundle, providing a continuous global frame; the frame is then rotated to agree with the reference frame on the diagonal, so that the unitary completion $W(t,s)$ is strongly continuous with $W(s,s) = I_{\mathcal{H}'}$. (b) Under (R3′), an absolutely continuous unitary completion is constructed for each fixed s by Kato parallel transport of the projection family along t, and the Hermitian family $\widehat{H}_s(t) := i(\partial_t W)W^{-1}$ is well defined a.e., strongly measurable, and integrable in operator norm. (c) Under divisibility, the composition rule for Stinespring dilations with a fresh environment factor per increment produces the grid cocycle of (iii); the obstruction to a fixed-environment continuum cocycle—after a step the environment is not returned to its reference state, so a second application on the *same* factor does not implement the increment channel—is exactly why the statement is at grid level. (d) Minimal-dilation uniqueness is Stinespring's theorem [18, Theorem 4.2]; the residual freedom of arbitrary unitary completions is exhibited explicitly.

Remark 5.5. The construction makes no detailed-balance assumption on the family $\{\Gamma(t \leftarrow s)\}$. This matters: the irreversible dynamics generic to indivisible families typically violate detailed balance, and a direct unitary representation on the configuration-space Hilbert space $\mathcal{H}$ alone exists only for unistochastic matrices (Proposition 2.11), a strict subset even of the doubly stochastic matrices for $n \geq 3$ [20, 21]—and carries no composition structure in time even where it exists pairwise (Remark 2.12). Routing through Stinespring dilation circumvents this restriction at the cost of working with the larger Hilbert space $\mathcal{H}' = \mathcal{H} \otimes \mathcal{K}$. The route through CPTP maps $\Phi(t,s)$ on $\mathcal{B}(\mathcal{H})$ also makes contact with the Lindblad–Gorini–Kossakowski–Sudarshan generator structure for divisible quantum dynamical semigroups [47, 48, 49], although the present construction is more general in that no time-translation invariance, semigroup property, or microscopic master equation is assumed.

Remark 5.6. The level of rigour established by Theorem 5.4 is the following. Part (i)—existence of a strongly continuous Stinespring lift—is rigorous under (R1)–(R2). Part (ii)—existence of an integrable formal Hamiltonian generator, via Kato parallel transport of the projection family—is rigorous under the strict-positivity and absolute-continuity hypothesis (R3′). Part (iii) is a grid-level statement: divisibility yields an exact cocycle dilation on every finite partition, with a fresh environment factor per step; both continuum directions (existence of a fixed-dimension continuum cocycle lift for divisible families, and divisibility as a consequence of such a lift) are open. Part (iv)—uniqueness for minimal dilations at each (t,s) pair—is rigorous, but joint (t,s)-continuity of the minimal-dilation equivalences, and the extension of the construction to infinite-dimensional configuration spaces, require additional regularity not pursued here. Where the construction is referred to in subsequent sections as a "Hilbert-space lift" or "unitary dilation dynamics" rather than a Schrödinger evolution, the intent is to mark the operator-theoretic content (existence and structure of the lift) as distinct from any interpretive content.

Remark 5.7. Theorem 5.4 should be understood operationally: from empirical transition statistics $\{\Gamma(t \leftarrow s)\}$ alone, one constructs a unitary lift $\{W(t,s)\}$ on a dilated Hilbert space whose partial-trace squared moduli reproduce the empirical statistics. Under (R3′), a formal Hamiltonian generator can be constructed for a selected continuous dilation. The generator depends on the selection: under a change of unitary completion $W(t,s) \mapsto (I_{\mathcal{H}} \otimes U(t))\, W(t,s)$ with U unitary on the environment and absolutely continuous in t, the channel is unchanged while $\widehat{H}_s$ becomes $(I \otimes U)\widehat{H}_s(I \otimes U)^* + i\,(I \otimes \dot{U}U^*)$, so no feature of $\widehat{H}_s$ is invariant under such changes; the invariant content is the channel family $\Phi(t,s)$, the populations it reproduces, and the deficit. Divisibility of the underlying family is sufficient for an exact cocycle dilation on every finite time grid, with a fresh environment factor per step, via part (iii); the fixed-dimension continuum cocycle question, in both directions, is open.

Remark 5.8 (Behavior of $\widehat{H}_s(t)$ at the diagonal $t = s$). The formal Hamiltonian generator $\widehat{H}_s(t)$ is constructed under (R3′) as a strongly measurable, a.e.-defined Hermitian family on $[s,T]$,

integrable but in general not essentially bounded in operator norm. Hypothesis (R3′) does not guarantee a finite limit of $\widehat{H}_s(t)$ as $t \downarrow s$: if $\Gamma_{ij}(t \leftarrow s) \to 0$ as $t \downarrow s$ (which is generic for off-diagonal entries given the boundary condition (R1) and the strict-positivity assumption on $\{s < t\}$), then $t \mapsto \sqrt{\Gamma_{ij}(t \leftarrow s)}$ behaves like $\sqrt{c\,(t-s)}$ near the diagonal—absolutely continuous, with derivative $\sim 1/\sqrt{t-s}$, locally integrable but not bounded near the diagonal. The Stinespring isometry $V(t,s)$ is therefore absolutely continuous in t but its L^∞ norm of derivative diverges at $t = s$, and the same is true of $\widehat{H}_s(t)$. In Hille–Yosida terms, $\widehat{H}_s$ is best read as an unbounded operator-valued function with domain $\{(s,t) : 0 \le s < t \le T\}$ that is L^1_{loc} in t on $(s,T]$. The integrated form $W(t,s) = T\text{–}\exp\bigl(-i\int_s^t \widehat{H}_s(\tau)\,d\tau\bigr)$ holds in the sense of Bochner integrals over compact subintervals of $(s,T]$, and the boundary condition $W(s,s) = I_{\mathcal{H}'}$ is recovered by absolute continuity of $W(t,s)$ at $t = s$. The pointwise singularity of $\widehat{H}_s(t)$ at the diagonal is therefore a feature of the differential form of the equation rather than a defect of the construction.

6 A two-state worked example

To exhibit the constructions of Sections 4–5 on a concrete system, we work out an explicit two-state non-divisible two-time kernel family for which the deficit operator, the subspace pair, the indivisibility tensor, the gauge-relative cross-term decomposition, and the integrated kernel deficit can be computed in closed form. Per Remark 2.5, we work at the level of the two-time transition family (Definition 2.3): the family below violates Chapman–Kolmogorov factorization at the pairwise transition level (Definition 2.2), which is exactly what the spectral framework analyses, and realizability by a process is a separate question the framework does not need. The example shows what the framework extracts that ordinary divisible analysis does not capture.

6.1 Construction

Let $\Omega = \{1,2\}$ and write the doubly-stochastic two-state column-stochastic matrix in the parametric form

$$\Gamma(t \leftarrow s) \;=\; \tfrac{1+\alpha(t,s)}{2}\, I \;+\; \tfrac{1-\alpha(t,s)}{2}\, X, \qquad X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \tag{12}$$

with parameter $\alpha(t,s) \in [-1,1]$ measuring stickiness (probability of remaining in the current state minus probability of switching). The boundary condition (R1) requires $\alpha(s,s) = 1$. Each $\Gamma(t \leftarrow s)$ is doubly stochastic and symmetric, so under counting measure every operator in this example is a self-adjoint contraction and Assumption 3.3 holds (Remark 3.4).

The two-state matrix product reduces to a scalar product in this parametrization:

$$\Gamma(t \leftarrow r)\,\Gamma(r \leftarrow s) \;=\; \tfrac{1+\alpha(t,r)\alpha(r,s)}{2}\, I \;+\; \tfrac{1-\alpha(t,r)\alpha(r,s)}{2}\, X, \tag{13}$$

using $X^2 = I$. The CK condition $\Gamma(t \leftarrow s) = \Gamma(t \leftarrow r)\Gamma(r \leftarrow s)$ therefore reduces to the multiplicative scalar identity $\alpha(t,s) = \alpha(t,r)\,\alpha(r,s)$.

Divisible reference. The time-homogeneous Markov family $\alpha_0(t,s) = e^{-\lambda(t-s)}$ with rate $\lambda > 0$ satisfies $\alpha_0(t,s) = \alpha_0(t,r)\alpha_0(r,s)$ identically and hence is divisible.

Non-divisible perturbation. Consider the family

$$\alpha_\varepsilon(t,s) \;=\; e^{-\lambda(t-s)}\bigl(1 - \varepsilon\,(t-s)^2\bigr), \qquad 0 < \varepsilon < 1/T^2 \tag{14}$$

(the upper bound $\varepsilon < 1/T^2$ ensures $1 - \varepsilon\tau^2 \in (0,1]$ for all $\tau \in [0,T]$, hence $\alpha_\varepsilon(t,s) \in (0,1]$ throughout $[0,T]^2_{\le}$, which is sufficient for column-stochasticity of $\Gamma(t \leftarrow s) = \frac{1+\alpha_\varepsilon}{2} I + \frac{1-\alpha_\varepsilon}{2} X$

since the entries are then in $[0,1)$ and 1). The boundary condition $\alpha_\varepsilon(s,s) = 1\cdot 1 = 1$ holds. The family α_ε is continuous in (t,s), smooth in t for each s, and reduces to the divisible reference α_0 at $\varepsilon = 0$. For $\varepsilon > 0$ the family is non-divisible: α_ε is time-homogeneous in elapsed time but not multiplicative, $\alpha_\varepsilon(t,s) \neq \alpha_\varepsilon(t,r)\alpha_\varepsilon(r,s)$, so Chapman–Kolmogorov factorization at the level of the two-time transition matrices fails for some intermediate r, which is the operator-theoretic content the spectral framework of Sections 4–5 analyses.

6.2 Computation of the deficit

By (13), the deficit at the kernel level is

$$\Gamma(t \leftarrow s) - \Gamma(t \leftarrow r)\,\Gamma(r \leftarrow s) \;=\; \tfrac{\alpha_\varepsilon(t,s) - \alpha_\varepsilon(t,r)\alpha_\varepsilon(r,s)}{2}\,(I - X),$$

and since $\mathcal{P}(t,s) = \Gamma(t \leftarrow s)$ as matrices under counting measure, the deficit operator is

$$\Delta(t,s,r) \;=\; \tfrac{\alpha_\varepsilon(t,s) - \alpha_\varepsilon(t,r)\alpha_\varepsilon(r,s)}{2}\,(I - X). \tag{15}$$

A direct expansion of the scalar deficit yields, with $a := t - r > 0$, $b := r - s > 0$, $a + b = t - s$:

$$\begin{aligned}
\alpha_\varepsilon(t,s) - \alpha_\varepsilon(t,r)\alpha_\varepsilon(r,s) &= e^{-\lambda(t-s)}\Big[(1 - \varepsilon(a+b)^2) - (1-\varepsilon a^2)(1-\varepsilon b^2)\Big] \\
&= e^{-\lambda(t-s)}\Big[-\varepsilon(a+b)^2 + \varepsilon(a^2+b^2) - \varepsilon^2 a^2 b^2\Big] \\
&= e^{-\lambda(t-s)}\Big[-2\varepsilon\, a\, b - \varepsilon^2\, a^2 b^2\Big] \\
&= -\varepsilon\,(t-r)(r-s)\, e^{-\lambda(t-s)}\,\big[2 + \varepsilon\,(t-r)(r-s)\big],
\end{aligned}$$

using $(a+b)^2 - (a^2+b^2) = 2ab$. The scalar deficit is strictly negative for $0 \le s < r < t$ and $\varepsilon > 0$.

The matrix $I - X = \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}$ is self-adjoint of rank 1 with operator norm 2 (eigenvalues 0 and 2), and its range is the one-dimensional subspace

$$V_* \;=\; \mathrm{span}\Big\{\, f^* := \tfrac{1}{\sqrt{2}}\,\big(|1\rangle - |2\rangle\big)\,\Big\}.$$

Self-adjointness gives $\mathrm{Range}(\Delta(t,s,r)^*) = \mathrm{Range}(\Delta(t,s,r)) = V_*$, so the input and output deficit subspaces coincide here. The indivisibility deficit (15) has operator norm

$$D(t,s,r) \;=\; \|\Delta(t,s,r)\|_{\mathrm{op}} \;=\; \varepsilon\,(t-r)(r-s)\, e^{-\lambda(t-s)}\,\big[2 + \varepsilon\,(t-r)(r-s)\big], \tag{16}$$

attained at the unit right-singular vector f^*, with $\Delta(t,s,r)f^* \in V_*$ and $\Delta(t,s,r)$ vanishing on $V_*^\perp = \mathrm{span}\{(|1\rangle + |2\rangle)/\sqrt{2}\}$.

The local-diagonal scaling result of Proposition 4.12 is verified directly in this example. From the parametric form (12) with $\alpha_\varepsilon(t,s) = e^{-\lambda(t-s)}(1 - \varepsilon(t-s)^2)$, the time-zero generator is $L(s) = \frac{1}{2}\partial_t \alpha_\varepsilon|_{t=s}\,(I-X) = -\frac{\lambda}{2}(I-X)$, and the second-order coefficient is $C(s) = \frac{1}{2}\partial_t^2 \alpha_\varepsilon|_{t=s}\,(I - X) = \frac{\lambda^2 - 2\varepsilon}{2}(I-X)$. Using $(I-X)^2 = 2(I-X)$, we have $L(s)^2 = \frac{\lambda^2}{2}(I-X)$ and $L'(s) = 0$, so the indivisibility tensor of Definition 4.11 is

$$\beta(s) \;=\; C(s) - L(s)^2 - L'(s) \;=\; -\varepsilon\,(I - X),$$

with $\|\beta(s)\|_{\mathrm{op}} = 2\varepsilon$. The leading-order prediction of Proposition 4.12(a) is therefore $D(t,s,r) \approx 2\varepsilon\,(t-r)(r-s)$ as $(t-s) \to 0$, in agreement with the exact formula (16) expanded at first order in ε and small $(t-s)$. The vanishing of β at $\varepsilon = 0$ recovers the divisible-reference case, consistently with Proposition 4.12(b).

6.3 Verification of Theorems 4.5 and 7.4

The spectral subspaces from Theorem 4.5 are

$$V_{\mathrm{in}} \;=\; \overline{\operatorname{span} \bigcup_{0\le s<r<t\le T} \operatorname{Range}\big(\Delta(t,s,r)^*\big)} \;=\; V_*, \qquad V_{\mathrm{out}} \;=\; \overline{\operatorname{span} \bigcup_{0\le s<r<t\le T} \operatorname{Range}\big(\Delta(t,s,r)\big)} \;=\; V_*,$$

both reducing to the one-dimensional symmetric-difference subspace because $\Delta(t,s,r)$ is self-adjoint at every triple in this example. Theorem 4.5(a) is therefore satisfied with $V_{\mathrm{in}} = V_{\mathrm{out}} = V_*$. Theorem 4.5(b) applies since Assumption 3.3 holds here, and is recovered by direct computation: any divisible two-state comparison family has $\tilde\alpha(t,s) = \tilde\alpha(t,r)\tilde\alpha(r,s)$ (the multiplicative law on its scalar parameter), hence $\tilde\Delta = 0$, and the operator-norm distance from $\mathcal{P}$ to $\tilde{\mathcal{P}}$ on V_* is bounded below by (16) minus the single-time-pair errors at (t,r) and (r,s), exactly matching (4).

The integrated kernel deficit from Theorem 7.4 is, to leading order in ε,

$$\begin{aligned} J(\mathcal{R}) \;&=\; \int_0^T\!\!\int_0^T\!\!\int_0^T \mathbf{1}_{\{s<r<t\}}\, D(t,s,r)^2 \, ds\, dr\, dt \\ &=\; 4\varepsilon^2 \iiint_{\{0\le s<r<t\le T\}} (t-r)^2 (r-s)^2\, e^{-2\lambda(t-s)}\, ds\, dr\, dt \;+\; O(\varepsilon^3). \end{aligned}$$

The triple integral is finite (the integrand is bounded on the bounded domain) and strictly positive, confirming $J(\mathcal{R}) > 0$ for every $\varepsilon > 0$ and the equivalence (i) $\Leftrightarrow$ (iii) of Theorem 7.4.

6.4 Gauge-relative cross-term decomposition in the example

For the two-state doubly-stochastic case, every $\Gamma(t \leftarrow s)$ is unistochastic (Proposition 2.11(b)); select the gauge

$$U(t,s) \;=\; \cos\theta(t,s)\, I \;+\; i\, \sin\theta(t,s)\, Y, \qquad Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix},$$

with $\cos^2\theta(t,s) = (1+\alpha_\varepsilon(t,s))/2$, so that $|U_{11}|^2 = |U_{22}|^2 = (1+\alpha)/2$ and $|U_{12}|^2 = |U_{21}|^2 = (1-\alpha)/2 = \Gamma_{12}$. Rotations about a common axis compose additively, $V(t,s;r) = e^{i(\theta(t,r)+\theta(r,s))Y}$, so the two summands of Theorem 4.8 are computable in closed form in this gauge: with $\theta_1 := \theta(t,r)$, $\theta_2 := \theta(r,s)$,

$$\begin{aligned} \mathfrak{D}(t,s,r) \;&=\; \tfrac{1}{2}\big[\cos 2\theta(t,s) - \cos(2\theta_1 + 2\theta_2)\big]\,(I - X), \\ \mathfrak{C}(t,s,r) \;&=\; \tfrac{1}{2}\big[\cos(2\theta_1 + 2\theta_2) - \cos 2\theta_1 \cos 2\theta_2\big]\,(I - X), \end{aligned}$$

using $\cos 2\theta(t,s) = \alpha_\varepsilon(t,s)$ and $[\mathcal{P}(t,r)\mathcal{P}(r,s)]$-diagonal $= (1 + \alpha_\varepsilon(t,r)\alpha_\varepsilon(r,s))/2$. Their sum telescopes to $\frac{1}{2}[\alpha_\varepsilon(t,s) - \alpha_\varepsilon(t,r)\alpha_\varepsilon(r,s)](I - X) = \Delta(t,s,r)$, matching (15) exactly, for every $\varepsilon \ge 0$. The individual summands illustrate Theorem 4.8(c) concretely: at $\varepsilon = 0$ the family is divisible, so $\mathfrak{C} + \mathfrak{D} = 0$, yet both terms are generically nonzero in this gauge—the multiplier $\alpha_0 = \cos 2\theta$ is multiplicative while the angle θ is not additive, so the selected gauge is not a cocycle and the interference term cancels the non-composition term rather than both vanishing. The vanishing of each term separately is available only in special gauges (Remark 4.9), and no conclusion about the family follows from the individual terms.

6.5 Hilbert-space lift dimensions

The Stinespring dilation of Theorem 5.4 uses the environment $\mathcal{K}$ of dimension $n^2 - n + 1 = 3$ (basis $\{|0\rangle, |1,2\rangle, |2,1\rangle\}$), so the dilated Hilbert space $\mathcal{H}'$ has dimension $n(n^2 - n + 1) = 6$. The

Kraus operators of the diagonal-retention lift (Definition 5.1) are

$$K_0(t,s) = \sqrt{\tfrac{1+\alpha_\varepsilon(t,s)}{2}}\, I, \qquad K_{(1,2)}(t,s) = \sqrt{\tfrac{1-\alpha_\varepsilon(t,s)}{2}}\, |1\rangle\langle 2|, \qquad K_{(2,1)}(t,s) = \sqrt{\tfrac{1-\alpha_\varepsilon(t,s)}{2}}\, |2\rangle\langle 1|,$$

all continuous in (t,s) via the explicit form (12). The Stinespring isometry (21) takes the explicit form

$$V(t,s)|m\rangle = \sqrt{\tfrac{1+\alpha_\varepsilon(t,s)}{2}}\, |m\rangle \otimes |0\rangle \;+\; \sqrt{\tfrac{1-\alpha_\varepsilon(t,s)}{2}}\, |m \oplus 1\rangle \otimes |m \oplus 1, m\rangle,$$

where $\oplus$ denotes addition mod 2 in $\{1,2\}$; the boundary $\alpha_\varepsilon(s,s) = 1$ gives $V(s,s)|m\rangle = |m\rangle \otimes |0\rangle$ exactly, and V extends by Step 1b to a strongly continuous unitary family $W(t,s) \in U(6)$ with $W(s,s) = I$ over $(t,s) \in [0,T]^2_{\leq}$. At $\varepsilon = 0$ the family $\{\Gamma(t \leftarrow s)\}$ is divisible, and Theorem 5.4(iii) supplies, on every finite grid $s = t_0 < \cdots < t_K = t$, an exact cocycle dilation built from one fresh three-dimensional environment factor per increment. For $\varepsilon > 0$ the family is non-divisible, so no grid coarser than the trivial one composes, and the formal Hamiltonian generator $\widehat{H}_s(t)$ on the absolutely continuous pairwise lift of Theorem 5.4(ii) carries genuine s-dependence inherited from the s-dependence of the Kraus operators $\sqrt{\Gamma_{ij}(t \leftarrow s)}$.

6.6 Comparison with divisible Koopman analysis

For the divisible reference family $\alpha_0(t,s) = e^{-\lambda(t-s)}$, ordinary Koopman semigroup analysis [10, 8] produces a generator $\mathcal{L} = -\lambda(I - X)/2$ with eigenvalues 0 (eigenvector $(|1\rangle + |2\rangle)/\sqrt{2}$) and $-\lambda$ (eigenvector f^*). The eigenvalue $-\lambda$ governs the symmetric-difference relaxation rate, and standard DMD applied to data sampled from the divisible family recovers λ as the dominant non-trivial mode.

For the non-divisible family α_ε with $\varepsilon > 0$, no such generator exists (Proposition 3.6). The action of $\mathcal{P}(t,s)$ on f^* has time dependence

$$\mathcal{P}(t,s)\, f^* \;=\; \alpha_\varepsilon(t,s)\, f^* \;=\; e^{-\lambda(t-s)}\big(1 - \varepsilon(t-s)^2\big) f^*,$$

which is non-exponential in $t-s$. The constant family $\varphi^{(u)} \equiv f^*$ with multiplier $\lambda(t,s) = \alpha_\varepsilon(t,s)$ is a covariant eigenfamily in the sense of Definition 4.2, and its multiplier is non-multiplicative for $\varepsilon > 0$: $\alpha_\varepsilon(t,s) \neq \alpha_\varepsilon(t,r)\,\alpha_\varepsilon(r,s)$. By Lemma 4.3 this witnesses indivisibility, identifying V_* as a one-dimensional indivisible mode. Ordinary Koopman analysis applied to data from the non-divisible family at fixed time origin would extract a single best-fit exponential rate, missing the $\varepsilon(t-s)^2$ correction; the present framework recovers the full non-exponential time dependence on V_* and identifies V_* as the subspace where the deviation lives. This is the structural content the divisible-approximation methods do not capture and the present framework recovers.

6.7 A three-state example exhibiting $V_{\text{in}} \neq V_{\text{out}}$

The two-state example above is doubly-stochastic, and as noted in Section 6.2, the deficit $\Delta(t,s,r)$ is self-adjoint there, so $V_{\text{in}} = V_{\text{out}}$ collapses to a single subspace. To exhibit a configuration in which the input and output deficit subspaces of Theorem 4.5 genuinely differ, we work out a non-doubly-stochastic three-state family.

Let $\Omega = \{1,2,3\}$ and consider the column-stochastic family with biased rows

$$\Gamma(t \leftarrow s) \;=\; \begin{pmatrix} 1 & p(t,s) & q(t,s) \\ 0 & 1 - p(t,s) & 0 \\ 0 & 0 & 1 - q(t,s) \end{pmatrix}, \tag{17}$$

parametrised by two "return" probabilities $p, q \in [0,1]$ with $p(s,s) = q(s,s) = 0$. State 1 is absorbing; state 2 returns to state 1 with probability $p(t,s)$ or remains in 2; state 3 returns to

state 1 with probability $q(t,s)$ or remains in 3. Direct computation gives

$$\Gamma(t \leftarrow r)\,\Gamma(r \leftarrow s) \;=\; \begin{pmatrix} 1 & p(t,r)+p(r,s)-p(t,r)p(r,s) & q(t,r)+q(r,s)-q(t,r)q(r,s) \\ 0 & (1-p(t,r))(1-p(r,s)) & 0 \\ 0 & 0 & (1-q(t,r))(1-q(r,s)) \end{pmatrix},$$

so CK at (t,s,r) is equivalent to the multiplicative law $1-p(t,s) = (1-p(t,r))(1-p(r,s))$ on $1-p$ (and analogously on $1-q$).

Divisible reference. The family $1-p_0(t,s) = e^{-\lambda(t-s)}$ and $1-q_0(t,s) = e^{-\mu(t-s)}$ for rates $\lambda,\mu>0$ satisfies the multiplicative law and yields a divisible family.

Non-divisible perturbation. Set $1-p_\varepsilon(t,s) = e^{-\lambda(t-s)}(1+\varepsilon(t-s)^2)$ and $1-q_\varepsilon(t,s) = e^{-\mu(t-s)}(1+\varepsilon'(t-s)^2)$. The constraint $p_\varepsilon, q_\varepsilon \in [0,1]$ on $[0,T]^2_{\leq}$ is equivalent to $\varepsilon(t-s)^2 \leq e^{\lambda(t-s)}-1$ and $\varepsilon'(t-s)^2 \leq e^{\mu(t-s)}-1$ for all $0 \leq s \leq t \leq T$, i.e.,

$$\varepsilon \;\leq\; \min_{\tau\in(0,T]} \frac{e^{\lambda\tau}-1}{\tau^2}, \qquad \varepsilon' \;\leq\; \min_{\tau\in(0,T]} \frac{e^{\mu\tau}-1}{\tau^2}.$$

For $\lambda T \leq 1$ the minimum is achieved at the right endpoint $\tau = T$, giving the leading-order bound $\varepsilon \leq \lambda/T$ (and analogously $\varepsilon' \leq \mu/T$). The deficit operator $\Delta(t,s,r) = \Gamma(t \leftarrow s) - \Gamma(t \leftarrow r)\Gamma(r \leftarrow s)$ has columns

$$\Delta(t,s,r)\,|1\rangle = 0, \qquad \Delta(t,s,r)\,|2\rangle = \delta p\,(|1\rangle - |2\rangle), \qquad \Delta(t,s,r)\,|3\rangle = \delta q\,(|1\rangle - |3\rangle),$$

where $\delta p = p_\varepsilon(t,s) - [p_\varepsilon(t,r) + p_\varepsilon(r,s) - p_\varepsilon(t,r)p_\varepsilon(r,s)]$ and analogously for δq. At first order in ε, $\delta p = -2\varepsilon(t-r)(r-s)e^{-\lambda(t-s)}$, similarly for δq with ε' and μ.

Input and output subspaces. For generic $\varepsilon,\varepsilon' > 0$, δp and δq are linearly independent functions of (t,s,r), so

$$V_{\text{in}} \;=\; \overline{\operatorname{span} \bigcup_{(t,s,r)} \operatorname{Range}(\Delta(t,s,r)^*)} \;=\; \operatorname{span}\{|2\rangle, |3\rangle\},$$

$$V_{\text{out}} \;=\; \overline{\operatorname{span} \bigcup_{(t,s,r)} \operatorname{Range}(\Delta(t,s,r))} \;=\; \operatorname{span}\{|1\rangle - |2\rangle,\ |1\rangle - |3\rangle\}.$$

Both are two-dimensional subspaces of $\mathbb{C}^3$, and they are distinct: V_{in} is the transient-states span (states 2 and 3, the sources of indivisible content), while V_{out} is the relaxation-direction span (the differences between the transient-state populations and the absorbing target). The orthogonal complements are $V_{\text{in}}^\perp = \operatorname{span}\{|1\rangle\}$ (the absorbing target) and $V_{\text{out}}^\perp = \operatorname{span}\{(|1\rangle + |2\rangle + |3\rangle)/\sqrt{3}\}$ (the uniform state), which are also distinct.

The input/output asymmetry is structural: the indivisibility content enters $\mathcal{P}$ through densities supported on the transient states $\{2,3\}$ (the input subspace V_{in}), and the deviation manifests in the population transfer from those states toward the absorbing target (the output subspace V_{out}). This is the generic behaviour the framework is designed to expose; the doubly-stochastic two-state example is a degenerate special case in which the two subspaces happen to coincide because of the symmetry $\Delta = \Delta^*$.

A scope note, per Remark 3.4: this family is column-stochastic but not doubly stochastic, and its operator norm under counting measure exceeds one whenever the return probabilities are positive ($\|\Gamma^\top e_1\| = \sqrt{1+p^2+q^2} > 1$), so Assumption 3.3 fails for it. The example accordingly illustrates the boundedness-based content of Theorem 4.5(a)—the subspace concentration, which is all it invokes—while the quantitative bounds of Theorem 4.5(b) and Corollary 4.6 are not asserted for it.

6.8 Products of families and examples in higher dimension

The theorems of Sections 4, 5 and 7 carry no hypothesis on the cardinality of Ω beyond compactness and Assumption 3.3, and the two- and three-state families above witness that every object they define is nonvacuous and computable in closed form: the two-state family exhibits each construction and the scaling law, and the three-state family exhibits $V_{\text{in}} \neq V_{\text{out}}$ and the failure of Assumption 3.3. Families in higher dimension with closed-form deficits are supplied by products.

Proposition 6.1 (Product families)**.** *Let $\{\Gamma(t \leftarrow s)\}$ and $\{\tilde{\Gamma}(t \leftarrow s)\}$ be column-stochastic families on finite configuration spaces Ω and $\tilde{\Omega}$. Then $\{\Gamma(t \leftarrow s) \otimes \tilde{\Gamma}(t \leftarrow s)\}$ is a column-stochastic family on $\Omega \times \tilde{\Omega}$, and for every triple (t, s, r),*

$$\Delta_{\Gamma\otimes\tilde{\Gamma}}(t,s,r) \;=\; \Delta_{\Gamma}(t,s,r) \otimes \tilde{\Gamma}(t \leftarrow s) \;+\; \Gamma(t \leftarrow r)\,\Gamma(r \leftarrow s) \otimes \Delta_{\tilde{\Gamma}}(t,s,r). \tag{18}$$

If both families are jointly C^2 and satisfy (R1), then

$$\beta_{\Gamma\otimes\tilde{\Gamma}}(s) \;=\; \beta_{\Gamma}(s) \otimes I \;+\; I \otimes \beta_{\tilde{\Gamma}}(s). \tag{19}$$

In particular the product of two divisible families is divisible, and if both factors satisfy Assumption 3.3 under counting measure, then $D_{\Gamma\otimes\tilde{\Gamma}}(t,s,r) \le D_{\Gamma}(t,s,r) + D_{\tilde{\Gamma}}(t,s,r)$.

Proof. The Kronecker product of column-stochastic matrices is column-stochastic. Write $\Gamma(t \leftarrow s) = \Gamma(t \leftarrow r)\Gamma(r \leftarrow s) + \Delta$ and $\tilde{\Gamma}(t \leftarrow s) = \tilde{\Gamma}(t \leftarrow r)\tilde{\Gamma}(r \leftarrow s) + \tilde{\Delta}$. By $(A \otimes B)(C \otimes E) = AC \otimes BE$,

$$\begin{aligned}\Gamma(t \leftarrow s) \otimes \tilde{\Gamma}(t \leftarrow s) \;-\; \big(\Gamma(t \leftarrow r)\Gamma(r \leftarrow s)\big) \otimes \big(\tilde{\Gamma}(t \leftarrow r)\tilde{\Gamma}(r \leftarrow s)\big)\\ =\; \Delta \otimes \tilde{\Gamma}(t \leftarrow r)\tilde{\Gamma}(r \leftarrow s) \;+\; \Gamma(t \leftarrow r)\Gamma(r \leftarrow s) \otimes \tilde{\Delta} \;+\; \Delta \otimes \tilde{\Delta},\end{aligned}$$

and the first and third terms combine to $\Delta \otimes \tilde{\Gamma}(t \leftarrow s)$, which gives (18). For (19), let L, C and $\tilde{L}, \tilde{C}$ be the first- and second-order Taylor coefficients at the diagonal of the two families (Definition 4.11). Differentiating the product twice in t at $t = s$ and using (R1) gives $L_{\otimes} = L \otimes I + I \otimes \tilde{L}$ and $C_{\otimes} = C \otimes I + 2\,L \otimes \tilde{L} + I \otimes \tilde{C}$, while $L^2_{\otimes} = L^2 \otimes I + 2\,L \otimes \tilde{L} + I \otimes \tilde{L}^2$ and $L'_{\otimes} = L' \otimes I + I \otimes \tilde{L}'$. Subtracting, the cross terms cancel and $\beta_{\otimes} = (C - L^2 - L') \otimes I + I \otimes (\tilde{C} - \tilde{L}^2 - \tilde{L}')$. Divisibility of the product follows from (18) with $\Delta = \tilde{\Delta} = 0$, and the norm bound from $\|A \otimes B\|_{\text{op}} = \|A\|_{\text{op}}\,\|B\|_{\text{op}}$ and the contraction assumption. □

A four-state family. Take both factors equal to the two-state family (14), so the configuration space is $\{1,2\}^2$. With $v = (|1\rangle - |2\rangle)/\sqrt{2}$ and $\mathbf{1} = (|1\rangle + |2\rangle)/\sqrt{2}$, the factor is $\Gamma(t \leftarrow s) = I - (1 - \alpha_{\varepsilon}(t,s))\,vv^*$, and by (15) its deficit is $\Delta(t,s,r) = \delta\,vv^*$ with the scalar deficit $\delta := \alpha_{\varepsilon}(t,s) - \alpha_{\varepsilon}(t,r)\alpha_{\varepsilon}(r,s)$. By (18),

$$\Delta_{\otimes}(t,s,r) \;=\; \delta\,\big(vv^* \otimes \Gamma(t \leftarrow s) \;+\; \Gamma(t \leftarrow r)\Gamma(r \leftarrow s) \otimes vv^*\big).$$

In the orthonormal basis $\{\mathbf{1}\otimes\mathbf{1},\ \mathbf{1}\otimes v,\ v\otimes\mathbf{1},\ v\otimes v\}$ every factor is diagonal, since $\Gamma(t \leftarrow s)$ has eigenvalue 1 on $\mathbf{1}$ and $\alpha_{\varepsilon}(t,s)$ on v; so $\Delta_{\otimes}$ is diagonal with entries 0, δ, δ, and $\delta\,\big(\alpha_{\varepsilon}(t,s) + \alpha_{\varepsilon}(t,r)\alpha_{\varepsilon}(r,s)\big)$. Hence the deficit has rank three whenever $\delta \neq 0$, $V_{\text{in}} = V_{\text{out}} = (\mathbf{1}\otimes\mathbf{1})^{\perp}$ is the three-dimensional mean-zero subspace, the largest subspace compatible with vanishing column sums, and $D_{\otimes}(t,s,r) = |\delta|\,\max\{1,\ \alpha_{\varepsilon}(t,s) + \alpha_{\varepsilon}(t,r)\alpha_{\varepsilon}(r,s)\}$. By (19) and $\beta = -\varepsilon(I - X) = -2\varepsilon\,vv^*$ for the factor,

$$\beta_{\otimes}(s) \;=\; -2\varepsilon\,\big(vv^* \otimes I \;+\; I \otimes vv^*\big),$$

with singular values 4ε on $v \otimes v$, 2ε on $\mathbf{1} \otimes v$ and on $v \otimes \mathbf{1}$, and 0 on $\mathbf{1} \otimes \mathbf{1}$. The locally most deficit-active direction is $v \otimes v = \frac{1}{2}(|1\rangle - |2\rangle) \otimes (|1\rangle - |2\rangle)$, which mixes all four coordinates.

The leading-order prediction $D_{\otimes} \approx 4\varepsilon\,(t-r)(r-s)$ of Proposition 4.12 agrees with the exact expression, since near the diagonal $\alpha_\varepsilon(t,s) + \alpha_\varepsilon(t,r)\alpha_\varepsilon(r,s) \to 2$ and $|\delta| \approx 2\varepsilon\,(t-r)(r-s)$ by (16). Iterating the product gives closed-form families on 2^k states whose indivisibility tensor is the k-fold Kronecker sum.

7 Causal structure: one-sided operators on $L^2([0,T]\times\Omega)$

For empirical analysis, it is useful to consolidate the two-time family into a single operator on the space of square-integrable trajectories on $[0,T]$, and to make the causal structure—that future evolution is determined by past evolution, not vice versa—explicit at the operator level. The construction throughout this section operates with one-sided causal operators only; the propagation on observables is time-forward, and no symmetry under time reversal is assumed or used.

7.1 The retarded transition operator

Let $\mathcal{H}_T = L^2([0,T]\times\Omega, dt\otimes d\mu)$ denote the Hilbert space of square-integrable spacetime functions, with norm $\|f\|^2_{\mathcal{H}_T} = \int_0^T \int_\Omega |f(t,x)|^2\, d\mu(x)\, dt$.

Definition 7.1 (Retarded transition operator)**.** For a two-time transition family with kernel $P(t,x \mid s,y)$, the *retarded transition operator* $\mathcal{R}\colon \mathcal{H}_T \to \mathcal{H}_T$ is defined for $f \in \mathcal{H}_T$ by

$$\big(\mathcal{R}f\big)(t,x) \;=\; \int_0^t \int_\Omega P(t,x \mid s,y)\, f(s,y)\, d\mu(y)\, ds \;=\; \int_0^t \big(\mathcal{P}(t,s)\, f(s,\cdot)\big)(x)\, ds.$$

The integral kernel of $\mathcal{R}$ is $K_{\mathcal{R}}(t,x;s,y) = \mathbf{1}_{\{s\le t\}} P(t,x \mid s,y)$ on $([0,T]\times\Omega)^2$.

Boundedness of $\mathcal{R}$ is not automatic and is supplied by a uniform bound on the two-time family.

Lemma 7.2 (Boundedness of the retarded operator)**.** *If $M := \sup_{0\le s\le t\le T} \|\mathcal{P}(t,s)\|_{\mathrm{op}} < \infty$ (under Assumption 3.3, $M \le 1$), then $\mathcal{R}$ is bounded on $\mathcal{H}_T$ with $\|\mathcal{R}\| \le M\,T$.*

Proof. Write $f(s) := f(s,\cdot) \in L^2(\Omega,\mu)$. Then $\|(\mathcal{R}f)(t)\|_{L^2(\Omega)} \le \int_0^t \|\mathcal{P}(t,s)\|\, \|f(s)\|\, ds \le M \int_0^T \|f(s)\|\, ds \le M\sqrt{T}\, \|f\|_{\mathcal{H}_T}$ by Cauchy–Schwarz in s, and integrating the square over $t\in[0,T]$ gives $\|\mathcal{R}f\|_{\mathcal{H}_T} \le MT\, \|f\|_{\mathcal{H}_T}$. □

Proposition 7.3 (Causality)**.** *The retarded transition operator $\mathcal{R}$ is causal with respect to the filtration $\{\mathcal{F}_t = L^2([0,t]\times\Omega)\}_{t\in[0,T]}$: for all $\tau\in[0,T]$ and $f\in\mathcal{H}_T$,*

$$\mathbf{1}_{[0,\tau]}\cdot\mathcal{R}f \;=\; \mathbf{1}_{[0,\tau]}\cdot\mathcal{R}\big(\mathbf{1}_{[0,\tau]}\cdot f\big).$$

Equivalently, the future subspaces $\{f\in\mathcal{H}_T : f = 0$ a.e. on $[0,\tau]\times\Omega\}$, $\tau\in[0,T]$, are invariant under $\mathcal{R}$.

Proof. Direct from Definition 7.1: $(\mathcal{R}f)(t,x)$ depends only on $f(s,y)$ for $s\le t$, so multiplication of either side by $\mathbf{1}_{[0,\tau]}(t)$ produces identical results. If moreover f vanishes a.e. on $[0,\tau]\times\Omega$, the identity gives $\mathbf{1}_{[0,\tau]}\cdot\mathcal{R}f = 0$, which is the invariance of the future subspaces. □

7.2 Kernel-level cocycle defect and indivisibility on $L^2([0,T]\times\Omega)$

The two-time family at fixed time pairs satisfies the composition $\mathcal{P}(t,s)=\mathcal{P}(t,r)\mathcal{P}(r,s)$ if and only if the underlying kernel satisfies CK at (t,s,r) (Section 3). The same property lifts to $\mathcal{H}_T$ at the level of integral kernels.

Theorem 7.4 (Kernel cocycle and indivisibility on $\mathcal{H}_T$)**.** *Let $\{P(t,\cdot\mid s,\cdot)\}$ be a two-time transition family on Ω whose operator family satisfies*

(C1) (Uniform bound) $M:=\sup_{0\le s\le t\le T}\|\mathcal{P}(t,s)\|_{\mathrm{op}}<\infty$ (supplied with $M\le 1$ by Assumption 3.3);

(C2) (Norm continuity off the diagonal) $(t,s)\mapsto\mathcal{P}(t,s)$ is continuous in operator norm on the open region $\{(t,s):0\le s<t\le T\}$.

Let $\mathcal{R}$ be the retarded transition operator on $\mathcal{H}_T$, bounded by Lemma 7.2, with integral kernel $K_{\mathcal{R}}(t,x;s,y)=\mathbf{1}_{\{s\le t\}}P(t,x\mid s,y)$. Define the kernel deficit *at the triple (t,s,r) with $0\le s<r<t\le T$ as*

$$\delta K_{\mathcal{R}}(t,x;s,y;r)\;=\;K_{\mathcal{R}}(t,x;s,y)-\int_\Omega K_{\mathcal{R}}(t,x;r,z)\,K_{\mathcal{R}}(r,z;s,y)\,d\mu(z),$$

and the integrated kernel deficit

$$J(\mathcal{R})\;=\;\int_0^T\!\!\int_0^T\!\!\int_0^T\mathbf{1}_{\{s<r<t\}}\,D(t,s,r)^2\,ds\,dr\,dt,$$

where $D(t,s,r)=\|\delta K_{\mathcal{R}}(t,\cdot;s,\cdot;r)\|_{\mathrm{op}}$ is the indivisibility deficit (Definition 2.4). The following are equivalent:

(i) The family is divisible.

(ii) For every triple $0\le s<r<t\le T$, the kernel deficit $\delta K_{\mathcal{R}}(t,\cdot;s,\cdot;r)$ induces the zero operator on $L^2(\Omega,\mu)$ — equivalently (Lemma D.1), $\delta K_{\mathcal{R}}(t,x;s,y;r)=0$ for $\mu\otimes\mu$-a.e. (x,y), at every triple.

(iii) $J(\mathcal{R})=0$.

For indivisible families, $J(\mathcal{R})>0$; and $J(\mathcal{R})\le(M+M^2)^2\,T^3/6<\infty$ under (C1).

The proof, given in Appendix D, is a direct identification of $\delta K_{\mathcal{R}}$ with the kernel-level CK deficit at each triple. Two passages carry the analytic weight, and each has its own hypothesis: (C2) makes $D(t,s,r)$ continuous on the open simplex of triples, which licenses the passage between a.e. vanishing and identical vanishing in (ii)⇔(iii); and the passage from the operator-level statement in (ii) back to the pointwise kernel statement is *not* automatic for general integral operators — it is supplied here by Lemma D.1, which uses only the nonnegativity of the two kernel terms, the boundedness (C1) of the induced operators, and σ-finiteness of μ on the countably generated Borel σ-algebra of the Polish state space.

Remark 7.5 (Why (C2) excludes the diagonal). The restriction of (C2) to the open region $\{s<t\}$ is forced in infinite dimensions, not merely convenient. If $\mathcal{P}(t,s)$ is compact for $t>s$ (the standing hypothesis of Section 4), $\mathcal{P}(s,s)=I$, and $(t,s)\mapsto\mathcal{P}(t,s)$ were norm-continuous up to the diagonal, the identity would be a norm limit of compact operators and hence compact—impossible when $\dim L^2(\Omega,\mu)=\infty$. The compatible diagonal notion is strong continuity, $\mathcal{P}(t,s)\to I$ strongly as $t\downarrow s$, the standard boundary behavior of C_0-families. Nothing in Theorem 7.4 or its proof uses continuity at the diagonal: the deficit, the integrated deficit

$J(\mathcal{R})$, and the equivalences (i)–(iii) all live on the open simplex $\{s < r < t\}$. In finite dimensions, where compactness is vacuous, norm continuity extends to the closed simplex without tension, and (C2) may be read there in its closed form.

Remark 7.6. Theorem 7.4 is the trajectory-space shadow of the failure of $\mathcal{P}(t,s)$-level factorization for indivisible families. Where Theorem 4.5 operates at fixed time pairs and identifies the input/output deficit subspaces $V_{\text{in}}, V_{\text{out}}$ on which the deficit is concentrated, Theorem 7.4 operates on the full trajectory space and registers indivisibility as a single scalar $J(\mathcal{R})$ that vanishes exactly under divisibility.

8 Discussion

The operator-theoretic framework for stochastic dynamics, as classically developed, is calibrated for divisible dynamics: the semigroup property is built into its spine, and the spectral analysis of generators presupposes that this property holds. The central contribution of the present paper is spectral machinery that operates without semigroup structure, built on the singular structure of the composition deficit. The content not captured by standard generator-based analysis is recoverable via the deficit operator $\Delta(t,s,r)$ at two scales. Locally, the indivisibility tensor $\beta(s) = C(s) - L(s)^2 - L'(s)$ calibrates the deficit's exact second-order scaling at the diagonal and furnishes a pointwise sufficient condition for indivisibility computable from the family's Taylor coefficients alone (Proposition 4.12). Globally, the deficit concentrates on the input/output subspace pair $V_{\text{in}}, V_{\text{out}}$, and under the standing contraction assumption a universal three-pair obstruction bounds every divisible approximation away from the family (Theorem 4.5, Corollary 4.6). In any selected unistochastic gauge, the deficit's matrix elements decompose into a cross-term and a non-composition term whose sum—not the individual, gauge-dependent summands—vanishes exactly under divisibility (Theorem 4.8). Under explicit regularity hypotheses, the family lifts to a unitary dilation constructed directly from empirical transition data without detailed-balance hypotheses, with an exact grid cocycle dilation available when the underlying family is divisible (Theorem 5.4). And the operator-algebraic shadow on $L^2([0,T]\times\Omega)$ is the integrated kernel deficit $J(\mathcal{R})$ on the retarded transition operator (Theorem 7.4). The worked examples of Section 6 demonstrate these constructions in closed form on concrete non-divisible two-time kernel families. The framework is designed for dynamics beyond the divisible regime; divisibility constitutes a structurally restrictive subclass under which the deficit operator vanishes and the standard generator-based machinery suffices. Whether non-divisibility is generic in any precise topology or measure on a space of two-time kernel families, and whether augmentation reductions to divisibility are typically available or unavailable for natural classes of dynamics, are separate questions we do not address here.

The framework suggests a sharper empirical test for indivisibility than has previously been available. Given a candidate stochastic process with empirical transition data, the protocol is: estimate the kernel deficit $\delta K_{\mathcal{R}}(t,x;s,y;r)$ from the data and form the integrated kernel deficit $J(\mathcal{R})$; if $J(\mathcal{R}) > 0$ at statistical significance, identify the input/output subspaces $V_{\text{in}}, V_{\text{out}}$ from the closed spans of ranges of $\Delta(t,s,r)^*$ and $\Delta(t,s,r)$ respectively, and, under (R3′), compute the Stinespring dilation via Theorem 5.4 and analyse gauge-invariant functionals of it — the singular structure of the deficit family, the spectra of the channels $\Phi(t,s)$, or invariants of the minimal dilation — to extract dynamical information not visible to divisible-approximation methods. (Spectral data of a particular unitary completion $W(t,s)$ is not by itself intrinsic to the family: completions carry residual gauge freedom on the orthogonal complement of the input subspace, Theorem 5.4(iv).) Two caveats delimit what such a test establishes. First, a statistically significant $J(\mathcal{R}) > 0$ establishes that the *estimated* two-time kernels on the *chosen* configuration space fail to factorize; it does not by itself distinguish among the mechanisms that can produce such failure—genuinely indivisible dynamics, coarse-graining of a divisible

process on a larger space (omitted variables), nonstationarity across the estimation window, finite-sample estimation error in the kernels, or a two-time family not realizable by any process at all (Remark 2.5). Attribution among these requires modeling commitments external to the test. Second, the estimated subspaces $\widehat{V}_{\text{in}}, \widehat{V}_{\text{out}}$ are singular subspaces of estimated compact operators, and their stability under kernel-estimation error is governed by singular-subspace perturbation theory of Davis–Kahan/Wedin type [52, 53]: the subspace perturbation angle scales as the kernel estimation error divided by the relevant singular-value gap of the deficit family, so reliable recovery of $V_{\text{in}}, V_{\text{out}}$ requires both a controlled kernel estimator and a nondegenerate singular gap. Within these limits, a protocol of this kind allows quantitative comparison of the indivisibility deficit across systems—between, for instance, mesoscopic devices in coherent versus decoherent regimes, turbulent fluid dynamics across Reynolds-number transitions, or distinct biological dynamical regimes. Stated as an obstruction theorem: under Assumption 3.3, the framework rules out the existence of any divisible family approximating $\mathcal{P}(t, s)$ within $D(t, s, r)/3$ in operator norm at every single time pair, for any triple (t, s, r) with $D(t, s, r) > 0$ (Corollary 4.6); divisible-approximation methods are quantitatively bounded away from the data by a deficit-determined gap. The gap concerns uniform approximation of the two-time family in operator norm—a divisible model may still reproduce selected observables, marginals, or low-order statistics of the same data to high accuracy, and the obstruction says nothing against divisible fits used for such restricted purposes.

The transfer-operator program of Klus, Koltai, Schütte and collaborators [7, 8] treats stochastic dynamics through the Perron–Frobenius operator on densities under Markov hypotheses; the framework above operates on the same forward-operator side while removing the CK hypothesis, and by Remark 3.2 every result transfers to the Koopman side by adjunction. The two programs agree under CK and diverge precisely on the indivisible content identified by $\Delta(t, s, r)$ and $J(\mathcal{R})$.

Relation to non-Markovianity measures. The integrated kernel deficit $J(\mathcal{R})$ of Theorem 7.4 is structurally analogous to the divisibility-based non-Markovianity measures developed for open quantum systems by Wolf, Eisert, Cubitt, and Cirac [41], Breuer, Laine, and Piilo [42], and Rivas, Huelga, and Plenio [43, 44], with the closest formal kinship to the RHP measure based on integrating a CP-divisibility deficit over time intervals. Two structural distinctions are worth noting. First, $J(\mathcal{R})$ operates on a column-stochastic kernel family directly rather than on its CP-divisible quantum lift; the corresponding "classical RHP measure" that one might write down by analogy on the unistochastic skeleton of Γ would in general yield a different scalar than $J(\mathcal{R})$ because the mapping $\Gamma \mapsto |U|^2$ is not isometric on operator-norm-induced metrics. Second, $J(\mathcal{R})$ integrates over triples (s, r, t) with $s < r < t$ on the trajectory space $\mathcal{H}_T = L^2([0, T] \times \Omega, dt \otimes d\mu)$ and registers the failure of the kernel cocycle property explicitly through the intermediate time r, whereas RHP-style measures integrate the cb-norm of the deviation of a one-parameter dynamical map from CP-divisibility along a single time axis. The two formulations agree on the qualitative content—vanishing iff divisible, strictly positive iff indivisible—but differ in scaling and in the way they distribute weight across triples versus pairs. A precise quantitative correspondence between $J(\mathcal{R})$ and the RHP measure on the Stinespring lift of Section 5 is a natural follow-up question we do not pursue here.

Several questions remain open. First, the extension from finite to separable infinite-dimensional configuration spaces is straightforward in form—the Stinespring framework extends to infinite dimensions under nuclear-operator hypotheses on $\Phi(t, s)$, and the minimal-dilation classification remains valid—but requires careful tracking of trace-class regularity for the deficit operators. Second, the product rule of Proposition 6.1 supplies closed-form families in dimension 2^k with additive indivisibility tensor; higher-dimensional families that are not products, and the scaling of the subspaces $V_{\text{in}}, V_{\text{out}}$ with the dimension n for them, are not treated. Third, an

EDMD-type [4, 7] numerical estimator for $\Delta(t,s,r)$ from finite trajectory data would convert the obstruction theorem of Corollary 4.6 into a computational test for indivisibility, complementing the RHP and BLP-style numerical procedures developed in the open-quantum-systems literature [43, 42, 45]. Fourth, the constant $1/3$ of Corollary 4.6 cannot be improved over the class of all divisible comparison families (Proposition 4.7), and it is approached by the extremal families exhibited in Proposition 4.7, whose intermediate kernels are deterministic; whether it can be improved on subclasses, for instance families whose intermediate kernels are bounded away from deterministic, is open. Fifth, the fixed-dimension continuum cocycle question of Theorem 5.4(iii) is open in both directions: neither the existence of a fixed-dimension continuum cocycle dilation for divisible families, nor divisibility as a consequence of one, is established.

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A Proof of Theorem 4.5

We prove parts (a) (deficit concentration) and (b) (obstruction to divisible approximation) of Theorem 4.5. Part (a) uses only boundedness and compactness of the two-time operators; part (b), from Step 3 onward, additionally invokes Assumption 3.3 for both families, and the steps are flagged accordingly.

Step 1: Construction of V_{in} and V_{out}. For each triple (s,r,t) with $0 \le s < r < t \le T$, the deficit operator $\Delta(t,s,r) \in \mathcal{B}(L^2(\Omega,\mu))$ is bounded with operator norm $\|\Delta(t,s,r)\|_{\mathrm{op}} = D(t,s,r)$. Since each $\mathcal{P}(t,s)$ is compact for $t > s$ and bounded, and compactness is preserved under composition with bounded operators and under differences, $\Delta(t,s,r)$ is compact, and so is its adjoint $\Delta(t,s,r)^*$. The sets

$$V_{\mathrm{in}} := \overline{\operatorname{span} \bigcup_{0\le s<r<t\le T} \operatorname{Range}\big(\Delta(t,s,r)^*\big)}, \qquad V_{\mathrm{out}} := \overline{\operatorname{span} \bigcup_{0\le s<r<t\le T} \operatorname{Range}\big(\Delta(t,s,r)\big)}$$

are closed subspaces of $L^2(\Omega,\mu)$.

Step 2: Deficit concentration on V_{in} and V_{out} (part (a)). For divisible families, every $\Delta(t,s,r) = 0$ (Definition 2.4), so $V_{\mathrm{in}} = V_{\mathrm{out}} = \{0\}$. For indivisible families, there exists at least one triple (s_0,r_0,t_0) with $\Delta(t_0,s_0,r_0) \neq 0$, so $\operatorname{Range}(\Delta(t_0,s_0,r_0)^*)$ and $\operatorname{Range}(\Delta(t_0,s_0,r_0))$ both have positive dimension, and so do V_{in} and V_{out}.

For each fixed triple (s,r,t) with $D(t,s,r)>0$, compactness of $\Delta(t,s,r)$ gives the discrete Schmidt decomposition

$$\Delta(t,s,r) \;=\; \sum_{n\geq 1} \sigma_n(t,s,r)\;\langle e_n(t,s,r),\,\cdot\rangle\; f_n(t,s,r),$$

where $\{e_n\}$ and $\{f_n\}$ are orthonormal systems in $L^2(\Omega,\mu)$ and $\sigma_1 \geq \sigma_2 \geq \cdots \geq 0$ are the singular values, with $\sigma_n \to 0$ and $\sigma_1 = \|\Delta(t,s,r)\|_{\mathrm{op}} = D(t,s,r)$ (see [9, Theorem A.4] or any standard reference on compact operators). For $\sigma_n > 0$, $e_n \in \mathrm{Range}(\Delta(t,s,r)^*) \subseteq V_{\mathrm{in}}$ and $f_n \in \mathrm{Range}(\Delta(t,s,r)) \subseteq V_{\mathrm{out}}$ by construction. The unit vector $f^*(t,s,r) := e_1(t,s,r) \in V_{\mathrm{in}}$ is therefore a leading right-singular vector with $\|\Delta(t,s,r)f^*(t,s,r)\|_{L^2} = D(t,s,r)$ and $\Delta(t,s,r)f^*(t,s,r) = D(t,s,r)\,f_1(t,s,r) \in V_{\mathrm{out}}$.

Finally, since $\mathrm{Range}(\Delta(t,s,r)^*) \subseteq V_{\mathrm{in}}$ and $\ker(\Delta(t,s,r)) = \mathrm{Range}(\Delta(t,s,r)^*)^{\perp}$ in $L^2(\Omega,\mu)$, we have $V_{\mathrm{in}}^{\perp} \subseteq \ker(\Delta(t,s,r))$, i.e., $\Delta(t,s,r)$ vanishes on $V_{\mathrm{in}}^{\perp}$. This establishes part (a).

Step 3: Triangle decomposition of the divisible-class error (part (b); Assumption 3.3 in force). Let $\{\tilde{\mathcal{P}}(t,s)\}$ be the operator family of any divisible two-time transition family on Ω, with both families satisfying Assumption 3.3. By divisibility, $\tilde{\mathcal{P}}(t,s) = \tilde{\mathcal{P}}(t,r)\,\tilde{\mathcal{P}}(r,s)$ for any intermediate $r \in (s,t)$. We decompose

$$\begin{aligned}\mathcal{P}(t,s) - \tilde{\mathcal{P}}(t,s) \;&=\; \big[\mathcal{P}(t,s) - \mathcal{P}(t,r)\mathcal{P}(r,s)\big] + \big[\mathcal{P}(t,r)\mathcal{P}(r,s) - \tilde{\mathcal{P}}(t,r)\tilde{\mathcal{P}}(r,s)\big]\\ &=\; \Delta(t,s,r) + \mathcal{P}(t,r)\big[\mathcal{P}(r,s) - \tilde{\mathcal{P}}(r,s)\big] + \big[\mathcal{P}(t,r) - \tilde{\mathcal{P}}(t,r)\big]\tilde{\mathcal{P}}(r,s).\end{aligned}$$

For $f \in V_{\mathrm{in}}$ with $\|f\| = 1$, the reverse triangle inequality together with the contraction bounds of Assumption 3.3 ($\|\mathcal{P}(t,r)\|_{\mathrm{op}} \leq 1$ on the second term, $\big\|\tilde{\mathcal{P}}(r,s)\big\|_{\mathrm{op}} \leq 1$ on the third) gives

$$\Big\|[\mathcal{P}(t,s) - \tilde{\mathcal{P}}(t,s)]f\Big\|_{L^2} \;\geq\; \|\Delta(t,s,r)f\|_{L^2} - \delta(r,s;\tilde{\mathcal{P}}) - \delta(t,r;\tilde{\mathcal{P}}),$$

where $\delta(\tau_1,\tau_2;\tilde{\mathcal{P}}) = \Big\|\mathcal{P}(\tau_1,\tau_2) - \tilde{\mathcal{P}}(\tau_1,\tau_2)\Big\|_{\mathrm{op}}$. Without Assumption 3.3, the same computation carries the operator-norm factors recorded in the unnumbered remark following Corollary 4.6.

Step 4: Optimal choice of f and supremum over r. By Step 2, for each triple (t,s,r) with $D(t,s,r)>0$ there exists a unit vector $f^*(t,s,r) \in V_{\mathrm{in}}$ with $\|\Delta(t,s,r)f^*(t,s,r)\|_{L^2} = D(t,s,r)$. Substituting $f = f^*(t,s,r)$ in Step 3 and taking the supremum over $f \in V_{\mathrm{in}}$ with $\|f\| = 1$:

$$\sup_{\substack{f\in V_{\mathrm{in}}\\ \|f\|=1}} \Big\|[\mathcal{P}(t,s) - \tilde{\mathcal{P}}(t,s)]f\Big\|_{L^2} \;\geq\; D(t,s,r) - \delta(t,r;\tilde{\mathcal{P}}) - \delta(r,s;\tilde{\mathcal{P}}).$$

Taking the supremum over r yields (4), which is part (b).

Step 5: Quantitative obstruction to uniform divisible approximation (Assumption 3.3 in force). Fix a triple (t,s,r) with $0 \leq s < r < t \leq T$. For any divisible comparison family $\{\tilde{\mathcal{P}}(\tau_1,\tau_2)\}$, the divisibility identity $\tilde{\mathcal{P}}(t,s) = \tilde{\mathcal{P}}(t,r)\,\tilde{\mathcal{P}}(r,s)$ combined with the triangle inequality through $\tilde{\mathcal{P}}$ gives

$$\begin{aligned}D(t,s,r) \;&=\; \|\mathcal{P}(t,s) - \mathcal{P}(t,r)\mathcal{P}(r,s)\|_{\mathrm{op}}\\ &\leq\; \Big\|\mathcal{P}(t,s) - \tilde{\mathcal{P}}(t,s)\Big\|_{\mathrm{op}} + \underbrace{\Big\|\tilde{\mathcal{P}}(t,s) - \tilde{\mathcal{P}}(t,r)\tilde{\mathcal{P}}(r,s)\Big\|_{\mathrm{op}}}_{=0 \text{ by divisibility}} + \Big\|\tilde{\mathcal{P}}(t,r)\tilde{\mathcal{P}}(r,s) - \mathcal{P}(t,r)\mathcal{P}(r,s)\Big\|_{\mathrm{op}}\\ &\leq\; \delta(t,s;\tilde{\mathcal{P}}) + 0 + \delta(t,r;\tilde{\mathcal{P}}) + \delta(r,s;\tilde{\mathcal{P}}),\end{aligned}$$

where the third operator-norm difference is bounded using the contraction bounds of Assumption 3.3 on the factors $\tilde{\mathcal{P}}(r,s)$ and $\mathcal{P}(t,r)$:

$$\left\|\tilde{\mathcal{P}}(t,r)\tilde{\mathcal{P}}(r,s)-\mathcal{P}(t,r)\mathcal{P}(r,s)\right\|_{\mathrm{op}} \le \left\|\tilde{\mathcal{P}}(t,r)-\mathcal{P}(t,r)\right\|\left\|\tilde{\mathcal{P}}(r,s)\right\|+\|\mathcal{P}(t,r)\|\left\|\tilde{\mathcal{P}}(r,s)-\mathcal{P}(r,s)\right\| \le \delta(t,r;\tilde{\mathcal{P}})+\delta($$

Hence for any divisible comparison family,

$$D(t,s,r) \;\le\; \delta(t,s;\tilde{\mathcal{P}})+\delta(t,r;\tilde{\mathcal{P}})+\delta(r,s;\tilde{\mathcal{P}}) \;\le\; 3\sup_{(\tau_1,\tau_2)}\delta(\tau_1,\tau_2;\tilde{\mathcal{P}}). \tag{20}$$

The contrapositive form of (20) is the quantitative obstruction asserted in part (b): at any triple (t,s,r) with $D(t,s,r)>0$, every divisible $\tilde{\mathcal{P}}$ satisfies $\sup_{(\tau_1,\tau_2)}\delta(\tau_1,\tau_2;\tilde{\mathcal{P}}) \ge D(t,s,r)/3$, so the operator-norm distance between $\mathcal{P}$ and $\tilde{\mathcal{P}}$ at the worst single time pair is at least $D(t,s,r)/3>0$. The constant $1/3$ is universal over divisible comparison families and cannot be improved (Proposition 4.7). The bound (4) on the action over V_{in} at the time pair (t,s) is non-trivial precisely on those triples for which $\delta(t,r;\tilde{\mathcal{P}})+\delta(r,s;\tilde{\mathcal{P}})<D(t,s,r)$. □

B Proof of Theorem 4.8

We establish the gauge-relative matrix-element decomposition (7) and its consequences (a)–(c).

Step 1: Matrix elements. For finite $\Omega=\{1,\dots,n\}$ with counting measure, $L^2(\Omega,\mu)\cong\mathbb{C}^n$, and the transition operator is the stochastic matrix itself: $\mathcal{P}(t,s)_{ij}=\Gamma_{ij}(t\leftarrow s)=|U_{ij}(t,s)|^2$ in the selected gauge. Matrix multiplication gives

$$\big[\mathcal{P}(t,r)\mathcal{P}(r,s)\big]_{ij}=\sum_z |U_{iz}(t,r)|^2\,|U_{zj}(r,s)|^2.$$

Step 2: Squared modulus of the composite unitary. With $V(t,s;r):=U(t,r)U(r,s)$, the squared modulus is

$$\begin{aligned}|V_{ij}(t,s;r)|^2 &= \Big|\sum_z U_{iz}(t,r)U_{zj}(r,s)\Big|^2\\ &= \sum_z |U_{iz}(t,r)|^2|U_{zj}(r,s)|^2+\sum_{z\neq w}U_{iz}(t,r)U_{zj}(r,s)\overline{U_{iw}(t,r)\,U_{wj}(r,s)}.\end{aligned}$$

The first sum equals $[\mathcal{P}(t,r)\mathcal{P}(r,s)]_{ij}$; the second sum is $\mathfrak{C}(t,s,r)_{ij}$ (8).

Step 3: The identity (part (a)). The deficit has matrix elements

$$\Delta(t,s,r)_{ij}=|U_{ij}(t,s)|^2-\sum_z|U_{iz}(t,r)|^2|U_{zj}(r,s)|^2.$$

Adding and subtracting $|V_{ij}(t,s;r)|^2$:

$$\begin{aligned}\Delta(t,s,r)_{ij} &= \big[|U_{ij}(t,s)|^2-|V_{ij}(t,s;r)|^2\big]+\Big[|V_{ij}(t,s;r)|^2-\sum_z|U_{iz}(t,r)|^2|U_{zj}(r,s)|^2\Big]\\ &= \mathfrak{D}(t,s,r)_{ij}+\mathfrak{C}(t,s,r)_{ij}\end{aligned}$$

by Step 2. This establishes (7). The computation used only the squared-modulus relations $\Gamma_{ij}(\cdot)=|U_{ij}(\cdot)|^2$ at the three time pairs; it holds for every gauge selection, which is part (a).

Step 4: The equivalence (part (b)). CK at the triple (t,s,r) is by definition the statement $\Delta(t,s,r)=0$, so part (a) gives immediately: CK at $(t,s,r) \iff \mathfrak{C}(t,s,r)+\mathfrak{D}(t,s,r)=0$. Nothing stronger is available: the identity constrains only the sum.

Step 5: Gauge dependence of the individual terms (part (c)). The counterexamples cited in the statement establish both halves. (i) *A perfect cocycle with maximal CK violation carried by $\mathfrak{C}$ alone:* for the indivisible triple $\Gamma(t,s)=X$, $\Gamma(t,r)=\Gamma(r,s)=F$, the gauge $(U(t,s),U(t,r),U(r,s))=(R(90°),R(45°),R(45°))$ satisfies $U(t,s)=V(t,s;r)$ exactly, so $\mathfrak{D}=0$, while $\mathfrak{C}=|V|^2-F^2=X-F=\Delta\neq 0$ (Remark 2.12). (ii) *A divisible triple whose terms are individually nonzero in one gauge and individually zero in another:* for $\Gamma(t,s)=\Gamma(t,r)=\Gamma(r,s)=F$ (divisible, $F^2=F$), the gauge $(H,R(45°),R(45°))$ gives $\mathfrak{D}=F-X\neq 0$ and $\mathfrak{C}=X-F\neq 0$ with $\mathfrak{C}+\mathfrak{D}=0$, while the gauge (KH,K,H) gives $|V|^2=|KH|^2=F$, hence $\mathfrak{D}=\mathfrak{C}=0$ (Remark 4.9). Individual vanishing of either term is therefore a property of the selected lifts, not of the family. □

C Proof of Theorem 5.4

We prove the four assertions of Theorem 5.4. The proof is organised so that each assertion is established under the hypotheses stated in the theorem and no stronger; auxiliary regularity is invoked only where it is needed, and explicitly so.

Step 1a: Kraus representation, Stinespring isometry, and reproduction (assertion (i)). For each fixed $(t,s)\in[0,T]^2_{\leq}$, the diagonal-retention lift $\Phi(t,s)$ of Definition 5.1 is CPTP: complete positivity follows from the Kraus form, and trace preservation from

$$K_0^*K_0+\sum_{i\neq j}K_{(i,j)}^*K_{(i,j)} \;=\; \sum_j \Gamma_{jj}\,|j\rangle\langle j| + \sum_j\Big(\sum_{i\neq j}\Gamma_{ij}\Big)|j\rangle\langle j| \;=\; \sum_j\Big(\sum_i\Gamma_{ij}\Big)|j\rangle\langle j| \;=\; I$$

by column-stochasticity. The Kraus operators depend continuously on (t,s) via the (continuous) square-root function applied to the continuous family $\{\Gamma_{ij}(t\leftarrow s)\}$ ((R2)).

Set $\mathcal{K}=\mathbb{C}^{n^2-n+1}$ with orthonormal basis $\{|0\rangle\}\cup\{|i,j\rangle : i\neq j,\ i,j\in\{1,\dots,n\}\}$, the label $|0\rangle$ carrying the collective diagonal Kraus index and the labels $|i,j\rangle$ the off-diagonal indices. Define the Stinespring isometry $V(t,s)\colon \mathcal{H}\to\mathcal{H}\otimes\mathcal{K}$ by

$$V(t,s)\,|j\rangle \;=\; \sqrt{\Gamma_{jj}(t\leftarrow s)}\,|j\rangle\otimes|0\rangle \;+\; \sum_{i\neq j}\sqrt{\Gamma_{ij}(t\leftarrow s)}\,|i\rangle\otimes|i,j\rangle. \tag{21}$$

Isometry: for $j\neq j'$ every term of $\langle V(t,s)j', V(t,s)j\rangle$ pairs orthogonal vectors (the $|0\rangle$-components carry orthogonal system factors $|j\rangle\perp|j'\rangle$; the environment labels $|i,j\rangle$ and $|i',j'\rangle$ are distinct whenever $j\neq j'$; and $\langle 0|i,j\rangle=0$), while $\|V(t,s)|j\rangle\|^2=\Gamma_{jj}+\sum_{i\neq j}\Gamma_{ij}=1$. Reading off the Kraus operators of the dilation, $(I_{\mathcal{H}}\otimes\langle 0|)\,V(t,s)=K_0(t,s)$ and $(I_{\mathcal{H}}\otimes\langle i,j|)\,V(t,s)=K_{(i,j)}(t,s)$, so

$$\mathrm{Tr}_{\mathcal{K}}\big(V(t,s)\,\rho\,V(t,s)^*\big) \;=\; K_0\rho K_0^* + \sum_{i\neq j}K_{(i,j)}\rho K_{(i,j)}^* \;=\; \Phi(t,s)(\rho),$$

the full-channel reproduction (10) for any unitary completion agreeing with V on $\mathcal{H}\otimes|0\rangle$. The transition-statistic reproduction (11) follows in the lifted form

$$\sum_k\big|\langle i|\otimes\langle k|V(t,s)|j\rangle\big|^2 \;=\; \Gamma_{jj}\,\delta_{ij}+\mathbf{1}_{\{i\neq j\}}\,\Gamma_{ij} \;=\; \Gamma_{ij}(t\leftarrow s).$$

Finally, the boundary: (R1) gives $\Gamma(s \leftarrow s) = I$, hence $V(s,s)|j\rangle = |j\rangle \otimes |0\rangle$ for every j—the canonical input embedding, exactly and uniformly in j. (This is the point of the diagonal-retention indexing: routing all diagonal transitions through the single environment vector $|0\rangle$ makes the isometry collapse to the input embedding at $t = s$, which is what a continuous unitary completion with $W(s,s) = I_{\mathcal{H}'}$ requires.)

Step 1b: Strongly continuous unitary completion with $W(s,s) = I$ under (R1)–(R2). Write $N := \dim(\mathcal{H} \otimes \mathcal{K}) = n(n^2 - n + 1)$. The image $V(t,s)(\mathcal{H})$ is an n-dimensional subspace of $\mathcal{H} \otimes \mathcal{K}$. The orthogonal projection $P(t,s) := V(t,s)V(t,s)^*$ is continuous in (t,s) and has rank n uniformly on $[0,T]^2_{\leq}$: off the diagonal by isometry of $V(t,s)$, and at the diagonal because Step 1a gives $V(s,s)|j\rangle = |j\rangle \otimes |0\rangle$, so $P(s,s) = P_0$, the projection onto $\mathcal{H} \otimes |0\rangle$. Thus the complementary projection $I - P(t,s)$ has constant rank $N - n$ throughout $[0,T]^2_{\leq}$ and determines a continuous rank-$(N-n)$ subbundle of the trivial bundle $[0,T]^2_{\leq} \times (\mathcal{H} \otimes \mathcal{K})$. The associated Stiefel bundle of orthonormal $(N-n)$-frames in $\mathrm{Range}(I - P(t,s))$ is a fibre bundle over the contractible base $[0,T]^2_{\leq}$. Any fibre bundle over a contractible paracompact base is trivial [22, §11], hence admits a continuous global section, equivalently a continuous orthonormal frame; alternatively, a continuous frame is supplied by Kato's perturbation framework [50, §II.6] applied to the C^0 family of projections $\{I - P(t,s)\}$. The dimension-independent continuity bound of Kretschmann, Schlingemann, and Werner [39] additionally shows that, since the diagonal-retention lift $\Phi(t,s)$ is continuous in cb-norm on $[0,T]^2_{\leq}$ (under (R2)), Stinespring isometries can be chosen close in operator norm whenever the underlying CPTP maps are close in cb-norm; this gives a quantitative continuity modulus for the construction.

Fix such a continuous frame $\{f_\alpha(t,s)\}_{\alpha=1}^{N-n}$ of $\mathrm{Range}(I - P(t,s))$ and the reference frame $\{e_\alpha\}_{\alpha=1}^{N-n}$, an orthonormal basis of $(\mathcal{H} \otimes |0\rangle)^\perp$. The frame must additionally be pinned to the reference frame on the diagonal, and this can always be arranged: since $P(s,s) = P_0$ for every s, both $\{f_\alpha(s,s)\}$ and $\{e_\alpha\}$ are orthonormal frames of the *same* space $\mathrm{Range}(I - P_0)$, so there is a unitary $(N-n) \times (N-n)$ matrix $G(s)$, continuous in s, with $f_\alpha(s,s) = \sum_\beta G_{\beta\alpha}(s)\, e_\beta$. Replace the frame by its rotation

$$f'_\alpha(t,s) \;:=\; \sum_\beta f_\beta(t,s)\, \big[G(s)^{-1}\big]_{\beta\alpha},$$

extending $G(s)^{-1}$ constantly in t; the rotated family is again a continuous orthonormal frame of $\mathrm{Range}(I - P(t,s))$, and now $f'_\alpha(s,s) = e_\alpha$ exactly. Define $W(t,s)$ on $\mathcal{H} \otimes \mathcal{K}$ by

- $W(t,s)\,(|j\rangle \otimes |0\rangle) := V(t,s)\,|j\rangle$ for $j = 1, \ldots, n$;
- $W(t,s)\, e_\alpha := f'_\alpha(t,s)$ for $\alpha = 1, \ldots, N-n$.

By construction $W(t,s)$ is unitary and agrees with $V(t,s)$ on the input subspace $\mathcal{H} \otimes |0\rangle$, so $\mathrm{Tr}_{\mathcal{K}}[W(t,s)(\rho \otimes |0\rangle\langle 0|)W(t,s)^*] = \mathrm{Tr}_{\mathcal{K}}[V(t,s)\rho V(t,s)^*] = \Phi(t,s)(\rho)$, which is (10); it is strongly continuous in (t,s); and the boundary condition holds exactly: at $t = s$, the input part is the canonical embedding ($V(s,s)|j\rangle = |j\rangle \otimes |0\rangle$) and the complement part fixes each e_α, so $W(s,s) = I_{\mathcal{H}'}$. We note for completeness that strong convergence of Stinespring isometries does not in general lift to strong convergence of unitary completions (Shirokov [40]); the construction above produces a particular continuous lift via the constant-rank projection family and the diagonal pinning, and it is this specific lift, not an arbitrary completion, on which the joint continuity is established. This establishes (i).

Step 2: Formal Hamiltonian generator under (R3′) (assertion (ii)). We now invoke hypothesis (R3′): for each fixed s, each map $t \mapsto \sqrt{\Gamma_{ij}(t \leftarrow s)}$ is absolutely continuous on

$[s,T]$ *by assumption.* (Absolute continuity must be assumed rather than derived from almost-everywhere differentiability with locally integrable derivative: singular Cantor-type functions are a.e.-differentiable with integrable derivative yet are not recovered by integrating it.) The Stinespring isometry $V(t,s)$ defined by (21) is therefore absolutely continuous in t for each fixed s (a finite linear combination of absolutely continuous entries), with $V(s,s)$ the canonical embedding, and the projection $P(t) := V(t,s)V(t,s)^*$ is absolutely continuous in t with $\partial_t P \in L^1([s,T];\mathcal{B}(\mathcal{H}'))$, by the product rule for absolutely continuous matrix functions and $\|V\| = 1$.

Kato parallel transport. The absolutely continuous unitary completion is supplied by parallel transport of the projection family rather than by an abstract frame selection. For the fixed s, solve the linear Carathéodory problem

$$\partial_t U(t) \;=\; \big[\partial_t P(t),\, P(t)\big]\, U(t), \qquad U(s) = I_{\mathcal{H}'},$$

on $[s,T]$. The coefficient lies in $L^1([s,T];\mathcal{B}(\mathcal{H}'))$, so a unique absolutely continuous solution exists (Picard iteration; the Dyson series converges absolutely). The coefficient is skew-Hermitian a.e. ($\partial_t P$ is Hermitian a.e.), so $\partial_t(U^*U) = 0$ a.e. and $U(t)$ is unitary for every t. Differentiating $P^2 = P$ gives $\partial_t P = (\partial_t P)P + P(\partial_t P)$ a.e.; multiplying this identity by P on both sides gives $P(\partial_t P)P = 0$ a.e.; and a direct computation then yields $[[\partial_t P, P], P] = (\partial_t P)P + P(\partial_t P) - 2P(\partial_t P)P = \partial_t P$ a.e. Consequently

$$\partial_t\big(U^* P\, U\big) \;=\; U^*\big(\partial_t P - [[\partial_t P, P], P]\big)U \;=\; 0 \quad \text{a.e.},$$

so $U(t)P(s)U(t)^* = P(t)$ for a.e. t, hence for all t by continuity: the transport intertwines the projections, and in particular $U(t)$ carries $\mathrm{Range}(I - P(s)) = (\mathcal{H} \otimes |0\rangle)^\perp$ onto $\mathrm{Range}(I - P(t))$.

Fix an orthonormal basis $\{e_\alpha\}$ of $(\mathcal{H} \otimes |0\rangle)^\perp$ and define $W(t,s)$ by

$$W(t,s)\,(|j\rangle \otimes |0\rangle) := V(t,s)\,|j\rangle, \qquad W(t,s)\,e_\alpha := U(t)\,e_\alpha.$$

The two blocks are orthonormal images landing in $\mathrm{Range}\, P(t)$ and $\mathrm{Range}(I - P(t))$ respectively, so $W(t,s)$ is unitary; it agrees with $V(t,s)$ on the input subspace, hence satisfies (10); it is absolutely continuous in t (both blocks are); and $W(s,s) = I_{\mathcal{H}'}$ since $V(s,s)$ is the canonical embedding and $U(s) = I_{\mathcal{H}'}$.

Define

$$\widehat{H}_s(t) := i\,(\partial_t W(t,s))\, W(t,s)^{-1},$$

well defined for almost every t, strongly measurable, and integrable in operator norm on $[s,T]$ (since $\partial_t W \in L^1$ and W is unitary); it need not be essentially bounded (Remark 5.8). Differentiating $W^*W = I$ a.e. yields $(\partial_t W)^* W + W^* \partial_t W = 0$, hence $W^* \partial_t W$ is anti-Hermitian a.e., so $\widehat{H}_s(t) = W(iW^*\partial_t W)W^*$ is Hermitian a.e. The relation $i\,\partial_t W(t,s) = \widehat{H}_s(t)\, W(t,s)$ then holds a.e. in the sense of absolute continuity, with $W(s,s) = I_{\mathcal{H}'}$. This establishes (ii).

Step 3: Grid cocycle dilation under divisibility (assertion (iii)). Suppose $\{\Gamma(t \leftarrow s)\}$ is divisible and fix a grid $\pi\colon s = t_0 < t_1 < \cdots < t_K = t$. For each increment (t_k, t_{k-1}), Steps 1a–1b produce an environment $\mathcal{K}_k$ with $\dim \mathcal{K}_k = n^2 - n + 1 \le n^2$, a reference vector $|0_k\rangle$, and a unitary W_k on $\mathcal{H} \otimes \mathcal{K}_k$ dilating the increment channel $\Phi(t_k, t_{k-1})$. Embed each as $\widetilde{W}_k$ acting on $\mathcal{H} \otimes \mathcal{K}_1 \otimes \cdots \otimes \mathcal{K}_K$ as W_k on $\mathcal{H} \otimes \mathcal{K}_k$ and as the identity on the remaining factors, and set $W_\pi(t_m, t_l) := \widetilde{W}_m \cdots \widetilde{W}_{l+1}$.

Cocycle. The law $W_\pi(t_m, t_l) = W_\pi(t_m, t_k)\, W_\pi(t_k, t_l)$ for $l \le k \le m$ holds by construction: the two sides are literally the same ordered product. The allocation of a fresh factor to each increment is what makes the products compose without interference.

Dilation of the composed channel. With all environments initialized in their reference vectors, induction on $m - l$ using the composition rule for Stinespring dilations [18, Ch. 4] gives

$$\mathrm{Tr}_{\mathcal{K}_1 \otimes \cdots \otimes \mathcal{K}_K}\big[\, W_\pi(t_m, t_l)\,\big(\rho \otimes |0_1 \cdots 0_K\rangle\langle 0_1 \cdots 0_K|\big)\, W_\pi(t_m, t_l)^*\,\big] \;=\; \big(\Phi(t_m, t_{m-1}) \circ \cdots \circ \Phi(t_{l+1}, t_l)\big)(\rho)\,:$$

when $\widetilde{W}_k$ is applied, its factor $\mathcal{K}_k$ is still in the reference state (no earlier unitary touched it), so tracing out $\mathcal{K}_k$ implements $\Phi(t_k, t_{k-1})$ on the system while the remaining factors are spectators.

Transition-statistic reproduction. Populations propagate autonomously through each increment channel—by Definition 5.1, the population vector of $\Phi(\rho)$ is Γ applied to the population vector of ρ—so the composed channel has population action $\Gamma(t_m \leftarrow t_{m-1}) \cdots \Gamma(t_{l+1} \leftarrow t_l) = \Gamma(t_m \leftarrow t_l)$ by divisibility. Reading off diagonal entries on basis-state inputs gives (11) for $W_\pi(t_m, t_l)$ with initial environment vector $|0_1 \cdots 0_K\rangle$.

Why the statement is at grid level. A fixed-environment continuum cocycle would require applying a second increment unitary to the *same* environment factor; after the first application that factor is no longer in its reference state (it is in general entangled with the system), so the second application does not implement its increment channel. Padding a doubled environment $\mathcal{K}_2 \otimes \mathcal{K}_1$ back into a fixed n^2-dimensional space destroys exactly the freshness the composition rule requires. Both continuum directions—existence of a fixed-dimension cocycle lift for divisible families, and divisibility as a consequence of one—are open, as stated in (iii).

Step 4: Uniqueness at the level of minimal dilations (assertion (iv)). For each fixed (t, s), Stinespring's uniqueness theorem [18, Theorem 4.2] states that any two *minimal* Stinespring dilations of the same completely positive map are unitarily equivalent; in the present finite-dimensional form, if $V_a \colon \mathcal{H} \to \mathcal{H} \otimes \mathcal{K}_a$ and $V_b \colon \mathcal{H} \to \mathcal{H} \otimes \mathcal{K}_b$ are minimal Stinespring isometries for $\Phi(t, s)$, there is a unitary $u \colon \mathcal{K}_a \to \mathcal{K}_b$ with $(I_{\mathcal{H}} \otimes u)\, V_a = V_b$. (The isometry of Step 1a is minimal at each pair (t, s) after deleting the environment directions of the Kraus operators that vanish *at that pair*: the nonzero operators $\{K_0(t, s)\} \cup \{K_{(i,j)}(t, s) : \Gamma_{ij}(t \leftarrow s) > 0\}$ are linearly independent, having disjoint matrix supports up to the diagonal block carried by K_0 alone. The deleted set is in general (t, s)-dependent; under (R3′), the dichotomy—each off-diagonal entry identically zero or strictly positive on $\{s < t\}$—makes it constant across the open region $s < t$.) Two points delimit the scope of the assertion. First, uniqueness pertains to the *isometries*: a unitary completion $W(t, s)$ of a given isometry is determined only up to an arbitrary unitary rotation of the orthogonal complement of the input subspace $\mathcal{H} \otimes |0\rangle$, and the pinned completion of Step 1b is one choice among these; no equivalence between arbitrary completions is asserted. Second, uniqueness applies to dilations of the same *channel*: families of unitaries satisfying only the basis-input relation (11) may dilate different channels and then admit no equivalence at all (Remark 5.3); this is why Definition 5.2 is stated at the channel level. Whether the minimal-dilation equivalences $u(t, s)$ can be chosen jointly continuous in (t, s) is a continuous-selection question over the bundle of equivalence unitaries between parameter-dependent dilations; we do not establish it here, and the theorem statement reflects the pointwise version. □

D Proof of Theorem 7.4

We first record the kernel-recovery lemma that converts operator-level vanishing into pointwise kernel vanishing; it is the ingredient that makes the pointwise formulation in (ii) legitimate, since for general bounded integral operators equality of induced operators does not by itself control the kernel representatives.

Lemma D.1 (Kernel recovery)**.** *Let (Ω, Σ, μ) be σ-finite with Σ countably generated, and let $K_1, K_2 \colon \Omega \times \Omega \to [0, \infty]$ be measurable kernels whose induced integral operators $(T_i g)(x) = \int_\Omega K_i(x, y)\, g(y)\, d\mu(y)$ are bounded on $L^2(\Omega, \mu)$. If $T_1 = T_2$, then $K_1 = K_2$ $\mu \otimes \mu$-a.e.*

Proof. Since Σ is countably generated and μ is σ-finite, there is a countable ring $\mathcal{R}_0$ of finite-μ-measure sets generating Σ and containing an exhausting sequence $\Omega = \bigcup_k B_k$. For $B \in \mathcal{R}_0$ we have $\mathbf{1}_B \in L^2$, so by Tonelli $(T_i \mathbf{1}_B)(x) = \int_B K_i(x, y)\, d\mu(y)$ is finite for μ-a.e. x, and $T_1 \mathbf{1}_B = T_2 \mathbf{1}_B$ in L^2 gives $\int_B K_1(x, y)\, d\mu(y) = \int_B K_2(x, y)\, d\mu(y)$ for μ-a.e. x. Discarding a single μ-null set of x (a countable union over $\mathcal{R}_0$), for every remaining x the two set functions $B \mapsto \int_B K_i(x, y)\, d\mu(y)$ are measures agreeing on the generating π-system $\mathcal{R}_0$, finite on $\mathcal{R}_0$, and σ-finite along the exhausting sequence; by the uniqueness theorem for measures they agree on Σ, so $K_1(x, \cdot) = K_2(x, \cdot)$ μ-a.e. (a.e.-uniqueness of densities of a σ-finite measure). Fubini then gives $K_1 = K_2$ $\mu \otimes \mu$-a.e. □

At each triple, the two nonnegative kernel terms of $\delta K_{\mathcal{R}}$ are $K_1 := P(t, \cdot \mid s, \cdot)$ and $K_2 := \int_\Omega P(t, \cdot \mid r, z) P(r, z \mid s, \cdot)\, d\mu(z)$; K_1 induces $\mathcal{P}(t, s)$, and by Tonelli K_2 induces $\mathcal{P}(t, r)\mathcal{P}(r, s)$, both bounded under (C1). Lemma D.1 therefore applies verbatim.

Step 0: Boundedness, continuity of D, and finiteness of J. Boundedness of $\mathcal{R}$ under (C1) is Lemma 7.2. Under (C1)–(C2), the map $(t, s, r) \mapsto \Delta(t, s, r) = \mathcal{P}(t, s) - \mathcal{P}(t, r)\mathcal{P}(r, s)$ is operator-norm continuous on the open simplex $\{0 \le s < r < t \le T\}$, which is all that the argument below uses: sums and products of norm-continuous, uniformly bounded operator families are norm-continuous ($\|\mathcal{P}(t, r)\mathcal{P}(r, s) - \mathcal{P}(t', r')\mathcal{P}(r', s')\| \le M \|\mathcal{P}(r, s) - \mathcal{P}(r', s')\| + M \|\mathcal{P}(t, r) - \mathcal{P}(t', r')\|$), and hence $D = \|\Delta\|_{\mathrm{op}}$ is continuous. Moreover $D(t, s, r) \le \|\mathcal{P}(t, s)\| + \|\mathcal{P}(t, r)\|\, \|\mathcal{P}(r, s)\| \le M + M^2$, so

$$J(\mathcal{R}) \;\le\; (M + M^2)^2 \cdot \mathrm{vol}\{0 \le s < r < t \le T\} \;=\; (M + M^2)^2\, T^3/6 \;<\; \infty.$$

Step 1: Identity of $\delta K_{\mathcal{R}}$ with the kernel-level CK deficit. By Definition 7.1, $K_{\mathcal{R}}(t, x; s, y) = \mathbf{1}_{\{s \le t\}} P(t, x \mid s, y)$. For $0 \le s < r < t \le T$:

$$\delta K_{\mathcal{R}}(t, x; s, y; r) \;=\; P(t, x \mid s, y) - \int_\Omega P(t, x \mid r, z)\, P(r, z \mid s, y)\, d\mu(z),$$

the kernel-level CK deficit. By Definition 2.4, $D(t, s, r) = \|\delta K_{\mathcal{R}}(t, \cdot; s, \cdot; r)\|_{\mathrm{op}}$.

Step 2: Equivalence (i) ⇔ (ii) ⇔ (iii). (i)⇒(ii): divisibility is by Definition 2.4 the kernel identity (2) at every triple, so $\delta K_{\mathcal{R}}$ vanishes $\mu \otimes \mu$-a.e. at every triple and a fortiori induces the zero operator: $\Delta(t, s, r) = 0$ and $D(t, s, r) = 0$ everywhere. (ii)⇒(i): if $\delta K_{\mathcal{R}}$ induces the zero operator at a triple, Lemma D.1 applied to its two nonnegative kernel terms yields $\delta K_{\mathcal{R}} = 0$ $\mu \otimes \mu$-a.e. at that triple, which is (2) there; over all triples this is divisibility. (ii)⇔(iii): D is continuous (Step 0), and the integrated deficit $J(\mathcal{R}) = \iiint \mathbf{1}_{\{s<r<t\}} D^2\, ds\, dr\, dt$ is the integral of a non-negative continuous function on a bounded domain, so it vanishes iff $D \equiv 0$ on the open simplex — a.e. vanishing of D upgrades to identical vanishing because a nonzero value of the continuous D would persist on an open neighborhood of positive measure.

Step 3: Strict positivity for indivisible families. For an indivisible family there exists a triple (t_0, s_0, r_0) with $D(t_0, s_0, r_0) > 0$. By continuity of D (Step 0), there is a neighborhood of this triple on which $D > c > 0$, and integrating gives $J(\mathcal{R}) > 0$. □

References

[1] Koopman, B.O.: Hamiltonian systems and transformations in Hilbert space. Proc. Natl. Acad. Sci. USA **17**(5), 315–318 (1931)

[2] Schmid, P.J.: Dynamic mode decomposition of numerical and experimental data. J. Fluid Mech. **656**, 5–28 (2010)

[3] Rowley, C.W., Mezić, I., Bagheri, S., Schlatter, P., Henningson, D.S.: Spectral analysis of nonlinear flows. J. Fluid Mech. **641**, 115–127 (2009)

[4] Williams, M.O., Kevrekidis, I.G., Rowley, C.W.: A data-driven approximation of the Koopman operator: extending dynamic mode decomposition. J. Nonlinear Sci. **25**, 1307–1346 (2015)

[5] Mezić, I.: Spectral properties of dynamical systems, model reduction and decompositions. Nonlinear Dyn. **41**, 309–325 (2005)

[6] Budišić, M., Mohr, R., Mezić, I.: Applied Koopmanism. Chaos **22**, 047510 (2012)

[7] Klus, S., Nüske, F., Koltai, P., Wu, H., Kevrekidis, I., Schütte, C., Noé, F.: Data-driven model reduction and transfer operator approximation. J. Nonlinear Sci. **28**, 985–1010 (2018)

[8] Klus, S., Koltai, P., Schütte, C.: On the numerical approximation of the Perron–Frobenius and Koopman operator. J. Comput. Dyn. **3**(1), 51–79 (2016)

[9] Eisner, T., Farkas, B., Haase, M., Nagel, R.: Operator Theoretic Aspects of Ergodic Theory. Graduate Texts in Mathematics, vol. 272. Springer, Cham (2015)

[10] Lasota, A., Mackey, M.C.: Chaos, Fractals, and Noise: Stochastic Aspects of Dynamics, 2nd edn. Springer, New York (1994)

[11] Barandes, J.A.: The stochastic-quantum correspondence. Philos. Phys. **3**, 186 (2025)

[12] Barandes, J.A.: The stochastic-quantum theorem. arXiv:2309.03085 (2025)

[13] Bell, J.S.: On the Einstein Podolsky Rosen paradox. Physics **1**(3), 195–200 (1964)

[14] Fine, A.: Hidden variables, joint probability, and the Bell inequalities. Phys. Rev. Lett. **48**(5), 291–295 (1982)

[15] Kochen, S., Specker, E.P.: The problem of hidden variables in quantum mechanics. J. Math. Mech. **17**(1), 59–87 (1967)

[16] Summers, B.D.: The Physics of the Instant: Analysis from Limits, Relativity, and Renormalization. SSRN preprint, DOI 10.2139/ssrn.6922380 (first version March 2026; revised September 2026); earlier version at PhilSci-Archive, University of Pittsburgh, ID 28724, https://philsci-archive.pitt.edu/id/eprint/28724

[17] Stinespring, W.F.: Positive functions on C^*-algebras. Proc. Amer. Math. Soc. **6**(2), 211–216 (1955)

[18] Paulsen, V.: Completely Bounded Maps and Operator Algebras. Cambridge Studies in Advanced Mathematics, vol. 78. Cambridge University Press (2002)

[19] Nielsen, M.A., Chuang, I.L.: Quantum Computation and Quantum Information, 10th anniversary edn. Cambridge University Press (2010)

[20] Bengtsson, I., Ericsson, Å., Kuś, M., Tadej, W., Życzkowski, K.: Birkhoff's polytope and unistochastic matrices, $N = 3$ and $N = 4$. Comm. Math. Phys. **259**, 307–324 (2005)

[21] Bengtsson, I., Życzkowski, K.: Geometry of Quantum States: An Introduction to Quantum Entanglement, 2nd edn. Cambridge University Press (2017)

[22] Steenrod, N.E.: The Topology of Fibre Bundles. Princeton University Press, Princeton (1951)

[23] Mori, H.: Transport, collective motion, and Brownian motion. Prog. Theor. Phys. **33**(3), 423–455 (1965)

[24] Zwanzig, R.: Ensemble method in the theory of irreversibility. J. Chem. Phys. **33**(5), 1338–1341 (1960)

[25] Ephraim, Y., Merhav, N.: Hidden Markov processes. IEEE Trans. Inform. Theory **48**(6), 1518–1569 (2002)

[26] Breuer, H.-P., Laine, E.-M., Piilo, J., Vacchini, B.: Colloquium: Non-Markovian dynamics in open quantum systems. Rev. Mod. Phys. **88**, 021002 (2016)

[27] Mauroy, A., Mezić, I., Susuki, Y. (eds.): The Koopman Operator in Systems and Control: Concepts, Methodologies, and Applications. Lecture Notes in Control and Information Sciences, vol. 484. Springer, Cham (2020)

[28] Surana, A., Banaszuk, A.: Linear observer synthesis for nonlinear systems using Koopman operator framework. IFAC-PapersOnLine **49**(18), 716–723 (2016)

[29] Giannakis, D.: Data-driven spectral decomposition and forecasting of ergodic dynamical systems. Appl. Comput. Harmon. Anal. **47**(2), 338–396 (2019)

[30] Maćešić, S., Črnjarić-Žic, N., Mezić, I.: Koopman operator family spectrum for non-autonomous systems. SIAM J. Appl. Dyn. Syst. **17**(4), 2478–2515 (2018)

[31] Črnjarić-Žic, N., Maćešić, S., Mezić, I.: Koopman operator spectrum for random dynamical systems. J. Nonlinear Sci. **30**, 2007–2056 (2020)

[32] Froyland, G.: An analytic framework for identifying finite-time coherent sets in time-dependent dynamical systems. Physica D **250**, 1–19 (2013)

[33] Froyland, G., Lloyd, S., Santitissadeekorn, N.: Coherent sets for non-autonomous dynamical systems. Physica D **239**, 1527–1541 (2010)

[34] Froyland, G., Koltai, P.: Detecting the birth and death of finite-time coherent sets. Comm. Pure Appl. Math. **76**(12), 3642–3684 (2023)

[35] Pollock, F.A., Rodríguez-Rosario, C., Frauenheim, T., Paternostro, M., Modi, K.: Non-Markovian quantum processes: complete framework and efficient characterization. Phys. Rev. A **97**, 012127 (2018)

[36] Milz, S., Modi, K.: Quantum stochastic processes and quantum non-Markovian phenomena. PRX Quantum **2**, 030201 (2021)

[37] Costa, F., Shrapnel, S.: Quantum causal modelling. New J. Phys. **18**, 063032 (2016)

[38] Milz, S., Egloff, D., Taranto, P., Theurer, T., Plenio, M.B., Smirne, A., Huelga, S.F.: When is a non-Markovian quantum process classical? Phys. Rev. X **10**, 041049 (2020)

[39] Kretschmann, D., Schlingemann, D., Werner, R.F.: A continuity theorem for Stinespring's dilation. J. Funct. Anal. **255**(8), 1889–1904 (2008)

[40] Shirokov, M.E.: Strong convergence of quantum channels: continuity of the Stinespring dilation and discontinuity of the unitary dilation. J. Math. Phys. **61**, 082204 (2020)

[41] Wolf, M.M., Eisert, J., Cubitt, T.S., Cirac, J.I.: Assessing non-Markovian quantum dynamics. Phys. Rev. Lett. **101**, 150402 (2008)

[42] Breuer, H.-P., Laine, E.-M., Piilo, J.: Measure for the degree of non-Markovian behavior of quantum processes in open systems. Phys. Rev. Lett. **103**, 210401 (2009)

[43] Rivas, Á., Huelga, S.F., Plenio, M.B.: Entanglement and non-Markovianity of quantum evolutions. Phys. Rev. Lett. **105**, 050403 (2010)

[44] Rivas, Á., Huelga, S.F., Plenio, M.B.: Quantum non-Markovianity: characterization, quantification and detection. Rep. Prog. Phys. **77**, 094001 (2014)

[45] Chruściński, D., Maniscalco, S.: Degree of non-Markovianity of quantum evolution. Phys. Rev. Lett. **112**, 120404 (2014)

[46] Buscemi, F., Datta, N.: Equivalence between divisibility and monotonic decrease of information in classical and quantum stochastic processes. Phys. Rev. A **93**, 012101 (2016)

[47] Lindblad, G.: On the generators of quantum dynamical semigroups. Comm. Math. Phys. **48**(2), 119–130 (1976)

[48] Gorini, V., Kossakowski, A., Sudarshan, E.C.G.: Completely positive dynamical semigroups of N-level systems. J. Math. Phys. **17**(5), 821–825 (1976)

[49] Davies, E.B.: Quantum Theory of Open Systems. Academic Press, London (1976)

[50] Kato, T.: Perturbation Theory for Linear Operators. Classics in Mathematics. Springer, Berlin (1995)

[51] Feller, W.: Non-Markovian processes with the semigroup property. Ann. Math. Statist. **30**(4), 1252–1253 (1959)

[52] Davis, C., Kahan, W.M.: The rotation of eigenvectors by a perturbation. III. SIAM J. Numer. Anal. **7**(1), 1–46 (1970)

[53] Wedin, P.-Å.: Perturbation bounds in connection with singular value decomposition. BIT **12**, 99–111 (1972)

[54] Ciccarello, F., Lorenzo, S., Giovannetti, V., Palma, G.M.: Quantum collision models: open system dynamics from repeated interactions. Phys. Rep. **954**, 1–70 (2022)

[55] Dyson, F.J.: The radiation theories of Tomonaga, Schwinger, and Feynman. Phys. Rev. **75**(3), 486–502 (1949)

[56] Blanes, S., Casas, F., Oteo, J.A., Ros, J.: The Magnus expansion and some of its applications. Phys. Rep. **470**(5–6), 151–238 (2009)

[57] Canturk, B.: Unification of stochastic matrices and quantum operations for N-level systems. Ann. Phys. **487**, 170367 (2026). arXiv:2503.09333

[58] Doukas, J.: On the emergence of quantum mechanics from stochastic processes. arXiv:2602.22095 (2026)

[59] Gubinelli, M.: Controlling rough paths. J. Funct. Anal. **216**(1), 86–140 (2004)

[60] Feyel, D., de La Pradelle, A.: Curvilinear integrals along enriched paths. Electron. J. Probab. **11**, 860–892 (2006)

[61] Brault, A., Lejay, A.: The non-linear sewing lemma I: weak formulation. Electron. J. Probab. **24**, paper no. 59 (2019)

[62] Brault, A., Lejay, A.: The non-linear sewing lemma II: Lipschitz continuous formulation. J. Differential Equations **293**, 482–519 (2021)

[63] Friz, P.K., Hairer, M.: A Course on Rough Paths, 2nd edn. Universitext. Springer, Cham (2020)